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Econometrics II (Econ 3062)

MODULE

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CHAPTER ONE

REGRESSION ANALYSIS WITH QUALITATIVE INFORMATION

Introduction

Both the dependent and explanatory variables that we have encountered in the preceding chapters were essentially quantitative/continuous. Regression models can also handle qualitative variables as dependent and explanatory variables. In this chapter, we shall consider models that may involve not only quantitative variables but also qualitative variables. Such variables are also known as **indicator variables**, **categorical variables**, **nominal scale variables** or **dummy variables**.

To be specific, in this chapter we shall discuss two main topics: (1) regression with qualitative explanatory variables and (2) regression with qualitative dependent variables.

1.1. REGRESSION WITH DUMMY EXPLANATORY VARIABLES

1.1.1 The Nature of Dummy Explanatory Variables

In regression analysis the dependent variable, or regressand, is frequently influenced not only by ratio scale variables (e.g. income, output, prices, costs, height, temperature) but also by variables that are essentially qualitative, or nominal scale, in nature. Examples are sex, race, color, religion, nationality, geographical region and political party membership. The importance of such variables is evidenced by several studies. For instance, studies found that female workers earn less than male workers, holding other factors constant. Also, in multicultural contexts, black workers are found to earn less than whites. This clearly shows that (whatever the reasons) qualitative variables such as sex and race can influence the dependent variable and should explicitly be included among explanatory variables.

How do you think will these variables be included in regression analysis?

Qualitative variables can be used in regression models just as easily as quantitative variables. However, to include qualitative variables into empirical analysis, one has to change them into quantitative variables. One way to quantify nominal scale variables is by creating artificial variables that take on values of 1 or 0, where 1 indicates the presence of a certain attribute/quality and 0 indicates the absence of that attribute. This artificial representation is possible since nominal scale variables usually indicate the presence or absence of a certain quality or an attribute. For example, if an investigator encounters "the sex of a household head" as a nominal variable in a regression analysis, he/she can use 1 to indicate that the household head is male and 0 to designate that the household head is female. Variables that assume 0 and 1 artificially are called **dummy variables**. Hence, dummy variables can be considered as a method to classify data on qualitative variables into mutually exclusive categories such as presence or absence of an attribute.

In a given regression model, the qualitative and quantitative variables may also occur together, i.e., some variables may be qualitative and others are quantitative. When all explanatory variables are

- **quantitative**, then the model is called a **regression model**,
 - **qualitative**, then the model is called an **analysis of variance model (ANOVA)** and
 - **quantitative and qualitative both**, then the model is called **analysis of covariance model (ANCOVA)**.
- Such models can be dealt within the framework of regression analysis. The usual tools of regression analysis can be used in case of dummy variables.

1.1.2 Regression with only Qualitative Variables (ANOVA Models)

To exemplify the application of dummy variables in regression analysis, let's suppose that an investigator wants to find out if average wheat farm productivity (yield) differs among male and female headed households. This objective can be accomplished within the framework of regression analysis.

To see this, consider the following regression model:

$$Y_i = \beta_0 + \beta_1 D_i + U_i \dots \dots \dots (1.1)$$

Where, Y_i = Wheat farm productivity of the i^{th} Wheat producing household

$$D_i = \begin{cases} 1 & \text{If Household Head is Male} \\ 0 & \text{If Household Head is Female} \end{cases}$$

Note that equation 1.1 has only one dummy variable since the grouping/qualitative variable in equation 1.1 has two categories. Similarly, if a qualitative variable has three categories, we introduce only two dummy variables. For instance, suppose that the investigator we considered above wants to see whether **wheat farm productivity** depends on **education attainment of household head**, where education attainment of household heads' are divided into three mutually exclusive categories: illiterate, elementary school and high school and above. The model can be stated as follows:

$$Y_i = \beta_0 + \beta_1 D_{1i} + \beta_2 D_{2i} + U_i \dots \dots \dots (1.2)$$

$$\text{Where, } D_{1i} = \begin{cases} 1 & \text{If Household Head has Elementary Education} \\ 0 & \text{otherwise} \end{cases}$$

$$D_{2i} = \begin{cases} 1 & \text{If Household Head has High school and above Education} \\ 0 & \text{otherwise} \end{cases}$$

What does the model (1.2) tell us? Assuming that the error term satisfies the usual assumptions of OLS, taking expectation on both sides of (1.2), we obtain:

- Mean wheat farm productivity of households with elementary education:

$$E(Y_i / D_{1i} = 1, D_{2i} = 0) = \beta_0 + \beta_1$$

- Mean wheat farm productivity of households with high school and above schooling:

$$E(Y_i / D_{1i} = 0, D_{2i} = 1) = \beta_0 + \beta_2$$

- Mean wheat farm productivity of illiterate households:

$$E(Y_i/D_{1i} = 0, D_{2i} = 0) = \beta_0$$

In other words, in the multiple regression equation (1.2), the mean wheat farm productivity of illiterate households is given by the intercept, β_0 , and the slope coefficients β_1 and β_2 tell by how much the mean wheat farm productivity of households with elementary education and high school and above schooling differ from the mean wheat farm productivity of illiterate households. But how do we know that these differences are statistically significant? To address this question, let's consider a numerical example.

Numerical Example-1:

Given data of the following form on wheat farm productivity (yield) and education attainment of household heads based on a sample of 120 farm households, we can estimate yield as a function of education attainment which is given in three categories: *illiterate, elementary and high school & above*.

Table-1.1: data on wheat farm productivity and educational attainment

Obs	Yield	D1	D2	Obs	Yield	D1	D2
1	12	0	0	8	12	0	0
2	9	1	0	9	12.8	1	0
3	10.6	0	0	10	12.8	1	0
4	13.4	0	1	.	.	.	.
5	12	1	0	.	.	.	.
6	12	1	0	119	15.6	0	1
7	10.6	1	0	120	18	0	1

Based on the above data, if we estimate equation 1.2, we obtain the following regression result

$$\hat{Y}_i = 11.81 + 0.86D_1 + 2.19D_2 \dots \dots \dots (1.3)$$

t: (49.52)(7.21) (3.22)

P – value: (0.002)(0.000) (0.000) $R^2 = 0.33$ & $n = 120$

Since we are treating illiterate household heads as the benchmark, the coefficients attached to the various dummies are **differential intercepts**, showing by how much the *average value* of *Y*(yield) in the category that receives 1 differs from that of the benchmark category. That is, as the regression result shows, the coefficient of D_1 implies that average yield of households with elementary education and households with high school and above schooling is larger than average yield level of households with no schooling by about **0.86** and **2.19** quintals, respectively. Therefore, the actual mean yield of households with elementary education and high school and above schooling can be obtained by adding the differential yields. Doing so, we obtain that the mean yield of households with elementary education is **12.66 quintals** and that of households with high school and above is **14 quintals**.

But how do we know that these mean yields are statistically different from the mean yield of illiterate households, the benchmark category? This is straightforward. All we have to do is to find out if each of the slope coefficients in the above equation are statistically significant. As can be seen from the regression result, the estimated slope coefficients both households with elementary and high school and above are **statistically significant** as implied by the P-values of the t-statistics.

1.1.3 Regression with a mixture of Qualitative and Quantitative Explanatory Variables: ANCOVA

Sometimes an investigator may deal with a regression analysis to assess the statistical significance of the relationship between quantitative dependent variable and qualitative explanatory variables. In such cases one often uses dummy variable models with some assumptions about their use. For instance, the implicit assumption in this case is that the regression lines for the different groups differ only in the intercept term but have the same slope coefficients.

To exemplify the application of dummy variables with the above implicit assumption, let us suppose that the investigator we considered above wants to analyze the relationship between **wheat farm productivity** and **chemical fertilizer application** for two groups: **male and female headed households**. The relationship can be represented by the following equation:

$$Y_i = \beta_0 + \beta_1 D_i + \beta_2 X_i + U_i \dots \dots \dots (1.4)$$

Where, Y_i = the i^{th} household wheat farm productivity in quintal

$$D_i = \begin{cases} 1 & \text{If Household Head is Male} \\ 0 & \text{If Household Head is Female} \end{cases}$$

X_i = chemical fertilizer applied in Kg

What does the model (1.4) tell us? Assuming that the error term satisfies the usual assumptions of OLS, taking expectation on both sides of (1.4), we obtain:

- Mean wheat farm productivity of male headed households:

$$E(Y_i/D_i = 1) = (\beta_0 + \beta_1) + \beta_2 X_i$$

This is a straight line relationship with intercept $(\beta_0 + \beta_1)$ and slope β_2 .

- Mean wheat farm productivity of female headed households:

$$E(Y_i/D_i = 0) = \beta_0 + \beta_2 X_i$$

This is a straight-line relationship with intercept β_0 and slope β_2 .

Graphical representation of equation 1.4 is given as:

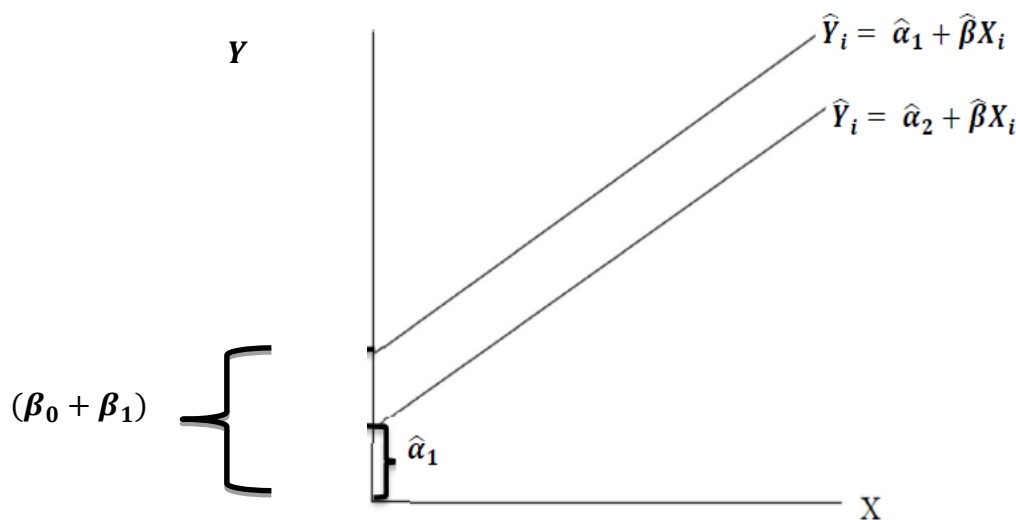


Figure 1.1: Regression lines with a common slope and different intercept

It is evident from equation 1.4 that the coefficient of the dummy variable, β_1 measures the differences in the two intercept terms. Hence, it is called **differential intercept coefficient** because it tells by how much the value of the intercept that receives the value of 1 differs from the intercept coefficient of the benchmark category.

Note that as we did above, *if there is a constant term in the regression equation, the number of dummies defined should be one less than the number of the categories of the variable.* This is because the constant term is the intercept for the base group. As it can be seen from equation 1.4, the constant term, β_0 measures the intercept for female headed households. Furthermore, the constant term plus the coefficient of D_i (i.e., $\beta_0 + \beta_1$) measures the intercept for the male headed households. This interpretation holds as long as the base group is female headed households. But this should not connote that the base group is always female headed households. Any one group may be chosen as the base group, depending on the preference of the investigator.

On the other hand, if we do not introduce a constant term in the regression equation, we can define a dummy variable for each group, and in this case the coefficient of the dummy variables measures the intercepts for the respective groups. However, if we include both the constant term and dummies for all categories of a variable, we will encounter the problem of **perfect multicollinearity** and the regression program either will not run or will omit one of the dummies automatically.

In general, when we are introducing dummy variables, we have to follow the following rule. If the qualitative variable has m categories and the regression equation has a constant intercept, introduce only $m - 1$ dummy variables, otherwise we will fall into what is called **the dummy variable trap**, that is, the problem of perfect multicollinearity.

As you might have noted, so far we have been dealing with regression analysis where there is only one qualitative variable. In practice, however, we may have more than one qualitative variable affecting the

dependent variable. **Then, how will you introduce more than one qualitative variable in your regression analysis?**

The introduction of dummy variables for more than one qualitative variable in a regression analysis is straightforward. That is, for each qualitative variable we introduce one or more than one dummy following the rule mentioned above. **For example**, suppose that we want to analyze the determinants of household consumption(C), and hence we have data on:

1. **Y**: income of households
2. **S**: the sex of the head of the household.
3. **A**: age of the head of the household, which is given in three categories: < 25 years, 25 to 50 years, and > 50 years.
4. **E**: education of the head of household, also in three categories: < high school, high school but < college degree, and college degree and above.

Following the rule of dummy variables, we can include these qualitative variables in the form of dummy variables as follows:

$$C_i = \beta_0 + \beta_1 D_1 + \beta_2 D_2 + \beta_3 D_3 + \beta_4 D_4 + \beta_5 D_5 + \beta_6 Y_i + U_i \dots \dots \dots (1.5)$$

$$\text{Where, } D_1 = \begin{cases} 1 - \text{If sex is Male} \\ 0 - \text{If sex is Female} \end{cases}$$

$$D_2 = \begin{cases} 1 - \text{If age is } < 25 \text{ years} \\ 0 - \text{otherwise} \end{cases}$$

$$D_3 = \begin{cases} 1 - \text{If age is b/n 25 and 50 years} \\ 0 - \text{otherwise} \end{cases}$$

$$D_4 = \begin{cases} 1 - \text{If } < \text{High school} \\ 0 - \text{otherwise} \end{cases}$$

$$D_5 = \begin{cases} 1 - \text{If } \geq \text{high school and } < \text{college degree} \\ 0 - \text{otherwise} \end{cases}$$

For each category the number of dummy variables is one less than the number of classifications.

The assumption made in the dummy-variable method is that it is only the intercept that changes for each group but not the slope coefficients (i.e., coefficients of **Y**). The intercept term for each individual is obtained by substituting the appropriate values for **D**₁ through **D**₅. For instance, for a male, age < 25, with college degree, we have **D**₁ = 1, **D**₂ = 1, **D**₃ = 0, **D**₄ = 0, **D**₅ = 0 and hence the intercept is **β**₀ + **β**₁ + **β**₂. For a female, age > 50 years, with a college degree, we have **D**₁ = 0, **D**₂ = 0, **D**₃ = 0, **D**₄ = 0, **D**₅ = 0 and hence the intercept term is just **β**₀.

Furthermore, the coefficients of the dummies are interpreted as the differences between the average consumption of the omitted (base) category and the category represented by the dummy under consideration, keeping other things constant. For instance, **β**₁ in equation 1.5 above is interpreted as, keeping other things

constant, the average consumption expenditure of male headed households is greater/less than that of their female headed counterparts.

The other most important question, perhaps, is how to check the statistical significance of the differences among groups (if any). For this purpose we simply test whether the coefficients of the dummies are statistically significant or not by using the usual procedure of hypothesis testing, i.e., by using the standard error test or the students t-test procedures.

1.1.4 Dummy Variables in Seasonal Analysis

Many economic time series that are based on monthly or quarterly data exhibit seasonal patterns, i.e., they show regular oscillatory movements. Examples are sales of department stores at Christmas and other major holiday times, demand for money by households at holiday times, prices of crops right after harvesting season, demand for air travel, etc.

Often it is desirable to remove that seasonal factor from time series, so that one can concentrate on the other components, such as the trend. The process of removing the seasonal component from a time series is known as **deseasonalization** or **seasonal adjustment**, and the time series thus obtained is called the **deseasonalized**, or **seasonally adjusted** time series. Usually the important economic series, such as the unemployment rate, the consumer price index (*CPI*), the producer price index (*PPI*), and the industrial production index that we see in different reports are published in their seasonally adjusted form.

One of the methods used to **deseasonalize** a time series is the method of dummy variables. On the other hand, regression on seasonal dummy variables such as quarterly dummy or monthly dummy is a simple way to estimate seasonal effects in time series data sets.

Example:

To illustrate the dummy variables technique in seasonal analysis, suppose that we have quarterly data on sales of refrigerators over the years 1978 through 1985 given in the following table.

Table-1.2: Quarterly Data on Sales of refrigerators (in thousands) (1978-I to 1985-IV)

Year	Quart	FRIG	Year	Quart	FRIG	Year	Quart	FRIG	Year	Quart	FRIG
1978	I	1317	1980	I	1277	1982	I	943	1984	I	1429
	II	1615		II	1258		II	1175		II	1699
	III	1662		III	1417		III	1269		III	1749
	IV	1295		IV	1185		IV	973		IV	1117
1979	I	1271	1981	I	1196	1983	I	1102	1985	I	1242

	II	1555		II	1410		II	1344		II	1684
	III	1639		III	1417		III	1641		III	1764
	IV	1238		IV	919		IV	1225		IV	1328

But first let us look at the data, which is shown in Figure 1.2 below.

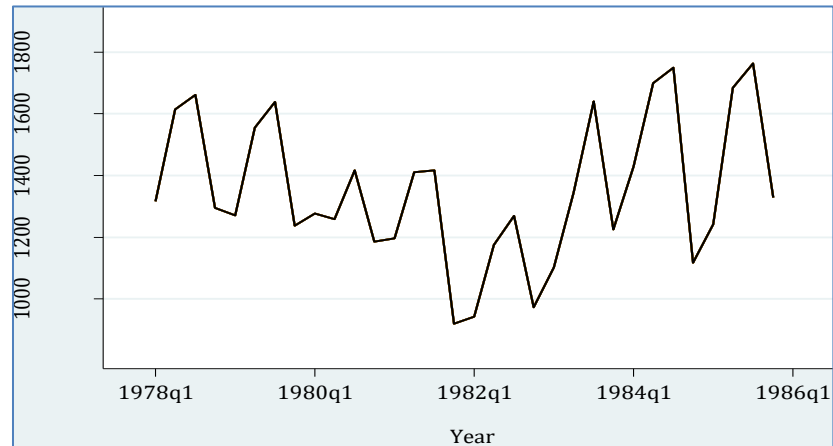


Figure 1.2: Quarterly Data on Sales of refrigerators (in thousands) (1978-I to 1985-IV)

This figure suggests that perhaps there is a seasonal pattern in the data associated with the various quarters. **How can we measure such seasonality in refrigerator sales?** We can use the seasonal dummy regression to estimate the seasonal effect. To do this, let's treat the first quarter as the reference quarter and assign dummies to the second, third, and fourth quarters (i.e., the data setup looks the table below). That is, to have the following model:

$$Y_t = \beta_1 + \beta_2 D_2 + \beta_3 D_3 + \beta_4 D_4 + U_t \dots \dots \dots (1.6)$$

Where, Y_t = sales of refrigerators (in thousands) and D_2, D_3 & D_4 are seasonal dummies defined as,

Where, $D_2 = \begin{cases} 1 & \text{in quarter II} \\ 0 & \text{in other quarters} \end{cases}$

$D_3 = \begin{cases} 1 & \text{in quarter III} \\ 0 & \text{in other quarters} \end{cases}$

$D_4 = \begin{cases} 1 & \text{in quarter IV} \\ 0 & \text{in other quarters} \end{cases}$

Table-1.3: Quarterly Data on Sales of refrigerators (in thousands) with seasonal dummies

Quarter	FRIG	D2	D3	D4	Quarter	FRIG	D2	D3	D4
1978-I	1317	0	0	0	1984-I	1277	0	0	0
-II	1615	1	0	0	-II	1258	1	0	0
-III	1662	0	1	0	.	.	.	.	.
-IV	1295	0	0	1	.	.	.	.	.
1979-I	1271	0	0	0	1985-I	1242	0	0	0
-II	1555	1	0	0	-II	1684	1	0	0
-III	1639	0	1	0	-III	1764	0	1	0
-IV	1238	0	0	1	-IV	1328	0	0	1

From the data on refrigerator sales given in Table 1.3, we obtain the following regression results:

$$Y_t = 1222.125 + 245.375D_2 + 347.625D_3 - 62.125D_4 \dots \dots \dots (1.7)t: (20.37)^{***}(2.89)^{***}(4.10)^{***}(-0.73)^*$$

$$R^2 = 0.53$$

Where, the values in the brackets are student's t-statistics for the respective coefficients.

➤ Interpretation:

Since we are treating the first quarter as the benchmark, the coefficients attached to the various dummies are *differential intercepts*, showing by how much the *average value* of Y (*FRIG Sales*) in the quarter that receives a dummy value of 1 differs from that of the benchmark quarter. Put differently, the coefficients on the seasonal dummies will give the seasonal increase or decrease in the average value of Y relative to the base season. For instance, the coefficient of quarter-2 dummy; D_2 implies that average volume of refrigerator sales in season two is larger than it in season one by 245.375 units and the difference is statistically significant at 1% level of significance. If you add the various *differential intercept* values to the benchmark average value of 1222.125, you will get the average value for the various quarters.

1.2. Regression with Dummy Dependent Variables

So far in this chapter, we have been considering models where the explanatory variables are qualitative or dummy variables.

In this section, we shall look at models where the dependent variable, Y is itself a qualitative variable. Such models are called **limited dependent variable models** or **qualitative or categorical variable models**. This dummy variable can take on two or more values. However, we shall concentrate on the binary case where Y_i can take only two values. For example, we want to study the *determinants of house ownership* by households in Arba Minch town. Since a household may either own a house or not, house ownership is a **Yes** or **No** decision. Hence, the response variable, or regressand, can take only two values, say, 1 if the household owns a house and 0 if s/he does not own a house. A second example would be a model of women labor force participation (LFP). The dependent variable in this case is the labor force participation (LFP) which would take the value of one if the women do participate in the labor force and a value of zero if the women do not participate in the labor force. A third example would be the poverty status of households in rural Ethiopia. The dependent variable in this case is the poverty status of households which would take the value of one if the household is poor or zero if the household is non-poor. In all of the above instances, the dependent variable is a dichotomous/dummy/binary/qualitative/limited variable.

To explain the models of the above nature, various types of explanatory variables could be included: both continuous variables such as age and income and dichotomous variables such as sex and race or exclusively continuous or exclusively qualitative.

If the dependent variable or the response variable (Y) is qualitative/dichotomous/limited/dummy or binary in nature and the Y assumes only a restricted value, we call the model a **Limited Dependent Variable Model (LDVM)**.

Binary choice models are commonly used for micro analysis in social sciences and medical research. There are three types of **Limited Dependent Variable Models**:

- A. The Linear Probability Models
- B. The Logit Models
- C. The Probit Models

1.2.1 The Linear Probability Models

The linear probability model applies the linear model,

$$Y_i = X_i\beta + U_i \dots \dots \dots (1.8)$$

to the case where Y is a dichotomous variable taking the value zero or one, where we require the assumption that $E(U_i) = 0$.

- The conditional expectation of the i^{th} observation is given by:

$$E(Y_i/X_i) = X_i'\beta = P_i = \Pr(Y_i = 1/X_i)$$

The conditional expectation of the dependent variable is equal to the **probability of something happening**, $\Pr(Y_i = 1/X_i)$.

The conditional expectation $E(Y_i/X_i)$ in this model has to be interpreted as the probability that $Y_i = 1$ given the particular value of X_i . In practice, however, there is nothing in this model to ensure that these probabilities will lie in the admissible range (0, 1). Since Y_i can only take the two values of 0 or 1, it follows that the error term U_i can only take the two values of $-X_i'\beta$ or $1 - X_i'\beta$.

- **What do you think is the difference between a regression model where the regressand Y is quantitative and a model where it is qualitative?**

In a model where Y is quantitative, our objective is prediction of the average value of the dependent variable from the given values of explanatory variables. That means, in such models the very objective of regression analysis is to determine, $E(Y_i/X_1, X_2, X_3, \dots, X_k)$. On the other hand, in models where Y is qualitative, our objective is to **predict the probability of something happening**, such as owning a smartphone, or owning a house, or being non-poor, etc. In other words, in binary regression models,

$$E(Y_i/X_i) = X_i'\beta = P_i \rightarrow \text{probability of success.}$$

$$\text{and hence, } 1 - X_i'\beta = 1 - P_i \rightarrow \text{probability of failure}$$

✓ That is why qualitative response regression models are often called **probability models**.

From the assumption that $E(U_i) = 0$, it follows that the probabilities of these two events are given by $1 - X_i'\beta$ and $X_i'\beta$, respectively. Thus, the probability distribution of Y_i and U_i , can be given by the following table.

Value of Y_i	U_i	Probability of U_i	Probability of Y_i
1	$1 - X_i'\beta$	$X_i'\beta$	$X_i'\beta$
0	$-X_i'\beta$	$1 - X_i'\beta$	$1 - X_i'\beta$
Total	1	1	1

Both the dependent variable (Y_i) and the error term (U_i) assume only two values with some probabilities ($X_i'\beta$ and $1 - X_i'\beta$). If a variable assumes two values with a probability, that variable is called **Bernoulli variable** and its distribution is called **Bernoulli distribution**. Therefore, the dependent variables do not follow a normal distribution but they follow the Bernoulli distribution. The dependent variable follows the Bernoulli distribution with a mean values of P_i or $X_i'\beta$ and variance of $P_i(1 - P_i)$ or $X_i'\beta(1 - X_i'\beta)$. While the disturbance term follows the Bernoulli distribution with a mean value of zero and variance of $P_i(1 - P_i)$ or $X_i'\beta(1 - X_i'\beta)$. As can be seen, the variance of the disturbance term (U_i) is not constant. It is a function of explanatory variables (X_i). Thus, the disturbance term in the case of LPM is heteroscedastic. Clearly, since its variance depends on X_i , U_i is heteroscedastic. Therefore, the OLS estimation of LPMs are not efficient.

✚ Limitations of Linear Probability Models (LPM)

The problem with the linear probability model (LPM) is that it models the probability of $Y = 1$ as being linear. That means, $Pr(Y = 1|X) = \beta_0 + \beta_1 X$. If we were to use OLS regression line, we would get some straight line, perhaps at higher values of X we would get values of Y above 1 and for low values of X we would get values of Y below 0. But we cannot have probabilities that fall below 0 or above 1.

If the values of the dependent variable (Y_i) are limited or discrete, we cannot use ordinary least square method (OLS). *This is because, unless we also restrict the values of the explanatory variable (X_i), we may get negative values of dependent variable for small values of explanatory variable and values greater than one for large values of the explanatory variable.* The estimated probability lies outside the limit zero and one.

➤ The following are the main problems with linear probability models:

A. The dependent (Y_i) and the disturbance term (U_i) are not normally distributed

One of the assumption of linear regression model is that the disturbance term (U_i), the dependent variable (Y_i) and OLS estimates ($\hat{\beta}_i$) are all normally distributed. But, as we have noted above, both the dependent variable and the disturbance term follows **Bernoulli distribution**. The error term (U_i) follows Bernoulli distribution with average

value of zero and variance of $P_i(1 - P_i)$. Similarly, the dependent variable follows Bernoulli distribution with mean P_i and variance of $P_i(1 - P_i)$.

$$U_i \sim N[0, P_i(1 - P_i)]$$

$$Y_i \sim [X_i'\beta, P_i(1 - P_i)]$$

Both the variance of U_i and Y_i are the same, *i. e.*, $P_i(1 - P_i)$ and a function of explanatory variable (X_i). Thus, the variance of the error term, $\text{var}(U_i)$, is not constant or it varies with the values of explanatory variable (X_i). Since the error term (U_i) is not normally distributed, so does the distribution of our estimates. If the estimates are not normally distributed, we cannot use them for hypothesis testing and estimation (inference). But, this limitation of LPM is not a series problem because as sample size increases, the distribution of the error term (U_i) approaches normal distribution.

B. Violation of the assumption of a homoskedastic error term

One of the assumptions of linear regression model is that the variance of the disturbance term is constant (homoscedastic). Since the variance of the error term and the variance of the dependent variable is the same, the assumption of homoscedastic variance implies that the average deviation of the dependent variable (Y_i) from estimated value is constant. However, in LPM, the variance of the disturbance term (U_i) is given by the following expression:

$$\text{Var}(U_i) = P_i(1 - P_i) = (1 - X_i'\beta)X_i'\beta = f(X_i) \dots \dots \dots (1.9)$$

The variance of the error term is a function of the explanatory variable (X_i) which shows that the variance of U_i is heteroskedastic. That is, the variance of the error term varies with the explanatory variable, X_i .

C. Violation of one of the axioms of probability (the probability may lie outside the interval; $0 \leq P_i \leq 1$)

Since $E(Y_i/X_i)$ in the linear probability models measure the conditional probability (P_i) of the event **Y** occurring given **X**, it must necessarily lie between 0 and 1. Although this is true a priori, there is no guarantee that $\hat{Y}_i$ the estimators of $E(Y_i/X_i) = P_i$, will necessarily fulfill this restriction, and *this is the real problem with the estimation of the LPMs by OLS*. There are two ways to get rid of this problem. One is to estimate the LPM by the usual OLS method and find out whether the estimated $\hat{Y}_i$ lie between 0 and 1. If some are less than 0 (that is, negative), we equate $\hat{Y}_i$ with zero ($Y_i = 0$), if they are greater than 1, Y_i can be equated with one ($Y_i = 1$). The second solution is to devise an estimating technique that will guarantee that the estimated conditional probabilities $\hat{Y}_i$ will lie between 0 and 1. In this respect, the Logit and Probit models will guarantee that the estimated probabilities will indeed lie between the logical limits 0 and 1.

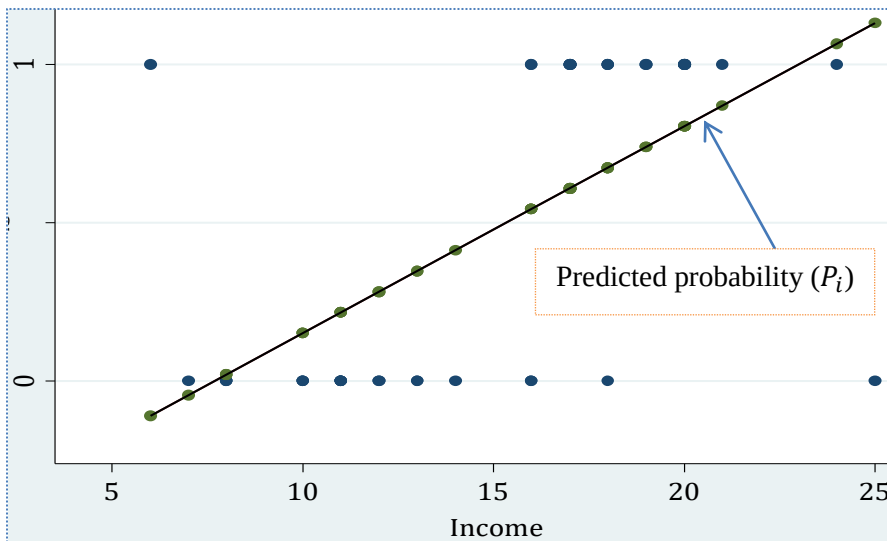


Figure 1.3: The scatter plot of the actual Y and predicted probabilities against the explanatory variables

As the above scatter plot shows (which is drawn based on the data on home ownership status given in table-1.4 below), the predicted probability (P_i) lies outside the limiting values, $0 \leq P_i \leq 1$ for some values of X_i . This violates one of the assumptions of the axioms of probability which states that probability lies only between 0 and 1.

D. The conventionally computed value of R^2 is not dependable

To see why, consider figure 1.3 above. Corresponding to a given X , Y is either 0 or 1. Therefore, all the Y values will either lie along the X axis or along the line corresponding to 1. That is, when we regress Y_i on the given explanatory variable (X_i) in the case of linear regression model, we will get the fitted regression line given above (i.e., Figure_1.3). Therefore, generally no LPM is expected to fit such a scatter well. As a result, the conventionally computed R^2 is likely to be much lower for such models. In most practical applications, the R^2 ranges between 0.2 to 0.6.

Numerical Example

To illustrate the method of Linear Probability Model and the points raised above about it, let's assume that we want to study home ownership status of households in a given town. Assume that we have data for 40 households on home ownership status and income of households (in thousands of Birr) as given below.

Table-1.4: Hypothetical data on home ownership status ($Y = 1$ if owns home, $0 =$ otherwise) and income, X (thousands of Birr)

Household	Y	X	Household	Y	X
1	0	8	21	1	24
2	1	20	22	0	16
3	1	18	23	0	12
4	0	11	24	0	11
5	0	12	25	0	16
6	1	19	26	0	11
7	1	20	27	1	20
8	0	13	28	0	18

9	1	20	29	0	11
10	0	10	30	0	10
11	1	17	31	1	17
12	1	18	32	0	13
13	0	14	33	1	21
14	0	25	34	1	20
15	1	6	35	0	11
16	1	19	36	0	8
17	1	16	37	1	17
18	0	10	38	1	16
19	0	8	39	0	7
20	1	18	40	1	17

Based on this data, the LPM estimated by OLS is as follows:

$$\begin{aligned} \hat{Y}_i &= -0.5 \\ &+ 0.0652 \text{Income} \dots \dots \dots (1.9) SE: (0.2096) (0.0134) \\ t: (-2.39)^{**} (4.88)^{***} \quad R^2 &= 0.3854 \end{aligned}$$

Interpret the Regression Result

The intercept of -0.5 gives the “probability” that a household with zero income will own a house. Since this value is negative, and since probability cannot be negative, we treat this value as zero, which is sensible in the present instance. The slope value of 0.0652 means that for a unit change in income (here *Birr 1000*), on average the probability of owning a house increases by 0.0652 or 6.52 percent.

Table-1.5 below shows the estimated probabilities, $\hat{Y}_i$, for the various income levels. The most noticeable feature of this table is that two estimated values are negative and two values are in excess of 1, demonstrating clearly the point made earlier that, although $E(Y_i/X_i)$ is positive and less than 1, their estimators, $\hat{Y}_i$, need not be necessarily positive or less than 1. This is one reason that the LPM is not the recommended model when the dependent variable is dichotomous.

Table-1.5: Actual Y_i and predicted $\hat{Y}_i$

Y	$\hat{Y}_i$	Y	$\hat{Y}_i$	Y	$\hat{Y}_i$	Y	$\hat{Y}_i$
0	0.022	1	0.609	1	1.065	1	0.609
1	0.804	1	0.674	0	0.543	0	0.348
1	0.674	0	0.413	0	0.280	1	0.869
0	0.217	0	1.130	0	0.217	1	0.804
0	0.283	1	-0.109	0	0.543	0	0.217
1	0.739	1	0.739	0	0.217	0	0.022
1	0.804	1	0.543	1	0.804	1	0.609

0	0.348	0	0.152	0	0.674	1	0.543
1	0.804	0	0.022	0	0.217	0	-0.043
0	0.152	1	0.674	0	0.152	1	0.609

Even if the estimated Y_i were all positive and less than 1, the LPM still suffers from the problem of heteroscedasticity, which can be seen readily from (1.9). As a consequence, we cannot trust the estimated standard errors reported in (1.10) above.

1.2.2. Alternatives to the Linear Probability Models (LPM)

In the previous section, we have seen the simplest and the less commonly used type of binary regression model, called the linear probability model (LPM). Linear probability model (LPM) is plagued with the following problems.

- *Non-normality of the distribution of the disturbance term (U_i)*
- *Heteroscedastic disturbance term*
- *Low value of R^2*
- *Predicted probability lies outside the natural interval, $0 \leq P_i \leq 1$*

We can remove the problem of non-normality of the error term by increasing the sample size. As sample size increases, the distribution of the error term which follows Bernoulli distribution in LPM will approach normal distribution. Similarly, we can also remove the problem of heteroscedasticity in LPM.

But, the serious limitation with LPM is that the predicted probability lies outside the natural limit: 0 and 1, because LPM assumes a linear relationship between the predicted probability and the level of the explanatory variable (X_i). In other words, the problem with LPM is that it assumes that the probability of success or something happening (P_i) is a linear function of explanatory variable(s) (X_i). That is, in our house ownership example, the probability of owning a house (P_i) changes by the same amount at both low and high level of income.

Therefore, the fundamental problem with LPM is that $E(Y_i/X_i) = P_i = \text{pr}(Y_i = 1/X_i)$ increases linearly with X_i . That means, the marginal or incremental effect of X_i remains constant at all levels of income (X_i). Thus, in our home ownership example we found that as X increases by a unit (Birr 1000), the probability of owning a house increases by the same constant amount of **0.0652**. This is so whether the income level is Birr 8000, Birr 10,000, Birr 18,000, or Birr 24,000. As can be seen from figure-1.3 above, the predicted probability (P_i) is a linear function of explanatory variable (X_i) which seems patently unrealistic.

In reality one would expect that P_i is nonlinearly related to X_i : at very low income a household will not own a house but at a sufficiently high level of income, say, X^* , it most likely will own a house. Any increase in income beyond X^* will have little effect on the probability of owning a house. Thus, at both ends of the income distribution, the probability of owning a house will be virtually unaffected by a small increase in X .

Therefore, what we need is a (probability) model that has these two features: (1) As X_i increases, $P_i = E(Y_i = 1/X_i)$ increases but never steps outside the 0–1 interval, and (2) the relationship between P_i and X_i is non-linear, that is, “one which approaches zero at slower and slower rates as X_i gets small and approaches one at slower and slower rates as X_i gets very large.”

In this respect, two alternative nonlinear models (1) The Logit Model & (2) The Probit Model were proposed.

The very objective of the Logit and Probit models is to ensure/guarantee that the predicted probability of the event occurring given the value of explanatory variable remains within the natural [0, 1] bound. That means,

$$0 \leq \Pr(Y = 1|X) \leq 1 \text{ for all } X$$

This requires a *nonlinear functional form for the probability*. This can be possible if we assume that the dependent or the error term (U_i) follows some sorts of cumulative distribution functions (CDF). The two important nonlinear functions which are proposed for this are the logistic CDF and the normal CDF.

$$\Pr(Y_i = 1/X_i) = P_i = G(\beta_0 + \beta_1 X_i) = G(Z_i)$$

Where, G is a function taking on values strictly between 0 and 1. That means, $0 \leq G(Z_i) \leq 1$, for all real numbers Z_i . This ensures that the predicted probability (P_i) strictly lies between 0 and 1.

The logistic distribution function and the Cumulative Normal Distribution Function can be represented graphically as follows.

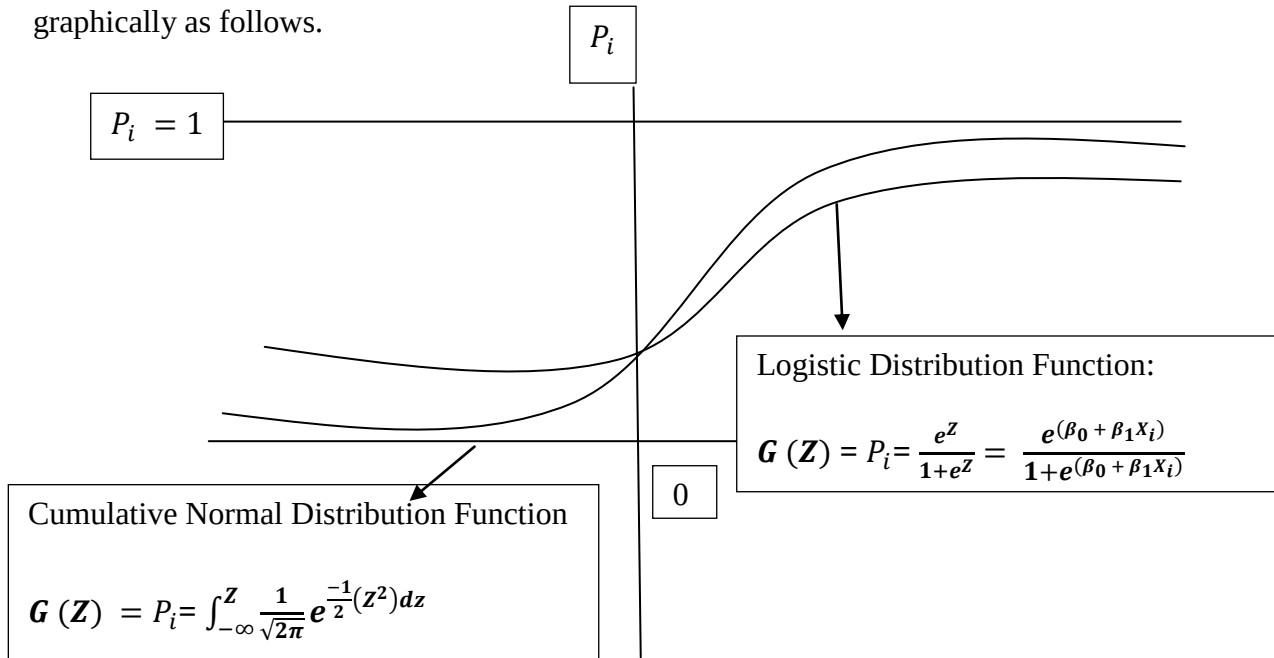


Figure1.4: Comparison of the two distribution functions which are commonly used in binary regression analysis

1.2.2.1. The Logit Model

For Logit model, the logistic cumulative distribution function, $G(Z_i)$ is defined as follows:

$$\begin{aligned}
 G(Z_i) = P_i &= \frac{\exp(Z_i)}{1 + \exp(Z_i)} = \frac{e^{Z_i}}{1 + e^{Z_i}} = \frac{1}{1 + e^{-Z_i}} = \frac{1}{1 + e^{-(\beta_0 + \beta_1 X_i)}} \\
 &= \frac{e^{(\beta_0 + \beta_1 X_i)}}{1 + e^{(\beta_0 + \beta_1 X_i)}} \dots \dots \dots (1.10)
 \end{aligned}$$

Therefore, $P_i = \frac{e^{Z_i}}{1 + e^{Z_i}}$,

Where, $Z_i = \beta_0 + \beta_1 X_i$.

As Z_i ranges from $-\infty$ to $+\infty$, predicted probability of an event occurring (P_i) ranges from 0 to 1. In other words, as X_i ranges from $-\infty$ to $+\infty$, the predicted probability of an event occurring (P_i) ranges from 0 to 1. Moreover, the predicted probability of an event occurring (P_i) is non-linearly related with explanatory variable (X_i). Thus, the Logit model satisfies the two conditions; namely (1) $0 \leq P_i \leq 1$ and (2) P_i is **non-linearly related with X_i** .

The following scatter plot shows the relationship between the predicted probability of owning a house (P_i) and the level of explanatory variable (income). As can be seen from the figure, the predicted probability of owning a house lies only in the natural limit, $0 \leq P_i \leq 1$. The relationship between the predicted probability of an event occurring (P_i) and the explanatory variable is also nonlinear. That means, at lower and higher income level, the change in the predicted probability of owning a house is small for a given level of income.

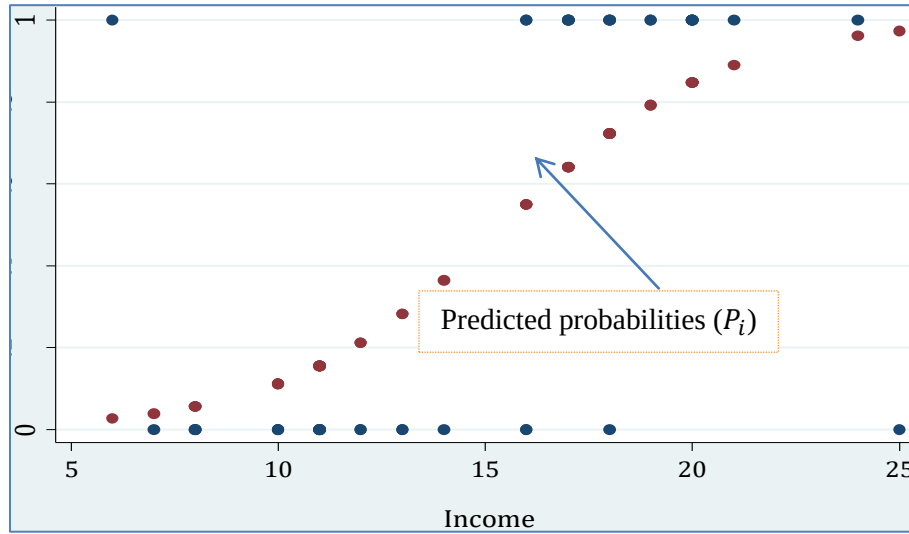


Figure 1.5: The scatterplot of the actual Y and predicted probabilities against the explanatory variable

Therefore, when we use Logit model, we can limit the estimated probabilities inside the 0-1 range as shown above.

But, while we are ensuring that the predicted probability of an event occurring (P_i) lies in the natural interval, $0 \leq P_i \leq 1$, we have created estimation problem because P_i is nonlinear in parameters and in explanatory variables and we cannot apply OLS. However, we can linearize the Logit model as follows.

If P_i = probability of an event occurring/perhaps probability of owning a house

$1 - P_i$ = the probability of not owning a house

$$P_i = \frac{e^{Z_i}}{1 + e^{Z_i}}$$

$$1 - P_i = \frac{1}{1 + e^{Z_i}}$$

Take the ratio of the probability of an event happening (P_i) to the probability of an event not happening ($1 - P_i$) and the resulting ratio is called **odds ratio**

$$\frac{P_i}{1 - P_i} = \frac{\frac{e^{Z_i}}{1 + e^{Z_i}}}{\frac{1}{1 + e^{Z_i}}} = e^{Z_i}$$

$$\frac{P_i}{1 - P_i} = e^{Z_i} \dots \dots \dots (1.11)$$

Take the natural log of the above **odds ratio** and the resulting equation is called **logit**.

$$\ln\left(\frac{P_i}{1 - P_i}\right) = L_i = Z_i$$

$$L_i = Z_i = \beta_0 + \beta_1 X_i \dots \dots \dots (1.12)$$

Where, L_i is called Logit which is linearly related with X_i

X_i is the explanatory variable (this can be extended to a matrix of multiple explanatory variables)

✚ Characteristics of Logit Model

- a. As Z_i goes from $-\infty$ to $+\infty$, the predicted probability (P_i) goes from 0 to 1. In other words, as X_i goes from $-\infty$ to $+\infty$, the predicted probability (P_i) goes from 0 to 1. This implies that in Logit model, the predicted probability (P_i) lies in the natural limit, $0 \leq P_i \leq 1$.
- b. Even if the Logit (L) is linear in X , the probabilities themselves are not. This property is in contrast with the LPM model where the probabilities increase linearly with X .
- c. Although we have included only a single regressor, in the preceding model, one can add as many regressors as may be dictated by the underlying theory.
- d. If the Logit (L) is positive, it means that when the value of the regressor increases, the odds that the regressand equals 1 (meaning some event of interest happens) increases. If the logit (L) is negative, the odds that the regressand equals 1 decrease as the value of X increases. To put it differently, the Logit becomes negative and increasingly large in magnitude as the odds ratio decreases from 1 to 0 and becomes increasingly large and positive as the odds ratio increases from 1 to infinity.
- e. More formally, the interpretation of the Logit model given above is as follows: β_1 , the slope, measures the change in L for a unit change in X , that is, it tells how the log-odds in favor of owning a house change as income changes by a unit (Birr 1000) in our example. The intercept β_0 is the value of the log odds in favor of owning a house if income is zero.

- f. Given a certain level of income, say, X^* , if we actually want to estimate not the odds in favor of owning a house but the probability of owning a house itself, this can be done directly from $P_i = \frac{e^{Z_i}}{1 + e^{Z_i}}$, once the estimates of $\beta_0 + \beta_1$ are available.
- g. Whereas the LPM assumes that P_i is linearly related to X_i , the Logit model assumes that the log of the odds ratio is linearly related to X_i .

Estimation and interpretation of the Logit model

The most common way to estimate binary response models is to use the **method of maximum likelihood (MML)**. ML estimation is maximizing the likelihood function with respect to the parameters. A parameter vector β at which the likelihood takes on its maximum value is called a **maximum likelihood estimate (MLE)** of the parameters. Let's first construct likelihood function, i.e., joint distributions.

The likelihood contribution of observation i with $Y_i = 1$ is given by $P\{Y_i = 1|X_i\}$ as a function of the unknown parameter vector β , and similarly for $Y_i = 0$. Assuming the observations are independent, the likelihood function (equation) for the entire sample is thus given by the joint probability. The joint density of the entire sample is just the product of the densities of the individual observations.

That is, suppose we have a *random sample* of n observations. Letting $f_i(Y_i)$ denote the probability that $Y_i = 1$ or 0, the joint probability of observing the n Y values, i.e., $f(Y_1, Y_2, \dots, Y_n)$ is given as:

$$f(Y_1, Y_2, \dots, Y_n) = \prod_1^n P_i^{Y_i} (1 - P_i)^{1-Y_i} \dots \dots \dots (1.13)$$

Where, $\prod$ is the product operator.

➤ **NB:** The likelihood of a Bernoulli variable is the probability of success to the power of Y_i times the probability of failure to the power of $1 - Y_i$.

The joint probability given in Eq. (1.13) is known as the **likelihood function (LF)**. If we take natural logarithm of eq. (1.13), we obtain what is called the **log likelihood function (LLF)**:

$$\begin{aligned} \ln f(Y_1, Y_2, \dots, Y_n) &= \sum_1^n [Y_i \ln P_i + (1 - Y_i) \ln(1 - P_i)] \\ \ln f(Y_1, Y_2, \dots, Y_n) &= \sum_1^n [Y_i \ln P_i - Y_i \ln(1 - P_i) + \ln(1 - P_i)] \\ \ln f(Y_1, Y_2, \dots, Y_n) &= \ln L = \sum_1^n \left[Y_i \ln \left(\frac{P_i}{1 - P_i} \right) \right] + \sum_1^n \ln(1 - P_i) \end{aligned}$$

But we know that $1 - P_i = \frac{1}{1 + e^{X_i\beta}}$ and $\ln \left(\frac{P_i}{1 - P_i} \right) = X_i\beta$

Therefore, we can rewrite the above LLF as:

$$\ln L = \sum_1^n [Y_i(X\beta)] + \sum_1^n \ln \left(\frac{1}{1 + e^{(X\beta)}} \right) \dots \dots \dots (1.14)$$

As you can see from (1.14), the log likelihood function is a function of the parameters β_0 and β_1 , since the X_i are known. In ML our objective is to maximize the LF (or LLF), that is, to obtain the values of the unknown parameters in such a manner that the probability of observing the given Y 's is as high (maximum) as possible. For this purpose, we differentiate (1.14) partially with respect to each parameter, set the resulting expressions to zero and solve the resulting expressions. But the resulting expressions become highly nonlinear in the parameters and no explicit solutions can be obtained. However, the estimates of the parameters can easily be computed with the aid of Software packages such as EViews, Stata or any other Software Package. Thus, here under we shall estimate Logit and Probit models using Stata.

Equally important is the interpretation of the results. Therefore, below we shall discuss on how to interpret the regression results of Binary Logit Models with the aid of numerical examples (estimated through the principle of Maximum Likelihood (MML) by Software packages). There are three ways of interpreting the regression results of a Logit model:

- A. Logit interpretation
- B. Odds ratio interpretation
- C. Probability interpretation (Marginal Effect Interpretation)

Example-1

Suppose that a Mathematical Economics instructor gave routine weekly exercises to her section “A” students in the previous semester. She wants to study the effect of the performance in these routine exercises on the final grade of students in the course, Mathematical Economics. At the end of the semester, she found average scores in exercise (ASE) for each student.

Table -1.6: Cross-sectional Data on GPA, ASE, PC and GRADE for 32 Students

Students	GPA	ASE	PC	GRADE (A=1, rest=0)
1	2.66	20	0	0
2	2.89	22	0	0
3	3.28	24	0	0
4	2.92	12	0	0
5	4	21	0	1
6	2.86	17	0	0
7	2.76	17	0	0
8	2.87	21	0	0
9	3.03	25	0	0
10	3.92	29	0	1
11	2.63	20	0	0

12	3.32	23	0	0
13	3.57	23	0	0
14	3.26	25	0	1
15	3.53	26	0	0
16	2.74	19	0	0
17	2.75	25	0	0
18	2.83	19	0	0
19	3.12	23	1	0
20	3.16	25	1	1
21	2.06	22	1	0
22	3.62	28	1	1
23	2.89	14	1	0
24	3.51	26	1	0
25	3.54	24	1	1
26	2.83	27	1	1
27	3.39	17	1	1
28	2.67	24	1	0
29	3.65	21	1	1
30	4	23	1	1
31	3.1	21	1	0
32	2.39	19	1	1

The dependent variable is the final grade of students for Mathematical Economics ($Y_i = 1$, if the student scores A and $Y_i = 0$, otherwise). Other explanatory variables include the Grade Point Average (GPA) of students and Personal –Computer ownership ($PC = 1$, if the student has PC and $PC = 0$, if the student does not own PC).

We are going to estimate the following Logit model:

$$L_i = \beta_0 + \beta_1 GPA + \beta_2 ASE + \beta_3 PC + U_i$$

✚ How can we interpret regression results?

In what follows, we present the estimation and interpretation of the model defined above. We can interpret the regression results using: **Logit, Odds ratio and marginal effect interpretations.**

A. Logit Interpretation of the Estimated Logit Model

Table-1.7 below presents the logit regression result obtained using Stata14.

Table-1.7: Logistic Regression Result

Logistic regression				Number of obs	=	32
				LR chi2(3)	=	15.40
				Prob > chi2	=	0.0015
Log likelihood = -12.889633				Pseudo R2	=	0.3740
grade	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
gpa	2.826113	1.262941	2.24	0.025	.3507938	5.301432
ase	.0951577	.1415542	0.67	0.501	-.1822835	.3725988
pc	2.378688	1.064564	2.23	0.025	.29218	4.465195
_cons	-13.02135	4.931325	-2.64	0.008	-22.68657	-3.35613

We have to interpret the coefficients of the regressors of the logit regression results as log of the odds ratio. Coefficients of GPA and ASE in the table above measures the change in the estimated logit for a unit change in the value of regressors (holding other regressors constant). Thus,

- ☛ The GPA coefficient of 2.826 means, with other variables held constant, that if GPA increases by a unit, on average the estimated logit increases by about 2.826 units, suggesting a positive relationship between the two. Or holding other factors constant, as GPA increases by one point, the average logit value goes up by about 2.83, that is, the log of odds in favor of scoring A grade increases by about 2.83 and it's statistically significant as implied by the P-value of the Z-test.
- ☛ The holding other factors constant, as ASE increases by one point, the log of the odds ratio increases by 0.095, but this effect is statistically insignificant.
- ☛ Among students, students who have a PC compared to those who do not have PC, the log of odds in favor of scoring "A" grade increases by about 2.38 and it's statistically significant as implied by the P-value of the Z-test.

As you can see, all the regressors have a positive effect on the logit, although statistically the effect of ASE is not significant. However, together all the regressors have a significant effect on scoring "A" grade, as the LR statistic (an equivalent to F-test in linear regression) is 15.40, whose **P-value** is about 0.0015, which is very small.

Note, however, that a more meaningful interpretation is in terms of odds, which are obtained by taking the antilog of the various coefficients of the logit model. Thus, if you take the antilog of the GPA coefficient of 2.826 you will get **16.8778** ($= \exp^{(2.826)}$). As GPA of a student rises by a unit, then the student is about 16.88 times likely to get an "A", other things remaining the same. Using Stata, we can easily estimate the odds ratio for all regressors as in the following table.

B. Odds Ratio Interpretation

Table-1.8 below presents the Odds ratio regression result of the logit model.

Table-1.8: Odds Ratio Regression Result

Logistic regression				Number of obs	=	32
				LR chi2(3)	=	15.40
				Prob > chi2	=	0.0015
Log likelihood = -12.889633				Pseudo R2	=	0.3740
grade	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
gpa	16.87972	21.31809	2.24	0.025	1.420194	200.6239
ase	1.099832	.1556859	0.67	0.501	.8333651	1.451502
pc	10.79073	11.48743	2.23	0.025	1.339344	86.93802
_cons	2.21e-06	.0000109	-2.64	0.008	1.40e-10	.03487

Therefore, the odds ratio interpretation is as follows:

- ☛ Holding other variables constant, as GPA increases by one point, the probability of scoring A is 16.87 times the probability of getting other grades (B, C, D, Fx, F).
- ☛ Holding other variables constant, as ASE increases by one point, the probability of scoring A is about 1.1 times the probability of getting other grades (B, C, D, Fx, F) but it is statistically insignificant.
- ☛ Holding other variables constant, students who have PC are 10.79 times more likely to score A than other grades compared to non-PC owners and it is statistically significant at 5% level of significance.

C. Marginal Effect Interpretation

Since the language of logit and odds ratio may be unfamiliar to some, we always compute the probability of something happening. Table-1.9 below presents the probability/marginal effect regression result of the logit model.

Table-1.9: Marginal effects after Logit Regression Result

Marginal effects after logit							
y = Pr(grade) (predict)							
= .25282025							
variable	dy/dx	Std. Err.	z	P> z	[95% C.I.]		X
gpa	.5338589	.23704	2.25	0.024	.069273	.998445	3.11719
ase	.0179755	.02624	0.69	0.493	-.033448	.069399	21.9375
pc*	.4564984	.18105	2.52	0.012	.10164	.811357	.4375
(*) dy/dx is for discrete change of dummy variable from 0 to 1							

Therefore, the marginal effect/probability interpretation of the estimated logit model given above is as follows:

- ☛ As GPA increases by one point, the probability of scoring grade “A” by an average student increases by 53%, assuming that the other factors score average.
- ☛ As ASE increases by one point, the probability of scoring grade “A” by an average student increases by about 1.8%, assuming that the other factors score average.
- ☛ For a student who owns PC the probability of scoring grade “A” is higher by 45.6% as compared to non-PC owners, assuming that the other factors score average.

1.2.2.2 The Probit Regression Model

As we have noted, to explain the behavior of a dichotomous dependent variable we will have to use a suitable CDF. The logit model uses the cumulative logistic function. But this is not the only CDF that one can use. In some applications, the normal CDF has been found useful. The model that emerges from the normal CDF is popularly known as the **probit model**, although sometimes it is also known as the **normit model**. As can be seen from Figure-1.4 above, the logistic distribution has a flatter tail than the normal distribution. This is because the variance of the logistic distribution, $\frac{\pi^2}{3}$ is greater than the variance of the standard normal distribution, 1. The difference between the coefficients of Logit model and that of Probit model is accounted to the difference in the variance of the two distributions.

The previous section, we have considered the Logit model. In this section, we shall present the Probit model. We will try to develop the Probit model based on utility theory, or rational choice perspective on behavior, as developed by McFadden.

Assume that in our house ownership example, the decision to own a house or not depends on an unobservable utility index I_i (also known as a **latent variable**), that is determined by one or more explanatory variables, say income X_i , in such a way that the larger the value of the index, I_i , the greater the probability of a household owning a house. We express the index I_i as

$$I_i = \beta_0 + \beta_1 X_i$$

Where, X_i is the income of the i^{th} household.

How is the (unobservable) index related to the actual decision to own a house? As before, let $Y_i = 1$ if a household owns a house and $Y_i = 0$ if it does not. Now, it is reasonable to assume that there is a critical or threshold level of the index, call it I_i^* such that if I_i exceeds I_i^* , the household will own house, otherwise it will not. The threshold I_i^* , like I_i , is not observable, but if we assume that it is normally distributed with the same mean and variance, it is possible not only to estimate the parameters of the index given above, but also to get some information about the unobservable index itself.

The probability that $I_i > I_i^*$ can be calculated as follows:

$$P_i = \Pr(Y_i = 1/X_i) = \Pr(I_i \geq I_i^*) = \Pr(\beta_0 + \beta_1 X_i \geq Z_i) = F(\beta_0 + \beta_1 X_i)$$

Where, $P_i = \Pr(Y_i = 1/X_i)$ is the probability that a household owns a house. Z_i is the standard normal variable which is normally distributed with a mean of 0 and variance of σ^2 . F is the standard normal CDF which can be explicitly written as follows.

$$F(Z) = P_i = \int_{-\infty}^Z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2)} dz$$

The area under the standard normal curve from $-\infty$ up to I_i measures the probability of owning a house and given by:

$$P_i = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{I_i} e^{-\frac{1}{2}(Z^2)} dz$$

$$P_i = \frac{1}{\sqrt{2\pi}} \int_{-\infty}^{\beta_0 + \beta_1 X_i} e^{-\frac{1}{2}(Z^2)} dz$$

In general, the Probit model can be specified as follows.

$$I_i = X\beta + u_i, \quad u_i \sim N(0, 1)$$

$$Y_i = \begin{cases} 1, & \text{if } I_i \geq I_i^* \\ 0, & \text{if } I_i < I_i^* \end{cases}$$

$$\Pr(Y_i = 1/X_i) = \Pr(I_i \geq I_i^*) = F(Z) = P_i = \int_{-\infty}^Z \frac{1}{\sqrt{2\pi}} e^{-\frac{1}{2}(z^2)} dz$$

This model is called **Probit model** and can be estimated using **Maximum Likelihood Method (MLM)**. The coefficients from the Probit model are difficult to interpret because they measure the change in the unobservable I_i associated with a change in one of the explanatory variables. A more useful measure is what we call the **marginal effects**.

Example

Let's revisit our grade example data given in table-1.6 above which gives data on 32 students about their final grade in Mathematical Economics examination in relation to the variables GPA, ASE, and PC. Let us see what the Probit results look like. The regression results estimated based on the method of maximum likelihood using Stata14 are given in the table below.

Table-1.10: Probit Regression Result

Probit regression				Number of obs	=	32
				LR chi2(3)	=	15.55
				Prob > chi2	=	0.0014
Log likelihood = -12.818803				Pseudo R2	=	0.3775
grade	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
gpa	1.62581	.6938825	2.34	0.019	.2658255	2.985795
ase	.0517289	.0838903	0.62	0.537	-.1126929	.2161508
pc	1.426332	.5950379	2.40	0.017	.2600795	2.592585
_cons	-7.45232	2.542472	-2.93	0.003	-12.43547	-2.469166

Interpretation of the estimated coefficients of the Probit model:

Estimated coefficients do not quantify the influence of the regressors on the probability that the regressand takes on the value one because these coefficients are parameters of the latent model. Thus, there is no any standard and convincing way to interpret the coefficients from the output of a Probit regression. We need to interpret the *marginal effects* of the regressors, that is, how much the (conditional) probability of the outcome variable changes when you change the value of a regressor, holding all other regressors constant at some values. As far as the coefficients of a Probit model are concerned, what we need to do is to explain the implication of the coefficients. For instance,

- ☛ The coefficient of GPA is significantly different from zero (at 5% level) indicating relevant relationship between scoring “A” grade and student’s GPA.
- ☛ The coefficient of PC is significantly different from zero (at 5% level) indicating relevant relationship between scoring “A” grade and PC ownership.

In short, students with higher GPA and PC are more likely to score “A” compared to their counterparts.

In general, “qualitatively”, the results of the Probit model are comparable with those obtained from the logit model in that GPA and PC are individually statistically significant while the variable ASE is statistically insignificant in both models. Collectively, all the coefficients are statistically significant, since the value of the **LR** statistic is 15.5458 with a *p* value of 0.0014.

Table-1.10: Marginal Effect of the Probit Model

Marginal effects after probit							
y = Pr(grade) (predict)							
= .26580809							
variable	dy/dx	Std. Err.	z	P> z	[95% C.I.]	X	
gpa	.5333471	.23246	2.29	0.022	.077726 .988968	3.11719	
ase	.0169697	.02712	0.63	0.531	-.036184 .070123	21.9375	
pc*	.464426	.17028	2.73	0.006	.130682 .79817	.4375	
(*) dy/dx is for discrete change of dummy variable from 0 to 1							

The marginal effect results and their interpretation in a Probit model are similar to that of logit model. Therefore, the marginal effect/probability interpretation of the estimated Probit model presented above is as follows:

- ☛ As GPA increases by one unit, the probability of scoring grade “A” by an average student increases by 53%, holding other factors constant.
- ☛ As ASE increases by one point, the probability of scoring grade “A” by an average student increases by about 1.7%, holding other factors constant.
- ☛ For a student who owns PC the probability of scoring grade “A” is higher by 45.6% as compared to non-PC owners, holding other factors constant.

1.2.2.3. Joint Significance in Qualitative Response Regression Models

As far as joint significance is concerned, as equivalent of the **F test** (in the linear regression model) there various ways of testing multiple restrictions in Probit and Logit models.

The most commonly used test, and most easily calculated test is the last, **Likelihood Ratio Test**. It is used when we wish to test our exclusion restrictions, i.e. whether we should or should not exclude a set of variables. The idea is a simple one, since what we are maximizing is the log likelihood function, and since as variables are excluded from the regression relationship, the objective function falls. The question then is whether we have a significant fall in the log likelihood function value. Like the F-test in linear regression models, in LR test the joint hypothesis to be tested is that all the explanatory variables are simultaneously irrelevant versus at least one of the regressors is relevant. The likelihood ratio statistic is just twice the difference in the log likelihood functions of the two models: the unrestricted (the model with all regressors included) and the restricted (the model with only intercept term):

$$i.e., \quad LR = 2(L_{UR} - L_R)$$

where L_{UR} and L_R are the log likelihoods for the unrestricted and restricted models.

Given the null hypothesis, the LR statistic follows the χ^2 distribution asymptotically with df equal to the number of explanatory variables.

🌈 Measuring Goodness-of-Fit in Logit and Probit Models

A goodness-of-fit measure is a summary statistic indicating the accuracy with which the model approximates the observed data. However, the conventional measure of goodness of fit, R^2 , is not particularly meaningful in binary regressand models. In the case in which the dependent variable is qualitative, accuracy can be judged either in terms of the fit between the calculated probabilities and observed response frequencies or in terms of the model's ability to forecast observed responses. Measures similar to R^2 , called **pseudo R^2** , are available. Note, however, that contrary to the linear regression model, there is no single measure for the goodness-of-fit in binary choice models and a variety of measures exists. The most common goodness of fit measure is the one that is proposed by McFadden(1974), which is defined as:

$$\text{Pseudo } R^2 = 1 - \frac{L_{UR}}{L_R}$$

Where, as before L_{UR} is the log likelihood value you obtain from the unrestricted model, and L_R is that generated by a regression (either Probit or Logit. But note that this goodness of fit measure is also used and reported often by statistical packages for any regression that involves MLE) with only the intercept.

- ☛ The **Pseudo R^2** measures the fit using the likelihood function and measures the improvement in the value of the log likelihood relative to having no explanatory variables (X_i). For instance, in our estimated Logit and Probit models above, we see that the **Pseudo $R^2 \cong 0.37$** . This suggests that the log-likelihood value increases by about 37% with the introduction of the set of regressors in the models.
- ☛ Similar to the R^2 of the linear regression model, it holds that $0 \leq \text{Pseudo } R^2 \leq 1$. An increasing Pseudo R^2 may indicate a better fit of the model, whereas no simple interpretation like for the R^2 of the linear regression model is possible.

🌈 Remarks:

Between Logit and Probit, which model is preferable? In most applications the models are quite similar, the main difference being that the logistic distribution has slightly fatter tails. That is to say, the conditional probability P_i approaches zero or one at a slower rate in Logit than in Probit. Therefore, there is no compelling reason to choose one over the other. In practice many researchers choose the Logit model because of its comparative mathematical simplicity.

Though the models are similar, one has to be careful in interpreting the coefficients estimated by the two models. For example, for our grade example, the coefficient of GPA of 1.6258 of the Probit model and 2.826 of the logit model are not directly comparable. The reason is that, although the standard logistic (the basis of logit)

and the standard normal distributions (the basis of probit) both have a mean value of zero, their variances are different; 1 for the standard normal and $\pi^2/3$ for the logistic distribution, where $\pi \approx 22/7$. Therefore, if you multiply the Probit coefficient by about 1.81 (which is approximately $= \pi/\sqrt{3}$), you will get approximately the logit coefficient. For our example, the Probit coefficient of GPA is 1.6258. Multiplying this by **1.81**, we obtain **2.94**, which is close to the Logit coefficient. Alternatively, if you multiply a logit coefficient by 0.55 ($= 1/1.81$), you will get the Probit coefficient. Amemiya, however, suggests multiplying a logit estimate by 0.625 to get a better estimate of the corresponding Probit estimate. Conversely, multiplying a Probit coefficient by 1.6 ($= 1/0.625$) gives the corresponding logit coefficient.

In general, in practical applications, we follow the suggestion of Amemiya that Logit coefficients are about **1.6** the corresponding Probit coefficients and Probit coefficients are about **0.625** the corresponding Logit coefficients. Similarly, the coefficients of LPM are about **0.25** the corresponding Logit coefficient except for the intercept. The intercept of LPM is about **0.25** the intercept of Logit plus **0.5**. Note, however, that the two models are the same in magnitude of marginal effect as well as sign of the coefficients of the independent variables.

Exercise

Assume that we want to study the impact of family size (below 18 years) of a household and the average annual hours of work by household on the poverty level of a household in a given Village. Suppose that we take a sample of 20 households from a particular village and obtained the following data on the family size (FS), hours of work (HRS) and poverty level (Y_i) of households ($Y_i = 0$, if the household is above the poverty line and $Y_i = 1$, if the household is below the poverty line). We use Probit regression to see the effect of family size and hours of work on the poverty level of a household.

Households	Family Size (FS)	Hours of Work (HRS)	Poverty Status (Y)
1	4	200	0
2	1	300	0
3	2	600	0
4	1	1000	0
5	8	400	1
6	9	0	1
7	3	900	0
8	2	0	0
9	1	0	1
10	2	0	0
11	1	1000	0
12	4	2000	0
13	3	1000	0
14	6	300	1

15	3	1000	0
16	4	1000	1
17	5	200	0
18	8	100	1
19	9	0	1
20	10	0	1

Regression results from Probit Model

```
. probit Y FS HRS

Iteration 0:  log likelihood = -12.93196
Iteration 1:  log likelihood = -6.3706877
Iteration 2:  log likelihood = -6.2174937
Iteration 3:  log likelihood = -6.215108
Iteration 4:  log likelihood = -6.2151078
```

```
Probit regression               Number of obs   =        19
                                LR chi2(2)      =        13.43
                                Prob > chi2     =        0.0012
Log likelihood = -6.2151078     Pseudo R2   =        0.5194
```

Y	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
FS	.4252467	.1920025	2.21	0.027	.0489287	.8015647
HRS	-.0009706	.0008239	-1.18	0.239	-.0025853	.0006441
_cons	-1.441228	.8421944	-1.71	0.087	-3.091899	.2094426

```
. mfx
```

```
Marginal effects after probit
y = Pr(Y) (predict)
= .45756724
```

variable	dy/dx	Std. Err.	z	P> z	[95% C.I.]		X
FS	.1686884	.07972	2.12	0.034	.012445	.324932	4.31579
HRS	-.000385	.00033	-1.18	0.238	-.001025	.000255	515.789

Since it is difficult to estimate the coefficient of Probit model, we will interpret the coefficient of the marginal effects.

Regression results from Logit Model for the above Poverty data

```
. logit Y FS HRS, nolog
```

Logistic regression		Number of obs =		19	
		LR chi2(2) =		13.20	
		Prob > chi2 =		0.0014	
Log likelihood = -6.3299557		Pseudo R2 =		0.5105	

Y	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
FS	.7221587	.3533792	2.04	0.041	.0295482	1.414769
HRS	-.0015087	.0014825	-1.02	0.309	-.0044145	.001397
_cons	-2.518599	1.611066	-1.56	0.118	-5.676231	.6390321

```
. xi:logit Y FS HRS, or nolog
```

Logistic regression		Number of obs =		19	
		LR chi2(2) =		13.20	
		Prob > chi2 =		0.0014	
Log likelihood = -6.3299557		Pseudo R2 =		0.5105	

Y	Odds Ratio	Std. Err.	z	P> z	[95% Conf. Interval]	
FS	2.058873	.7275628	2.04	0.041	1.029989	4.115536
HRS	.9984924	.0014803	-1.02	0.309	.9955953	1.001398
_cons	.0805724	.1298074	-1.56	0.118	.0034264	1.894646

```
. mfx
```

Marginal effects after logit

y = Pr(Y) (predict)

= .45509545

variable	dy/dx	Std. Err.	z	P> z	[95% C.I.]		X
FS	.1790835	.09312	1.92	0.054	-.003424	.361591	4.31579
HRS	-.0003741	.00037	-1.02	0.308	-.001094	.000346	515.789

Assignments:

- Interpret the Probit Model
- Interpret the Logit Model
- Compare the regression results of the two models

CHAPTER TWO

INTRODUCTION TO BASIC TIME SERIES DATA ANALYSIS

Introduction

Time series data is one of the important types of data used in empirical analysis. Time series analysis comprises methods for analyzing time series data in order to extract meaningful statistics and other characteristics of the data. It is an attempt to understand the nature of the time series and developing appropriate models useful for forecasting. This chapter is devoted to the discussion of introductory concepts in time series analysis for two major reasons:

1. **Time series** data is frequently used in practice.
2. **Time series** data analysis poses several challenges to econometricians and practitioners.

The following are some of the challenges in time series analysis:

- A. **Serial correlation/nonindependence of observations:** Autocorrelation is common in time series analysis, because the underlying time series is often nonstationary.
- B. **Spurious or false regression** -in regressing a time series variable on (an) other time series variable(s), one often obtains a very high R^2 (in excess of 0.9) even though there is no meaningful relationship between the two variables. Sometimes we expect no relationship between two variables, yet a regression of one on the other variable often shows a significant relationship. This situation exemplifies the problem of **spurious, or nonsense, regression**. It is therefore very important to find out if the relationship between economic variables is spurious or false.
- C. Some financial time series, such as stock prices, exhibit what is known as the **random walk phenomenon**, that is, such series are non-stationary. Therefore, forecasting in such series would be a futile exercise.
- D. Causality tests (such as the Granger and Sims) assume that the time series involved in analysis are stationary. Therefore, tests of stationarity should precede tests of causality.

Note that almost all of the above problems arise mainly due to non-stationarity of time series data sets. Therefore, our major emphasis in this chapter is on the discussion of the nature and tests of stationarity, and the remedial measures for non-stationary data sets.

2.1 The Nature of Time Series Data

An obvious characteristic of time series data which distinguishes it from cross-sectional data is that a time series data set comes with a **temporal ordering**. For instance, in this chapter, we will discuss a time series data set on GDP(Y), trade balance (TB), money supply (MS2), exchange rate (EX) and government expenditure (G) of Ethiopia for the periods 1963-2003 E.C.. In this data set, we must know that the data of 1963 immediately precedes the data of 1962. For analyzing time series data, we must recognize that the past can affect the future,

but not vice versa. Table2.1.1 below gives a listing of the data on five macroeconomic variables in Ethiopia for the period 1963-2003.

In Econometrics I, we studied the statistical properties of the OLS estimators based on the notion that samples were randomly drawn from the appropriate population. Understanding why cross-sectional data should be viewed as random outcomes is fairly straightforward: a different sample drawn from the population will generally yield different values of the variables. Therefore, the OLS estimates computed from different random samples will generally differ, and this is why we consider the OLS estimators to be random variables.

How should we think about randomness in time series data? Certainly, economic time series satisfy the intuitive requirements for being outcomes of random variables. For example, today we do not know what the trade balance of Ethiopia will be at the end of this year. We do not know what the annual growth in output will be in Ethiopia during the coming year. Since the outcomes of these variables are not foreknown, they should clearly be viewed as random variables.

Table2.1.1

Time series Data on some macroeconomic variables in Ethiopia

Year	TB	MS2	GDP	EXR	G
1963	0.69	629.6	9,400	2.4000	303.99
1964	0.69	658.3	9,873	2.3000	316.19
1965	1.04	808	9,892	2.1900	348.14
1966	1.15	1066.6	10,353	2.0700	368.57
1967	0.71	1139.4	11,412	2.0700	442.89
1968	0.79	1421.8	11,145	2.0700	551.99
1969	0.86	1467.9	11,916	2.0700	654.73
1970	0.84	1682.2	13,221	2.0700	752.34
1971	0.61	1848	13,890	2.0700	942.56
1972	0.65	2053.2	15,143	2.0700	993.59
1973	0.62	2377.6	16,135	2.0700	1,017.56
1974	0.47	2643.7	16,530	2.0700	1,090.74
1975	0.29	3040.5	17,498	2.0700	1,244.54
1976	0.45	3383.7	19,655	2.0700	1,506.13
1977	0.42	3849	17,865	2.0700	1,454.12
1978	0.43	4448.2	21,517	2.0700	1,524.52
1979	0.36	4808.7	22,367	2.0700	1,636.04
1980	0.34	5238.7	23,679	2.0700	1,726.70
1981	0.43	5705	24,260	2.0700	2,066.50
1982	0.40	6708.2	25,413	2.0700	2,336.99
1983	0.29	7959.2	27,323	2.0700	2,467.53
1984	0.18	9010.7	31,362	2.0700	2,416.75
1985	0.26	10136.7	34,621	4.2700	2,819.51
1986	0.30	11598.7	43,171	5.7700	3,770.58
1987	0.43	14408.5	43,849	6.2500	4,220.57
1988	0.35	15654.8	55,536	6.3200	5,378.98
1989	0.46	16550.6	62,268	6.5000	5,671.26
1990	0.44	18585.3	64,501	6.8800	5,984.25
1991	0.31	19399.0	62,028	7.5100	7,069.36
1992	0.35	22177.8	66,648	8.1400	11,921.90
1993	0.31	24516.2	68,027	8.3300	9,963.85
1994	0.27	27322.0	66,557	8.5400	9,873.38

1995	0.26	30469.6	73,432	8.5809	9,849.58
1996	0.23	34662.5	86,661	8.6197	11,315.21
1997	0.23	40211.7	106,473	8.6518	13,203.04
1998	0.22	46377.4	131,641	8.6810	16,080.46
1999	0.23	56651.9	171,989	8.7943	18,071.82
2000	0.22	68182.1	248,303	9.2441	24,364.45
2001	0.19	82509.8	335,392	10.4205	27,592.06
2002	0.24	104432.4	382,939	12.8909	37,527.99
2003	0.33	145377.0	511,157	16.1178	50,093.38

Formally, a sequence of random variables indexed by time is called a **stochastic process** or a **time series process** (“stochastic” is a synonym for random). When we collect a time series data set, we obtain one possible outcome, or realization, of the stochastic process. We can only see a single realization, because we cannot go back in time and start the process over again (this is analogous to cross-sectional analysis where we can collect only one random sample). However, if certain conditions in history had been different, we would generally obtain a different realization for the stochastic process, and this is why we think of time series data as the outcome of random variables. This set of all possible realizations of a time series process plays the role of the population in cross-sectional analysis.

A **random** or **stochastic process** is a collection of random variables ordered in time. We let Y denote a random variable. We will use the notation Y_t to express the Y value in time t . **In what sense can we regard GDP as a stochastic process?** Consider for instance the GDP of 9.4 billion Birr of Ethiopia for the period 1963. In theory, the GDP figure for the period 1963 could have been any number, depending on the economic and political climate then prevailing. The figure of 9.4 billion Birr is a particular **realization** of all such possibilities. Therefore, we can say that GDP is a stochastic process and the actual values we observed for the period 1963-2003 are a particular realization of that process (i.e. the sample). The distinction between the stochastic process and its realization is akin to the distinction between population and sample in cross-sectional data. Just as we use sample data to draw inferences about a population, in time series we use the realization to draw inferences about the underlying stochastic process.

2.2 Stationary and Non-stationary Stochastic Processes

A type of stochastic process that has received a great deal of attention by time series analysts is the so-called **stationary stochastic process**. Stationarity means that the probabilistic character of the series must not change over time, i.e., that any section of the time series is “typical” for every other section with the same length. In other words, a stochastic process is said to be stationary if its mean and variance are constant over time and the value of the covariance between any two time periods depends only on the distance or gap or lag between the two time periods and not the actual time at which the covariance is computed. In the time series literature, such a stochastic process is known as a **weakly stationary**, or **covariance stationary**, or **second-order stationary**. This type of stationarity is sufficient for applied time series analysis, and strict stationarity is a

practically useless concept. It is a necessary condition for building a time series model that is useful for future forecasting. To explain weak stationarity, let Y_t be a stochastic time series with these properties:

$$\text{Mean: } E(Y_t) = \mu \quad (1)$$

$$\text{Variance: } \text{Var}(Y_t) = E(Y_t - \mu)^2 = \sigma^2 \quad (2)$$

$$\text{Covariance: } \text{Cov}(Y_t, Y_{t+k}) = \gamma_k = E[(Y_t - \mu)(Y_{t+k} - \mu)] \quad (3)$$

Where, γ_k the covariance (or autocovariance) at lag k , is the covariance between the values of Y_t and Y_{t+k} , that is, between two Y values k periods apart. If $k = 0$, we obtain γ_0 , which is simply the variance of Y ($= \sigma^2$); if $k = 1$, γ_1 is the covariance between two adjacent values of Y .

In short, if a time series is stationary, its mean, variance, and autocovariance (at various lags) remain the same no matter at what point we measure them; that is, **they are time invariant**. Such a time series will tend to return to its mean (called **mean reversion**) and fluctuations around this mean (measured by its variance) will have broadly constant amplitude. If a time series is not stationary in the sense just defined, it is called a **non-stationary time series**. In other words, **non-stationary time series** will have a time varying mean or a time-varying variance, or both.

✚ **Why are stationary time series so important?** There are two major reasons.

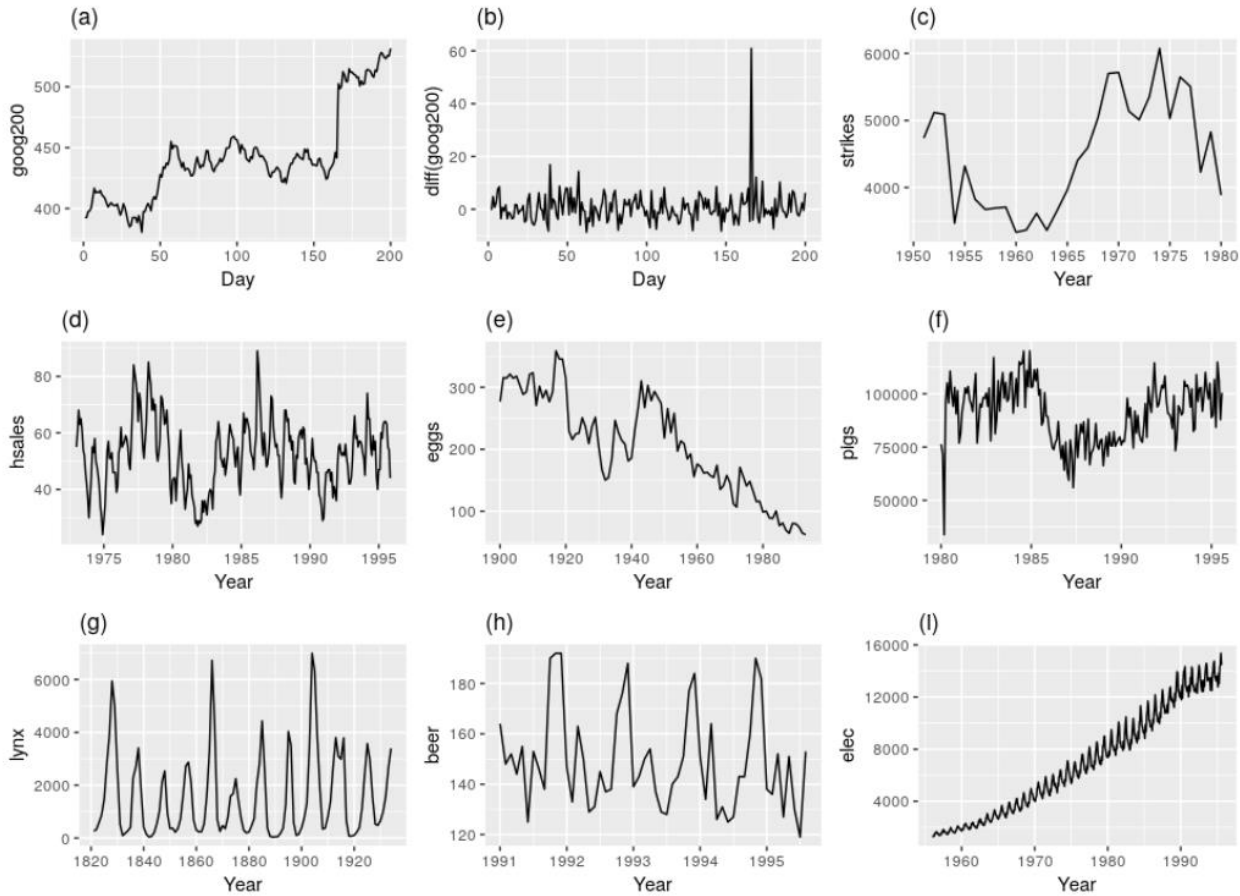
1. If a time series is non-stationary, we can study its behavior only for the time period under consideration. Each set of time series data will therefore be for a particular episode. As a result, it is not possible to generalize it to other time periods. Therefore, **for the purpose of forecasting or policy analysis**, such (non-stationary) time series may be of little practical value.
2. If we have two or more non-stationary time series, regression analysis involving such time series may lead to the phenomenon of **spurious** or **non-sense regression**.

✚ **How do we know that a particular time series is stationary?** In particular, are the time series shown in Figure 2.2.1 stationary? If we depend on common sense, it would seem that the time series depicted in the above five figures are non-stationary, at least in the mean values. This is because some of them are trending upward while others are trending downwards.

Exercise: for the graphs in Figure 2.2.1, indicate if the time series are stationary.

Figure 2.2.1

Nine examples of time series data; (a) Google stock price for 200 consecutive days; (b) Daily change in the Google stock price for 200 consecutive days; (c) Annual number of strikes in the US; (d) Monthly sales of new one-family house (e) Annual price of a dozen eggs in the US (constant dollars); (f) Monthly total of pigs slaughtered in Victoria, Australia; (g) Annual total of lynx trapped in the McKenzie River district of north-west Canada; (h) Monthly Australian beer production; (i) Monthly Australian electricity production. (Hyndman & Athanasopoulos, 2018)



Answer: b and g are the only stationary time series.

Although our interest is in stationary time series, we often encounter **non-stationary time series**, the classic example being the **random walk model** (RWM). It is often said that asset prices, such as stock prices follow a random walk; that is, they are non-stationary. We distinguish two types of random walks: (1) random walk without drift (i.e., no constant or intercept term) and (2) random walk with drift (i.e., a constant term is present).

A. Random Walk without Drift

A **random walk** is defined as a process where the current value of a variable is composed of the past value plus an error term defined as a white noise (a normal variable with zero mean and variance one). Algebraically a random walk is represented as follows:

$$Y_t = Y_{t-1} + u_t \quad (4)$$

In the **random walk model**, as (4) shows the value of Y at time t is equal to its value at time $(t - 1)$ plus a **random shock**. We can think of (4) as a regression of Y at time t on its value lagged one period.

Now from (4) we can write

$$Y_1 = Y_0 + u_1$$

$$Y_2 = Y_1 + u_2 = Y_0 + u_1 + u_2$$

$$Y_3 = Y_2 + u_3 = Y_0 + u_1 + u_2 + u_3$$

In general, if the process started at some time 0 with a value of Y_0 , we have

$$Y_t = Y_0 + \sum_{t=1}^t u_t \quad (5)$$

$$E(Y_t) = Y_0 \quad (6)$$

$$Var(Y_t) = t\sigma^2 \quad (7)$$

As the preceding expression shows, the mean of Y is equal to its initial, or starting, value, which is constant, but as t increases, its variance increases indefinitely, thus violating a condition of Stationarity. **In short, the RWM without drift is a non-stationary stochastic process.** In practice Y_0 is often set at zero, in which case $E(Y_t) = 0$. An interesting feature of **RWM** is the persistence of random shocks (random errors), which is clear from (5): Y_t is the sum of initial Y_0 plus the sum of random shocks. As a result, the impact of a particular shock does not die away. For example, if $u_2 = 2$ rather than $u_2 = 0$, then all Y_t 's from Y_2 onward will be 2 units higher and the effect of this shock never dies out. That is why random walk is said to have an infinite memory.

Interestingly, if we write (4) as

$$(Y_t - Y_{t-1}) = \Delta Y_t = u_t \quad (8)$$

Where, Δ is the **first difference operator**.

It is easy to show that, while Y_t is non-stationary, its first difference is stationary. In other words, the first differences of a random walk time series are stationary. But we will have more to say about this later.

B. Random Walk with Drift

Let us modify (4) as follows

$$Y_t = \delta + Y_{t-1} + u_t \quad (9)$$

Where, δ is known as the **drift parameter**. The name drift comes from the fact that if we write the preceding equation as

$$(Y_t - Y_{t-1}) = \Delta Y_t = \delta + u_t \quad (10)$$

It shows that Y_t drifts upward or downward, depending on δ being positive or negative. Following the procedure discussed for random walk without drift, it can be shown that for the random walk with drift model (9),

$$Y_t = Y_0 + \delta t + \sum_{t=1}^t u_t$$

$$E(Y_t) = Y_0 + t\delta \quad (11)$$

$$Var(Y_t) = t\sigma^2 \quad (12)$$

As can be seen from above, for RWM with drift the mean as well as the variance increases over time, again violating the conditions of (weak) Stationarity. **In short, RWM with or without drift, is a non-stationary stochastic process.**

Remark

- **The random walk model** is an example of what is known in the literature as a **unit root process**. Since this term has gained tremendous currency in the time series literature, we need to note **what a unit root process is**.
- Let's rewrite the RWM (4) as:

$$Y_t = \rho Y_{t-1} + u_t - 1 \leq \rho \leq 1 \quad (13)$$

- If $\rho = 1$, (13) becomes an **RWM (without drift)**. If ρ is in fact **1**, we face what is known as the **unit root problem**, that is, a situation of non-stationarity; we already know that in this case the variance of Y_t is not stationary. The name **unit root** is due to the fact that $\rho = 1$. As noted above, the first differences of a random walk time series/a **unit root process** are stationary. **Thus, the terms non-stationarity, random walk, and unit root can be treated as synonymous.**

2.3 Trend Stationary- and Difference Stationary Stochastic Process

The distinction between stationary and non-stationary stochastic processes (or time series) has a crucial bearing on whether the **trend** (the slow long-run evolution of the time series under consideration) in the economic time series is **deterministic** or **stochastic**. If the trend in a time series is completely predictable and not variable, we call it a **deterministic trend**, whereas if it is not predictable, we call it a **stochastic trend**. To make the definition more formal, consider the following model of the time series Y_t .

 **Pure random walk:** Consider the following model

$$Y_t = Y_{t-1} + u_t \quad (14)$$

Which is nothing but an RWM without drift and is therefore non-stationary. But note that, if we write (14) as

$$\Delta Y_t = Y_t - Y_{t-1} = u_t \quad (15)$$

it becomes stationary, as noted before. Hence, an RWM without drift is a **difference stationary process (DSP)**.

 **Random walk with drift:**

$$Y_t = \delta + Y_{t-1} + u_t \quad (16)$$

which is a random walk with drift and is, therefore, non-stationary. If we write it as

$$\Delta Y_t = Y_t - Y_{t-1} = \delta + u_t$$

this means Y_t will exhibit a positive (if $\delta > 0$) or negative (if $\delta < 0$) trend. Such a **trend** is called a **stochastic trend**. Equation (16) is a **difference stationary process (DSP)** because the non-stationarity in Y_t can be eliminated by taking first differences of the time series.

✚ **Deterministic trend:** a trend stationary process (TSP) has a data generating process of:

$$Y_t = \delta + \beta t + u_t \quad (17)$$

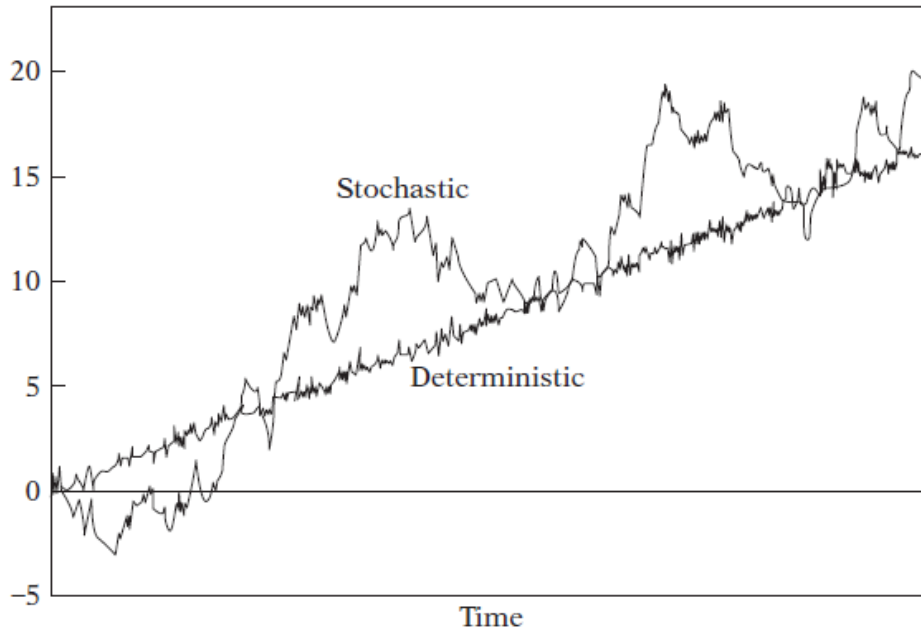
Although the mean of Y_t is $\delta + \beta t$, which is not constant, its variance ($= \sigma^2$) is. Once the values of δ and β are known, the mean can be forecast perfectly. Therefore, if we subtract the mean of Y_t from Y_t , the resulting series will be **stationary**, hence the name **trend stationary**. This procedure of removing the (deterministic) trend is called **detrending**.

To see the difference between stochastic and deterministic trends, consider Figure 2.3.1 (next page). The series named **stochastic** in Figure 2.3.1 is generated by a RWM: $Y_t = 0.5 + Y_{t-1} + u_t$, where 500 values of u_t were generated from a standard normal distribution and where the initial value of Y was set at 1. The series named **deterministic** is generated as follows: $Y_t = 0.5t + u_t$, where the u_t s were generated as above and where t is time measured chronologically.

As you can see in Figure 2.3.1, in the case of the **deterministic trend**, the deviations from the trend line (which represents **nonstationary mean**) are purely random and they die out quickly, i.e. they do not contribute to the long-run development of the time series, which is determined by the trend component $0.5t$. In the case of the **stochastic trend**, on the other hand, the random component u_t affects the long-run course of the series Y_t .

Figure 2.3.1

Deterministic Versus Stochastic Trend



Summarizing, a **deterministic trend** is a nonrandom function of time whereas a **stochastic trend** is random and varies over time. The simplest model of a variable with a **stochastic trend** is the **random walk**. According to Stock and Watson (2007), it is more appropriate to model economic time series as having stochastic rather than deterministic trends. Therefore, our treatment of trends in economic time series focuses mainly on stochastic rather than deterministic trends, and when we refer to “trends” in time series data, we mean **stochastic trends**.

2.4 Integrated Stochastic Process

The **random walk model** is a special case of a more general class of stochastic processes known as **integrated processes**. Recall that the RWM without drift is non-stationary, but its first difference is stationary. Therefore, we call the RWM without drift **integrated of order 1**, denoted as **I(1)**. Similarly, if a time series has to be differenced twice (i.e., take the first difference of the first differences) to make it stationary, we call such a time series **integrated of order 2**, denoted as **I(2)**. In general, if a (non-stationary) time series has to be differenced **d** times to make it stationary, that time series is said to be **integrated of order d**. The order of integration is the number of differencing operations it takes to make the series stationary. For the random walk without drift, there is one differencing operation it takes to make the series stationary, so it is an I(1) series. A time series Y_t integrated of order **d** is denoted as $Y_t \sim I(d)$. If a time series Y_t is stationary to begin with, it is said to be integrated of order zero, denoted by $Y_t \sim I(0)$. Thus, stationary time series and time series integrated of order zero are generally the same.

Most economic time series are generally **I(1)**; that is, they generally become stationary only after taking their first differences. In our example above, trade balance of Ethiopia is integrated of order one, **I(1)**. But, the data on government expenditure (G) and exchange rate (EXR) of Ethiopia are integrated of order two, **I(2)**.

2.5 Test Stationarity of Time Series Data

In many time series analyses, one of the most important preliminary steps in regression analysis is to uncover the characteristics of the data used in the analysis. The main goals of undertaking stationarity test are to get a variable which has a constant mean, variance and time invariant covariance of the variables called ***second order stationary*** or ***covariance stationary***. Following the stationary test, if the variables are non-stationary, we cannot use the data for forecasting purpose unless the series is transformed to a stationary one.

Thus, now we may have two important practical questions:

- (1) How do we find out if a given time series is stationary?
- (2) If we find that a given time series is not stationary, is there a way that it can be made stationary? We now discuss these two questions in turn.

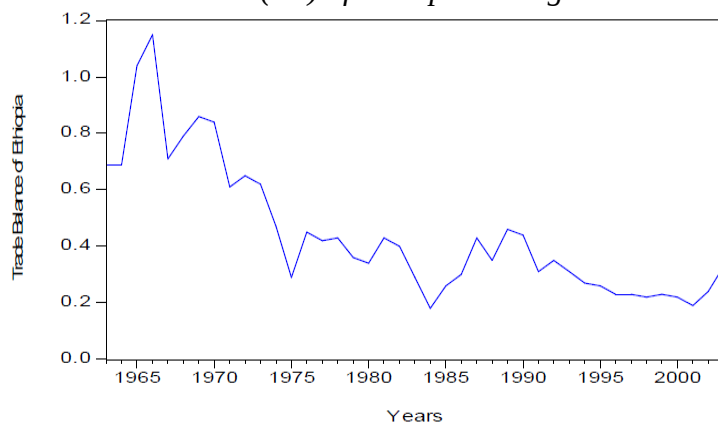
There are basically three ways to examine the stationarity of a time series, namely: (1) graphical analysis, (2) correlogram, and (3) the unit root test.

2.5.1 Graphic Analysis

As noted earlier, before one pursues formal tests, it is always advisable to plot the time series under study, as we have done above for the data given in Table-2.1.1. Such a plot gives an initial clue about the likely nature of the time series. Take, for instance, the trade balance time series shown in Figure-2.5.1. You will see that over the period of study, trade balance has been declining, that is, showing a downward trend, suggesting perhaps that the mean of the TB has been changing. This perhaps suggests that the TB series is not stationary. Such an intuitive feel is the starting point of more formal tests of stationarity.

Figure 2.5.1

The Trade Balance (TB) of Ethiopia through time



The trend of Money Supply (MS2) of Ethiopia through Time

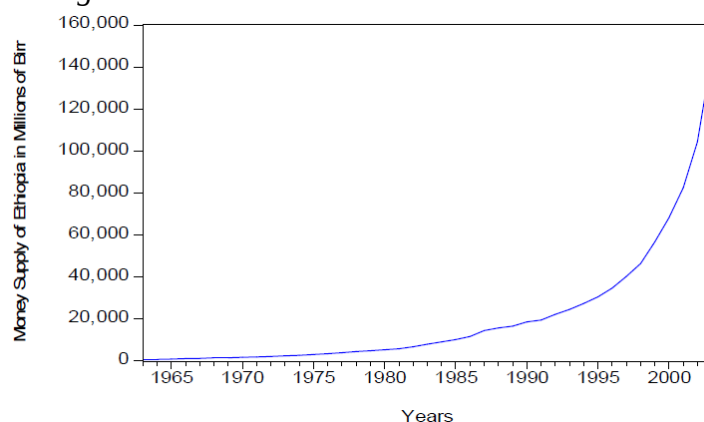


Figure 2.5.2

Figure 2.5.3

The Exchange Rate (ETH/\$US) of Ethiopia through time

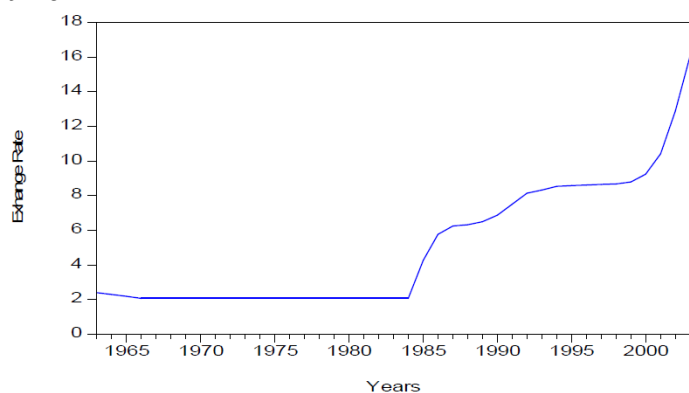
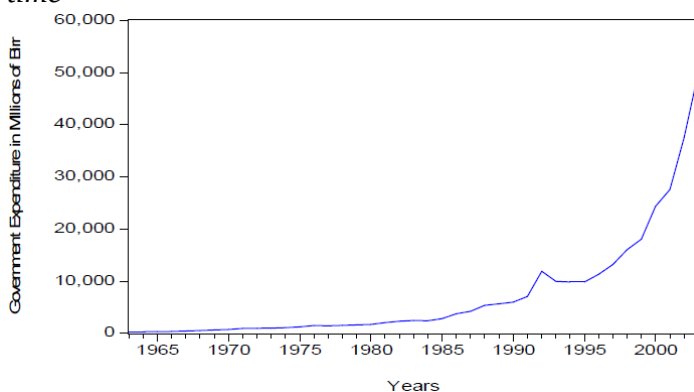


Figure2.5.4

The Government Expenditure of Ethiopia through time



Some of the above time series graph have an upward trending (MS2, EX and G) which may be an indication of the non-stationarity of these data sets. That means, the mean or variance or both may be increasing with the passage of time. But, the graph for Trade Balance (TB) of Ethiopia ($\frac{\text{value of Export}}{\text{value of Import}}$) showed a downward trend which may also be an indication of the non-stationarity of the trade balance (TB) series.

2.5.2 Autocorrelation Function (ACF) and Correlogram

Autocorrelation is the correlation between a variable lagged one or more periods and itself. **The correlogram or autocorrelation function** is a graph of the autocorrelations for various lags of a time series data.

The autocorrelation function (ACF) at lag k , denoted by ρ_k , is defined as:

$$\rho_k = \frac{\gamma_k}{\gamma_0} : \quad i.e., \quad \rho_k = \frac{\text{Covariance at lag } k}{\text{Variance}}$$

Since both covariance and variance are measured in the same units of measurement, ρ_k is a unit less, or pure, number. It lies between -1 and $+1$, as any correlation coefficient does. If we plot ρ_k against k , the graph we obtain is known as the population correlogram. Since in practice we only have a realization (i.e., sample) of a stochastic process, we can only compute the sample autocorrelation function (SAFC), $\hat{\rho}_k$. To compute this, we must first compute the sample covariance at lag k , $\hat{\gamma}_k$, and the sample variance, $\hat{\gamma}_0$, which are defined as

$$\hat{\gamma}_k = \text{Sample Covariance at lag } k = \frac{\sum (Y_t - \bar{Y})(Y_{t-k} - \bar{Y})}{n}$$

$$\hat{\gamma}_0 = \text{Sample Variance} = \frac{\sum (Y_t - \bar{Y})^2}{n}$$

Therefore, the sample autocorrelation function at lag k is simply the ratio of sample covariance (at lag k) to sample variance is given as

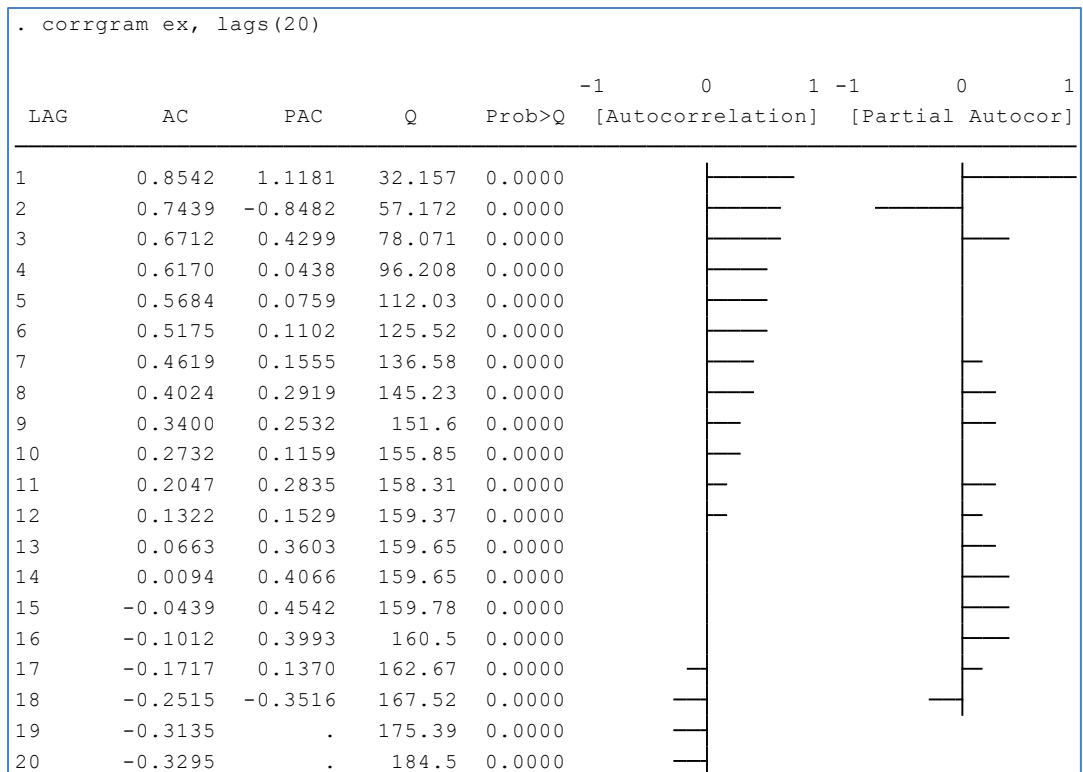
$$\hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}$$

A plot of $\hat{\rho}_k$ against k is known as the **sample correlogram**.

How does a sample correlogram enable us to find out if a particular timeseries is stationary? For this purpose, let us first present the sample correlogram of our exchange rate data set given in Table 2.1.1. The correlogram of the exchange rate data is presented for 20 lags in Figure 2.5.5.

Figure 2.5.5

Correlogram of Ethiopian Exchange Rate (EX), 1963-2003



Now, look at the column labeled AC, which is the sample autocorrelation function, and the first diagram on the left, labeled autocorrelation. The solid vertical line in this diagram represents the zero axis; observations above the line are positive values and those below the line are negative values. Given a correlogram, if the value of ACF is close to zero from above or below, the data are said to be **stationary**. In other words, for stationary time series, the autocorrelations (i.e., ACF) between Y_t and Y_{t-k} for any lag k are close to zero (i.e., the autocorrelation coefficient is statistically insignificant). That is, the successive values of a time series (such as Y_t and Y_{t-k}) are not related to each other. Moreover, the lines on the right-hand side of the correlograms tends to be shorter to the left or to the right from the vertical zero line for stationary time series.

On the other hand, if a series has a (stochastic) trend, i.e., nonstationary, successive observations are highly correlated, and the autocorrelation coefficients are typically significantly different from zero for the first several time lags and then gradually drop toward zero as the number of lags increases. The autocorrelation coefficient for time lag 1 is often very large (close to 1).

Thus, looking at Figure2.5.6, we can see that the autocorrelation coefficients for the exchange rate series at various lags are high (i.e., close to +1 from above) and low (i.e., close to -1 from below). Figure2.5.6 is an example of correlogram of a nonstationary time series.

For further illustration, the correlogram of the trade balance and money supply time series are presented for 10 lags in Figure2.5.7 & 2.5.8 below.

In both correlograms, if we look at the column labeled as **AC**, all the values are close to 1 from above. Moreover, the lines on the right-hand side of the correlograms are also longer to the right from the vertical zero line. That means, the lines are close to +1. Thus, both the money supply and trade balance time series are **non-stationary**.

Figure 2.5.6
Correlogram of Ethiopian Trade Balance, 1963-2003

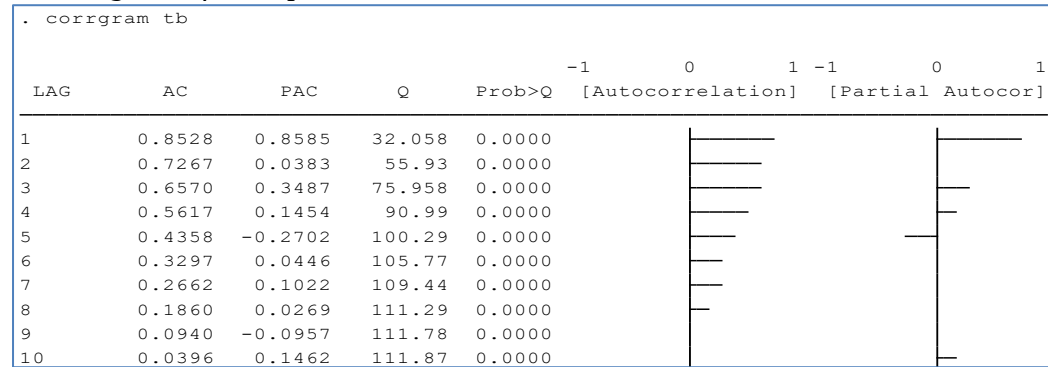
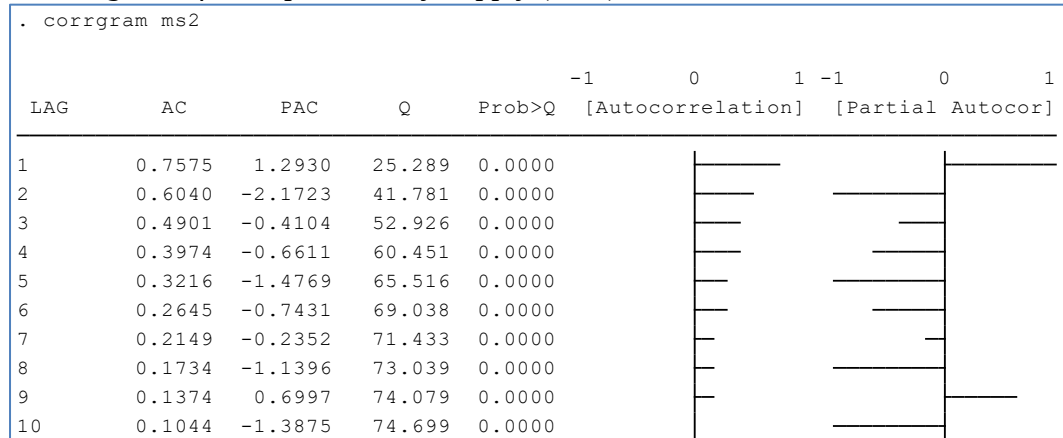


Figure 2.5.7

Correlogram of Ethiopian Money Supply (MS2), 1963-2003



➤ An important practical question for the above analysis is: how do we choose the lag length to compute the ACF? **Arule of thumb** is to compute ACF up to one-third to one-quarter the length of the time series.

2.5.3 The Unit Root Test (Augmented Dickey –Fuller Test)

A test of stationarity or non-stationarity that has become widely popular over the past several years is the **unit root test**. In this section, therefore, we shall address the unit root test, using the **Augmented Dickey-Fuller (ADF) test** of stationarity of the time series. Dickey and Fuller (1979, 1981) devised a procedure to formally test for non-stationarity. The ADF test is a modified version of the original **Dickey–Fuller (DF) Test**. The modification is the inclusion of extra lagged terms of the dependent variable to eliminate autocorrelation in the test equation.

The key insight of their test is that testing for non-stationarity is equivalent to testing for the existence of a unit root. Thus, the starting point is the unit root (stochastic) process given by:

$$Y_t = \rho Y_{t-1} + u_t - 1 \leq \rho \leq 1 \quad (18)$$

We know that if $\rho = 1$, that is, in the case of a unit root, (18) becomes a random walk model without drift, which we know is a **non-stationary** stochastic process. Therefore, why not simply regress Y_t on its one period lagged value Y_{t-1} and find out if the estimated ρ is statistically equal to 1? If it is, then Y_t is non-stationary. This is the general idea behind the unit root test of stationarity.

In a nutshell, what we need to examine here is $\rho = 1$ (unity and hence ‘**unit root**’). Obviously, the null hypothesis is $H_0: \rho = 1$, and the alternative hypothesis is $H_A: \rho < 1$.

We obtain a more convenient version of the test by subtracting Y_{t-1} from both sides of (18):

$$\begin{aligned} Y_t - Y_{t-1} &= \rho Y_{t-1} - Y_{t-1} + u_t \\ \Delta Y_t &= (\rho - 1) Y_{t-1} + u_t \\ \Delta Y_t &= \delta Y_{t-1} + u_t \end{aligned} \quad (19)$$

Where, Δ is the first-difference operator and $\delta = \rho - 1$. In practice, therefore, instead of estimating (18), we estimate (19) and test the null hypothesis $H_0: \delta = 0$, against the alternative hypothesis $H_A: \delta < 0$. In this case, if $\delta = 0$, then $\rho = 1$, i.e. we have a unit root, Y_t , that follows a pure random walk (and, of course, Y_t is nonstationary).

Now let us turn to the estimation of (19). This is simple enough; all we have to do is to take the first differences of Y_t and regress them on Y_{t-1} and see if the estimated slope coefficient in this regression ($= \hat{\delta}$) is zero or not. If it is zero, we conclude that Y_t is non-stationary. But if it is negative, we conclude that Y_t is stationary.

The modified version of the test equation given by (19), i.e., the test equation with extra lagged terms of the dependent variable is specified as:

$$\Delta Y_t = \delta Y_{t-1} + \sum_{i=1}^P \beta_i \Delta Y_{t-i} + u_t \quad (20)$$

Dickey and Fuller (1981) also proposed two alternative regression equations that can be used for testing for the presence of a unit root. The first contains a **constant** in the random walk process, and the second contains a **non-stochastic time trend** as in the following equations, respectively:

$$\Delta Y_t = \alpha + \delta Y_{t-1} + \sum_{i=1}^P \beta_i \Delta Y_{t-i} + u_t \quad (21)$$

$$\Delta Y_t = \alpha + \gamma T + \delta Y_{t-1} + \sum_{i=1}^P \beta_i \Delta Y_{t-i} + u_t \quad (22)$$

The difference between the three regressions concerns the presence of the deterministic elements α and γT .

Note that, P , in all the above three test equations is the lag length for the number of lagged dependent variables to be included. That is, we need to choose a lag length to run the ADF test so that the residuals are not serially correlated. To determine the number of lags, P , we can use one of the following procedures.

- General-to-specific testing:** Start with $P_{\max}$ and drop lags until the last lag is statistically significant, i.e., delete insignificant lags and include the significant ones.
- Use information criteria** such as the Schwarz information criteria, Akaike's information criterion (AIC), Final Prediction Error (FPE), or Hannan-Quinn criterion (HQIC).

In practice, we just click the '**automatic selection**' on the '**lag length**' dialog box in EViews.

Now, the only question is which test we use to find out if the estimated coefficient of Y_{t-1} in (20, 21 & 22) is statistically zero or not. The ADF test for stationarity is simply the normal ' t ' test on the coefficient of the lagged dependent variable Y_{t-1} from one of the three models (20, 21 & 22). This test does not, however, have

a conventional ‘*t*’ distribution and so we must use special critical values which were originally calculated by Dickey and Fuller which is known as the **Dickey-Fuller *tau* statistic**¹. However, most modern statistical packages such as Stata and EViews routinely produce the critical values for Dickey-Fuller tests at 1%, 5%, and 10% significance levels.

- In all the three test equations, the ADF test concerns whether $\delta = 0$. The ADF test statistic is the ‘*t*’ statistic for the lagged dependent variable. ***ADF statistic is a negative number and more negative ADF test statistic is the stronger the rejection of the hypothesis that there is a unit root.***
 - **Null Hypothesis (H0):** If accepted, it suggests the time series has a unit root, meaning it is non-stationary. It has some time dependent structure.
 - **Alternative Hypothesis (H1):** If accepted, the null hypothesis is rejected; it suggests the time series does not have a unit root, meaning it is stationary.

Equivalently, if

- ✓ **p-value > 0.05:** Accept H0; the data has a unit root and is non-stationary
- ✓ **p-value ≤ 0.05:** Reject H0; the data does not have a unit root and is stationary

In short, if the ADF statistical value is smaller in absolute terms than the critical value(s) or equivalently if the P-value for the ADF test statistic is significant then we reject the null hypothesis of a unit root and conclude that Y_t is a stationary process.

Note that the choice among the three possible forms of the ADF test equations depends on the knowledge of the econometrician about the nature of his/her data. Plotting the data² and observing the graph is sometimes very useful because it can clearly indicate the presence or not of deterministic regressors. However, if the form of the data-generating process is unknown, estimation of the most general model given by (22) and then answering a set of questions regarding the appropriateness of each model and moving to the next model is suggested (*i.e., general to specific procedure*).

Illustration: Unit Root Test of Some Macroeconomic Variables of Ethiopia using EViews.

As noted above, the ADF test is based on the null hypothesis that a unit root exists in the time series. Using the ADF test, some of the variables (TB, MS2, and EX) from our time series data set are examined for unit root as visible in Table 2.5.1-Table 2.5.3.

Table 2.5.1

¹The Dickey-Fuller *tau* statistic can be found from appendix section of any statistical book (Eg: you can find it in Gujarati).

²Sometimes if you have data that is **exponentially trending** then you might need to take the log of the data first before differencing it. In this case in your ADF unit root tests you will need to take the differences of the log of the series rather than just the differences of the series.

Augmented Dickey-Fuller Unit Root Test on Trade Balance

Null Hypothesis: TB has a unit root Exogenous: Constant, Linear Trend Lag Length: 0 (Automatic - based on SIC, maxlag=9)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-2.843336	0.1912
Test critical values: 1% level	-4.205004	
5% level	-3.526609	
10% level	-3.194611	
*MacKinnon (1996) one-sided p-values.		

Table2.5.2

Augmented Dickey-Fuller Unit Root Test on Exchange Rate

Null Hypothesis: EX has a unit root Exogenous: Constant, Linear Trend Lag Length: 1 (Automatic - based on SIC, maxlag=9)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-0.419050	0.9833
Test critical values: 1% level	-4.211868	
5% level	-3.529758	
10% level	-3.196411	

Table2.5.3

Augmented Dickey-Fuller Unit Root Test on Log of Money Supply

Null Hypothesis: LMS2 has a unit root Exogenous: Constant, Linear Trend Lag Length: 0 (Automatic - based on SIC, maxlag=9)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	0.645002	0.9994
Test critical values: 1% level	-4.205004	
5% level	-3.526609	
10% level	-3.194611	
*MacKinnon (1996) one-sided p-values.		

All the above test results showed that the data sets are non-stationary. The test statistics is lower than the critical values at 1%, 5% and 10% levels of significance for all the three data sets. This implies the acceptance of the null hypothesis which states that there is unit root in the data sets.

2.5.4. Transforming Non-stationary Time Series

Now that we know the problems associated with non-stationary time series, the practical question is what to do. To avoid the spurious regression problem that may arise from regressing a non-stationary time series on one or more non-stationary time series, we have to transform non-stationary time series to make them stationary. The transformation method depends on whether the time series are difference stationary process (**DSP**) or trend stationary process (**TSP**). We consider each of these methods in turn.

A. Difference-Stationary Processes

If a time series has a unit root, the first differences of such time series (*i.e. a series with stochastic trend*) are stationary. Therefore, the solution here is to take the first differences of the time series.

Returning to our Ethiopian Trade Balance (TB) time series, we have already seen that it has a unit root. Let us now see what happens if we take the first differences of the TB series. Will the first difference still have a unit root? Let's perform the Durbin Watson test.

Let $\Delta TB_t = (TB_t - TB_{t-1})$. Now consider the following regression result:

$$\begin{aligned}\Delta(\Delta TB_t) &= -1.099\Delta TB_{t-1} \\ t &: (-6.7626) \\ R^2 &= 0.546; \quad d = 1.85\end{aligned}$$

The 1 percent critical ADF τ value is about -3.5073 as given in APPENDIX-D –Table-7 of Gujarati. Since the computed $\tau (= t)$ is more negative than the critical value, we conclude that the first-differenced TB is stationary; that is, it is $I(0)$. In other words, the overall unit root test result for TB time series at first difference is as given in Table 2.5.4 (the test result from EViews).

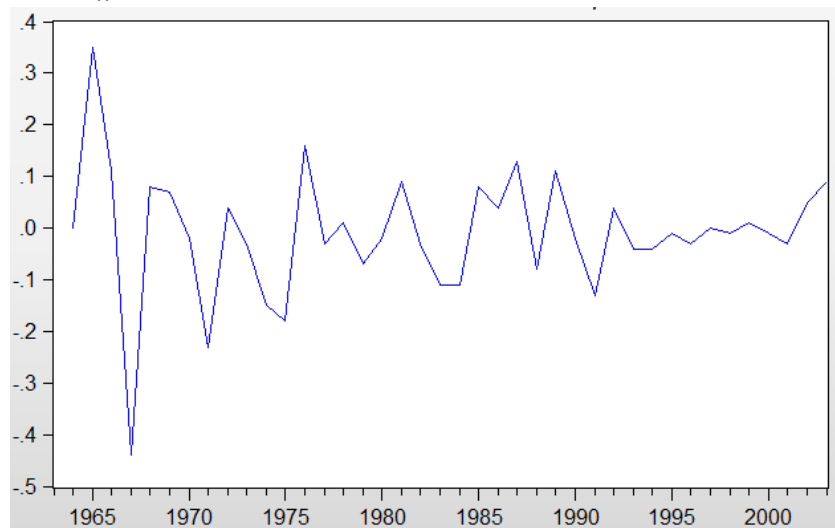
Table 2.5.4

The output of the unit root test (ADF test) for the variable TB

Null Hypothesis: D(TB) has a unit root				
Exogenous: None				
Lag Length: 0 (Automatic - based on SIC, maxlag=9)				
		t-Statistic	Prob.*	
Augmented Dickey-Fuller test statistic		-6.762648	0.0000	
Test critical values:	1% level	-2.625606		
	5% level	-1.949609		
	10% level	-1.611593		
*MacKinnon (1996) one-sided p-values.				
Augmented Dickey-Fuller Test Equation				
Dependent Variable: D(TB,2)				
Method: Least Squares				
Date: 04/18/19 Time: 06:06				
Sample (adjusted): 1965 2003				
Included observations: 39 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
D(TB(-1))	-1.099231	0.162545	-6.762648	0.0000
R-squared	0.546106	Mean dependent var		0.002308
Adjusted R-squared	0.546106	S.D. dependent var		0.184611
S.E. of regression	0.124376	Akaike info criterion		-1.305712
Sum squared resid	0.587835	Schwarz criterion		-1.263057
Log likelihood	26.46139	Hannan-Quinn criter.		-1.290408
Durbin-Watson stat	1.852202			

Figure 2.5.8

First differenced trade balance time series data



If you compare the above Figure 2.5.8 with Figure 2.5.1 (i.e. the Trade Balance figure at level), you will see the obvious difference between the two.

Table2.5.5*Unit Root test of the Log of MS2 data set at first difference*

Null Hypothesis: D(LMS2) has a unit root Exogenous: Constant, Linear Trend Lag Length: 0 (Automatic - based on SIC, maxlag=9)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-4.672461	0.0030
Test critical values: 1% level	-4.211868	
5% level	-3.529758	
10% level	-3.196411	
*MacKinnon (1996) one-sided p-values.		

Note that sometimes a series with stochastic trend may need to be differenced twice to be stationary. And such a series is said to be integrated of order two, $I(2)$. For instance, the Exchange Rate data set became stationary after second difference.

Table 2.5.6*Unit Root test of the Exchange Rate data set at second difference*

Null Hypothesis: D(EX,2) has a unit root Exogenous: Constant, Linear Trend Lag Length: 0 (Automatic - based on SIC, maxlag=9)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-5.077982	0.0010
Test critical values: 1% level	-4.219126	
5% level	-3.533083	
10% level	-3.198312	
*MacKinnon (1996) one-sided p-values.		

In general, the above test results in Table 2.5.4- Table 2.5.6 revealed that Trade balance (TB) and log of money supply (LMS2) became stationary at first difference while exchange rate data (EX) became stationary at the second difference. In all the above test results, the null hypothesis of Unit Root is rejected.

B. Trend-Stationary Process (TSP)

As we have noted before, aTSP is stationary around the trend line. Hence, the simplest way to make such a time series stationary is to regress it on time and the residuals from this regression will then be stationary. In other words, run the following regression:

$$Y_t = \delta + \beta t + u_t \quad (23)$$

where Y_t is the time series under study and where t is the trend variable measured chronologically.

Now,

$$\hat{u}_t = (\hat{Y}_t - \hat{\delta} - \hat{\beta}t)$$

will be stationary. $\hat{u}_t$ is known as a (linearly) **detrended time series**.

It is important to note that the trend may be **nonlinear**. For example, it could be

$$Y_t = \delta + \beta_1 t + \beta_2 t^2 + u_t \quad (24)$$

which is a quadratic trend series. If that is the case, the residuals from the above regression model will now be **(quadratically) detrended** time series. Note, however, that most macroeconomic time series are DSP rather than TSP.

2.5.5. How to run regression after unit-root testing?

The outcome of unit root testing matters for the empirical model to be estimated. The empirical model may consist of a **short run** part and a **long run** part. The **short run part** is to predict the behavior of the time series after a shock, the **long run part** is to predict the behavior of the time series in the long run. The following cases explain what is needed, dependent on the outcome of the unit root test.

CASE 1: All the series in the model under examination are stationary.

What if all the time series under consideration are stationary? Technically speaking, we mean they are $I(0)$ series (integrated of order zero).

- Under this scenario, any shock to the system in the short run quickly adjusts to the long run. Therefore, **only the long run model should be estimated**. Thus, the estimation of short run model is not necessary if series are $I(0)$.
- You can run a normal regression with the stationary time series, but do not forget to check the assumptions of the classical regression analysis (linear relation between the variables, no multicollinearity, homoskedasticity, no autocorrelation and a normal distribution of the disturbance term). Two examples of time series analysis with stationary data are given in 2.7.1 and 2.7.2: the ARDL and the VAR model.

CASE 2: All series in the model under consideration are $I(1)$.

Under this scenario, the series are non-stationary. One special feature of these series is that they are of the same order of integration, **$I(1)$** .

- To verify further the relevance of the model, there is need to test for cointegration.
- Cointegration is: can we assume a long run relationship in the model despite the fact that the series are trending either upward or downward?
- Cointegration implies that, even if there are shocks in the short run, which may affect movement in the individual series, they would converge with time (in the long run). However, there is no long run if series are not cointegrated. This implies that, if there are shocks to the system, the model is not likely to converge in the long run.
- Note that both **long run** and **short run models** must be estimated when there is cointegration. If there is no cointegration, there is no long run and therefore, **only the short run model** will be estimated.
- The test to be used are the Engle-Granger test or the Johansen test. These will be explained below.

CASE 3: The series are different order of integration.

Researchers are more likely to be confronted with this situation. For instance, some of the variables are $I(0)$ while others are $I(1)$.

- Like case 2, the cointegration test is also required under this scenario.
- Similar to case 2, if series are not cointegrated, we are expected to estimate **only the short run**. However, both **the long run** model and **short run** model are valid if there is cointegration.
- The appropriate test to use is the Bounds cointegration test. This test, however, is beyond the scope of this course. It may be covered in post graduate econometrics courses.

2.6 Spurious Regression Co-integration Test of Time Series Data

Why not just run the regression with non-stationary data? The regression of a nonstationary time series on another nonstationary time series may produce a **spurious regression**.

The Phenomenon of Spurious Regression

To see why stationary time series are so important, consider the following **two random walk models**:

$$Y_t = Y_{t-1} + u_t \quad (25)$$

$$X_t = X_{t-1} + v_t \quad (26)$$

Assume that u_t and v_t are serially uncorrelated as well as mutually uncorrelated. As you know by now, both these time series are non-stationary; that is, they are integrated of order one or exhibit stochastic trends. Suppose we regress Y_t on X_t . Since Y_t and X_t are uncorrelated $I(1)$ processes, the R^2 from the regression of Y_t on X_t should tend to zero; that is, there should not be any relationship between the two variables. But the regression results are as follows:

Table 2.6.1

Output of the linear regression $Y_t = \alpha + \beta X_t + u_t$

Y_t	Coefficients	t-Statistics	P-Value
X_t	0.54	3.7431	0.0012
Constants	110	2.0832	0.0232
$N = 500, R^2 = 0.712$		d-statistics = 0.241	

As you can see, the coefficient of t is highly statistically significant, and the R^2 value is also high and greater than the Durbin-Watson d-statistics. From these results, you may be tempted to conclude that there is a significant statistical relationship between Y and X , whereas a priori there should be none. This is in a *nutshell* the **phenomenon of spurious** or **non-senseregression**, first discovered by Yule. Yule showed that (spurious) correlation could persist in non-stationary time series, even if the sample is very large. That there is something wrong in the preceding regression is suggested by the extremely low Durbin-Watson d value,

which suggests very strong first-order autocorrelation. According to Granger and Newbold, when $R^2 > d$, it is a good **rule of thumb** to suspect that the estimated regression is **spurious**, as in the example above.

That the regression results presented above are meaningless can be easily seen from regressing the first differences of $Y_t (= \Delta Y_t)$ on the first differences of $X_t (= \Delta X_t)$, (remember that although Y_t and X_t are non-stationary, their first differences are stationary). In such a regression you will find that R^2 is practically zero, as it should be, and the Durbin–Watson d is about 2. Although dramatic, this example is a strong reminder that one should be extremely wary of conducting regression analysis based on time series that exhibit **stochastic trends**. And one should therefore be extremely cautious in reading too much in the regression results based on $I(1)$ variables.

Co-integration Test of Time Series Data

One solution to this spurious regression problem exists if the time series under investigation are cointegrated. A cointegration test is the appropriate method for detecting the existence of long-run relationship. Engle and Granger (1987) argue that, even though a set of economic series is not stationary, there may exist some linear combinations of the variables that are stationary, if the two variables are really related. **If the separate series are stationary only after differencing but a linear combination of their levels is stationary, the series are cointegrated.** Cointegration becomes an overriding requirement for any economic model using nonstationary time series data. If the variables do not co-integrate, we usually face the problems of spurious regression and econometric work becomes almost meaningless.

Suppose that, if there really is a genuine long-run relationship between any two variables: Y_t and X_t , although the variables will rise/decline overtime (because they are trended), there will be a common trend that links them together. For an equilibrium, or long-run relationship to exist, what we require, then, is a linear combination of Y_t and X_t that is a **stationary variable** [an $I(0)$ variable].

Assume that two variables Y_t and X_t are individually nonstationary time series. A linear combination of Y_t and X_t can be directly taken from estimating the following regression:

$$Y_t = \beta_0 + \beta_1 X_t + U_t$$

and taking the residuals:

$$U_t = Y_t - \beta_0 - \beta_1 X_t$$

We now subject U_t to a unit root test and hence if we find that $U_t \sim I(0)$, then the variables Y_t and X_t are said to be **co-integrated**. This is an interesting situation, for although Y_t and X_t are individually $I(1)$, that is, they have stochastic trends, their linear combination, $U_t = Y_t - \beta_0 - \beta_1 X_t$, is $I(0)$. So to speak, the linear combination cancels out the stochastic trends in the two series. If you take consumption and income as two $I(1)$ variables, savings defined as (income – consumption) could be $I(0)$. As a result, a regression of $Y_t = \beta_0 + \beta_1 X_t + U_t$ would be meaningful (i.e., not spurious). In this case we say that the two variables are

co-integrated. Economically speaking, two variables will be co-integrated if they have a long-term, or equilibrium, relationship between them.

In short, provided we check that the residuals from regressions like, $Y_t = \beta_0 + \beta_1 X_t + U_t$, are $I(0)$ or stationary, the traditional regression methodology (including the t and F tests) that we have considered extensively is applicable to data involving (nonstationary) time series. The valuable contribution of the concepts of unit root, co-integration, etc., is to force us to find out if the regression residuals are stationary. As Granger notes, “*A test for co-integration can be thought of as a pre-test to avoid ‘spurious regression’ situations*”.

In the language of co-integration theory, a regression such as, $Y_t = \beta_0 + \beta_1 X_t + U_t$, is known as a **co-integrating regression** and the slope parameter β_1 is known as the **co-integrating parameter**. The concept of co-integration can be extended to a regression model containing k regressors. In this case we will have k co-integrating parameters.

🚦 Testing for Cointegration

A number of methods for testing cointegration have been proposed in the literature. These include

1. Engle–Granger (EG)- *appropriate only when all variables are $I(1)$*
2. Johansen and Juselius (1990) co-integration test
3. Bounds cointegration test → *appropriate when variables are $I(0)$ and $I(1)$ or mutually $I(0)$ or mutually $I(1)$*

In this chapter we will only use the Engle-Granger (1987) two stage co-integration test procedures. The other procedures are discussed in post-graduate econometrics courses.

🚦 Engle-Granger Test for Cointegration

For a single equation, the simple test of cointegration is the ADF unit root tests on the residuals estimated from the cointegrating regression. This test is a modified test of unit root by the Engle-Granger (EG) and Augmented Engle-Granger (AEG) tests. Notice the difference between the unit root and cointegration tests. Tests for unit roots are performed on single time series, whereas cointegration test deals with the relationship among a group of variables, each having a unit root.

According to the **Engle-Granger two steps method**, first we have to estimate the static regression equation to get the long run multiplier. In the second step an error correction model is formulated and estimated using residuals from the first step as equilibrium error correction.

In other words, all we have to do is estimate a regression like, $Y_t = \beta_0 + \beta_1 X_t + U_t$, obtain the residuals, and use the ADF unit root test procedure. There is one precaution to exercise, however. Since the estimated residuals are based on the *estimated* co-integrating parameter β_1 , the Augmented Dickey Fuller critical

significance values are not quite appropriate. However, Engle and Granger have calculated these values, which can be found in the references of any statistical books. Therefore, the ADF tests in the present context are known as **Engle–Granger (EG)** test. However, several software packages now present these critical values along with other outputs.

Let us illustrate this test using a Trade Balance model.

1. Long run/Static Model of the Trade Balance of Ethiopia

Suppose we want to run the following model that predicts the log of the trade balance (LTB) based on the logs of MS2, GDP, EXR and LG:

$$LTB = \beta_0 + \beta_1 LMS2 + \beta_2 LGDP + \beta_3 LEXR + \beta_4 LG + U_i$$

Estimate the above model using least squares and save the residuals for stationarity test.

Table 2.6.2

Dependent Variable: LTB; Method: Least Squares; Included observations: 40 after adjustments

Variables	Coefficients	Standard Error	t-Statistics	P-value
LMS2	-0.734042	0.256962	-2.856615	0.0071
LGDP	0.219502	0.174400	1.258610	0.2163
LEXR	0.332259	0.137795	2.411259	0.0211
LG	0.175736	0.298237	0.589248	0.5594
Constant	1.500707	0.951265	1.577591	0.1234
$R^2 = 0.79$ Adjusted $R^2 = 0.77$; Durbin Watson d statistics = 1.18				

- Interpret the above regression results
- Which variables significantly affect the trade balance of Ethiopia in the long run?

2. Test of the Stationarity of the residuals obtained from the long run model

$$\Delta \text{Residual}_t = \beta_0 + \beta_1 \text{Residual}_{t-1} + v_t$$

Table 2.6.3

Augmented Dickey-Fuller Test Equation; Dependent Variable: D(RESIDUALS); Method: Least Squares; Included observations: 40 after adjustments

Variables	Coefficients	Standard Error	t-Statistics	P-value
Residuals _{t-1}	-0.650962	0.150385	-4.328652	0.0001
Constant	0.014651	0.031967	0.458332	0.6493
R^2 0.330246		F-statistic	18.73722	
Adjusted R^2 0.312621		Prob(F-statistic)	0.000105	
		Durbin-Watson stat	1.812471	

- Test the above model for stationarity
- Is the null hypothesis accepted or rejected?

This above regression result showed that the model is significant and we accept the alternative hypothesis. That means, the residuals from the long run model is stationary and this implies that regression of **LTB** =

$\beta_0 + \beta_1 LMS2 + \beta_2 LGDP + \beta_3 LEXR + \beta_4 LG + U_t$ is meaningful (i.e., not spurious). Thus, the variables in the model have a long run r/ships or equilibrium.

3. The Short Run Dynamics (Error Correction Model)

We just showed that LTB, LMS2, LGDP, LEXR and LG are cointegrated; that is, there is a long term, or equilibrium, relationship between these variables. Of course, in the short run there may be disequilibrium. Therefore, one can treat the error term in

$$LTB = \beta_0 + \beta_1 LMS2 + \beta_2 LGDP + \beta_3 LEXR + \beta_4 LG + U_t \quad (27)$$

as the “equilibrium error”. And we can use this error term to tie the short-run behavior of TB to its long-run value. This **error correction mechanism (ECM)**, first used by Sargan and later popularized by Engle and Granger, corrects for disequilibrium. An important theorem, known as the **Granger representation theorem**, states that if two variables Y and X are cointegrated, then the relationship between the two can be expressed as **ECM**. To see what this means, we will revert to our example of the trade balance of Ethiopia. But before that let's consider a typical example. Assume that Y_t and X_t are co-integrated and assuming further that X_t is exogenous, the ECM can be given as follows:

$$\Delta Y_t = \beta_0 + \beta_1 \Delta X_t + \delta u_{t-1} + \varepsilon_t \quad (28)$$

To account for short run dynamics, include lagged terms as:

$$\Delta Y_t = \beta_0 + \sum_{i=0}^p \beta_{1i} \Delta Y_{t-i} + \sum_{i=1}^k \beta_{2i} \Delta X_{t-i} + \delta u_{t-1} + \varepsilon_t \quad (29)$$

This is called **Error Correction Model (ECM) in time series econometrics**. u_{t-1} = *the lagged values of the residuals from the long run model is used in the short run regression analysis* and this is because the last periods disequilibrium or equilibrium error affects the direction of the dependent variable in the current period.

The coefficient of the lagged values of the residuals is expected to be less than one and negative. It is less than 1 because the adjustment towards equilibrium may not be 100%. If the coefficient is zero, the system is at equilibrium. Note that if δ is negative and significant, the adjustment is towards the equilibrium, but if δ is positive and significant, the adjustment is away from the equilibrium.

Illustration: Error Correction Model for the Trade Balance of Ethiopia

The **Error Correction Model (ECM)** for the Trade Balance of Ethiopia is specified as:

$$\begin{aligned} \Delta LTB_t = & \beta_0 + \sum_{i=0}^p \beta_{1i} \Delta LTB_{t-i} + \sum_{i=1}^q \beta_{2i} \Delta LEXR_{t-i} + \sum_{i=1}^q \beta_{3i} \Delta LMS2_{t-i} + \sum_{i=1}^q \beta_{4i} \Delta LGDP_{t-i} \\ & + \sum_{i=1}^q \beta_{5i} \Delta LG_{t-i} + \delta ECT_{t-1} + \varepsilon_t \end{aligned}$$

Where, p and q are the optimal lag length (which can be determined by minimizing the model selection information criteria) for the dependent and explanatory variables, respectively, while ECT_{t-1} is the

one period lagged values of the residuals obtained from the cointegrating regression, i.e., long run/static model.

Table 2.6.4

Short Run Determinants of the Trade Balance of Ethiopia (Output from EViews); Dependent Variable: DLTB

Method: Least Squares; Included observations: 39 after adjustments

Variables	Coefficients	Standard Error	t-Statistics	P-value
D(LTB(-1))	0.238526	0.162666	1.466352	0.1523
D(LEXR(-1))	-0.079162	0.274364	-0.288531	0.7748
D(LMS2(-1))	-0.540010	0.580852	-0.929685	0.3595
D(LGDP(-1))	-0.160748	0.351836	-0.456882	0.6508
D(LG(-1))	0.401352	0.277733	1.445103	0.1582
ECT(-1)	-0.888009	0.189013	-4.698145	0.0000
				<i>R – Square = 0.516354</i>
				<i>Adjusted R – Square = 0.425670</i>
				<i>Durbin Watson Statistics = 1.968382</i>

- Interpret the above regression results of ECM
- Which variables significantly affect the trade balance of Ethiopia in the short run?

The coefficient of the **error-correction term** of about **−0.89** suggests that about 89% of the discrepancy between long-term and short-term Trade Balance is corrected within a year (yearly data), suggesting a high rate of adjustment to equilibrium.

In summary, the Engle-Granger Two Stage Procedures:

- Run the Static Model at level

$$LTB = \beta_0 + \beta_1 LMS2 + \beta_2 LGDP + \beta_3 LEXR + \beta_4 LG + U_t$$

- Save the residuals from this long run or static model:

$$U_t = LTB - \beta_0 - \beta_1 LMS2 - \beta_2 LGDP - \beta_3 LEXR - \beta_4 LG$$

- Test the above residual for unit root:

$$\Delta U_t = \beta + \delta U_{t-1} + v_t$$

Where, $H_0 = U_t$ is non – stationary

$H_1 = U_t$ is stationary

- If the null hypothesis is accepted, the regression

$$LTB = \beta_0 + \beta_1 LMS2 + \beta_2 LGDP + \beta_3 LEXR + \beta_4 LG + U_t$$

is spurious. We don't have long run model and we have to run only the following short run model by disregarding the long run model.

$$\Delta LTB_t = \beta_1 \Delta LMS2_t + \beta_2 \Delta LY_t + \beta_3 \Delta LEXR_t + \beta_4 \Delta LG_t + U_t$$

(NB: one can also add the lagged values of the dependent & explanatory variables)

- If the null hypothesis is rejected, the regression

$$LTB = \beta_0 + \beta_1 LMS2 + \beta_2 LGDP + \beta_3 LEXR + \beta_4 LG + U_t$$

is meaningful (non-spurious). We do have both the long run model and short run dynamics. We have to include the lagged value of the residuals in the short run model as one explanatory variables.

2.7 Two other popular models to analyze time series data

Thus far, this chapter focused much on identifying problems related to time series, and a bit on solving these problems. What still lacks is some concrete examples of time series models, that can be run for the bachelor thesis next year. Therefore, this section discusses two models that are commonly used in time series analysis:

- The Autoregressive Distributed Lag (ARDL) model
- The Vector Auto Regression (VAR) model

2.7.1 The ARDL model

Autoregressive distributed lag models are models that contain the lagged values of the dependent variable and the current and lagged values of regressors. The model is used for forecasting and for policy analysis. For example, the ARDL model can visualize the effects of changes in government expenditure or taxation on unemployment and inflation (fiscal policy). Usually the effect of a change in a policy variable (e.g. public spending) does not immediately fully affect to other economic variables (e.g. GDP). The effect is working for a longer time period, (various time lags) for psychological, technical and institutional reasons. Mathematically, the model is expressed as follows:

$$y_t = \delta + \theta_1 y_{t-1} + \dots + \theta_p y_{t-p} + \delta_0 x_{t-1} + \dots + \delta_q x_{t-q} + v_t \quad (30)$$

- p is the number of lags of the dependent variable, y
- q is the number of lags of the additional predictor x

That means, y depends on its own lagged values, and on the lagged values of the explanatory variable, x. If the variables in the model are stationary, and the usual least squares assumptions are valid, we can estimate ARDL models using least squares. It is also possible to use non-stationary data for this model, but only if you add an error correction mechanism (like we discussed in the section above). To simplify the introduction of the ARDL, we will work with stationary data only in this section³. A challenge is to use the optimal number of lag lengths for p and q. Too many lags could increase the error in the forecasts, too few could leave out relevant information.

How should we choose the lag length p?

- Sufficient lags should be included to solve for serial correlation in the errors. It can be checked if sufficient lags are included if you make a correlogram.

³This is done in most econometric textbooks, for example in Hill, Griffiths & Lim (2011)

- Including too many lags may lead to insignificant parameter estimates. It is wise to exclude the longest lag (the final lag) if it is insignificant. Next, re-run the model and perform a hypothesis test on the new final lag. A drawback of this approach is that it can produce a model that is too large.
- An alternative way to determine the lag length p is to minimize the **BIC** (Bayes' information criterion, also called Schwarz information criterion), **AIC** (Akaike's information criterion). These criteria consider the pay off between the benefits of a larger model (with more lagged values, so more explanatory power) and the disadvantage of a greater model (less degrees of freedom). When running the model, Eviews asks you which information criterion you want to use. It's best to choose the Schwarz information criterion. The output will reveal the optimal lag length, both for the dependent (p) and the independent (q) variable.

Example short run Philips curve

When predicting future inflation, economic theory suggests that lagged values of the unemployment rate might be a good predictor. Therefore, we can use an ARDL(p, q) model to estimate the effect of unemployment (ILO estimate: U) on inflation (GDP deflator: I).

$$I_t = \delta + \theta_1 I_{t-1} + \dots + \theta_p I_{t-p} + \delta_0 U_{t-1} + \dots + \delta_q U_{t-q} + v_t$$

I. First we check if I and U are stationary

Table 2.7.1 and Table 2.7.2 show the outcome of the unit root tests for U and I . I has no unit root ($p=0.0066$, so H_0 is rejected). U is officially not stationary ($p=0.0501$, so H_0 cannot be rejected on a 95% confidence level), but it's close to stationarity. For the ease of the assignment we continue with the data.

Table 2.7.1

Unit root test I

Null Hypothesis: I has a unit root Exogenous: Constant, Linear Trend Lag Length: 0 (Automatic - based on SIC, maxlag=6)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-4.542348	0.0066
Test critical values: 1% level	-4.356068	
5% level	-3.595026	
10% level	-3.233456	
*MacKinnon (1996) one-sided p-values.		

Table 2.7.2

Unit root test U

Null Hypothesis: U has a unit root Exogenous: Constant, Linear Trend Lag Length: 4 (Automatic - based on SIC, maxlag=6)		
	t-Statistic	Prob.*
Augmented Dickey-Fuller test statistic	-3.610837	0.0501
Test critical values: 1% level	-4.394309	
5% level	-3.612199	
10% level	-3.243079	
*MacKinnon (1996) one-sided p-values.		

II. Next, we check the correlogram for the residuals of $I_t = \delta + \theta_1 I_{t-1} + \delta_0 U_t + v_t$, to see how many lags of $I(p)$ would be wise to include

The regression in Table 2.7.3 shows that unemployment has a significant effect on inflation ($p=0.0193$), but the lagged value of inflation has no significant effect on inflation ($p=0.5463$). The correlogram of the residuals (see Figure 2.7.1) reveals that there is no autocorrelation in the model, because the autocorrelation coefficients are all insignificant ($\text{Prob} > 0.05$). Based on the correlogram it is right to conclude that no lagged values of I must be included in the ARDL model. That means $p=0$.

Figure 2.7.1

Correlogram of the residual of the function
 $I_t = \delta + \theta_1 I_{t-1} + \delta_0 U_t + v_t$

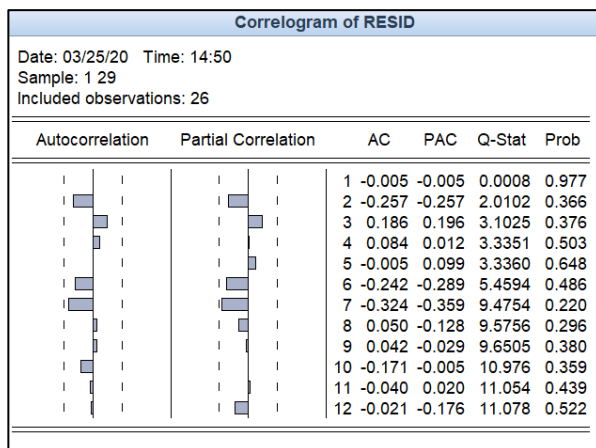
III. Next, we ask Eviews to run the model, using the number of lags (both for I and U) that minimize the BIC

Table 2.7.4 shows the output of the ARDL model. The output of Eviews reveals that, according to Bayes' information criterium (BIC=Schwarz information criterium), the optimal model is ARDL(1,0). This means that one lag is included for the dependent variable (inflation) and no lag is included for the independent variable (unemployment).

IV. Present and interpret the model

The last step is to interpret the model. We see that, though the lagged value for inflation is included, it is insignificant (prob.>0.05). Controlled for the previous value of inflation, the effect of a 1 unit increase in the unemployment rate leads to a 9.28 decrease in the inflation rate. This is a significant effect (p=0.0193).

V. Diagnostic testing of the model



ARDL is a linear regression model and therefore the underlying assumptions of CLRM have to be verified. These assumptions are, among others, that the model is correctly specified, homoscedasticity, there is no serial correlation and the error term is normally distributed.

Table 2.7.3

Output for the function $I_t = \delta + \theta_1 I_{t-1} + \delta_0 U_t + v_t$

Dependent Variable: I
Method: ARDL
Date: 03/25/20 Time: 15:04
Sample (adjusted): 2 27
Included observations: 26 after adjustments
Maximum dependent lags: 4 (Automatic selection)
Model selection method: Schwarz criterion (SIC)
Dyn: **Table 2.7.4**
Fixed: *Output of the ARDL model*
Number of models evaluated: 20
Selected Model: ARDL(1, 0)
Note: final equation sample is larger than selection sample

Variable	Coefficient	Std. Error	t-Statistic	Prob.*
I(-1)	0.106163	0.173358	0.612393	0.5463
U	-9.281618	3.690712	-2.514859	0.0193
C	33.40308	10.97821	3.042670	0.0058
R-squared	0.288695	Mean dependent var	9.427434	
Adjusted R-squared	0.226843	S.D. dependent var	11.12205	
S.E. of regression	9.779547	Akaike info criterion	7.506630	
Sum squared resid	2199.710	Schwarz criterion	7.651795	
Log likelihood	-94.58619	Hannan-Quinn criter.	7.548433	
F-statistic	4.667471	Durbin-Watson stat	1.984457	
Prob(F-statistic)	0.019890			

Dependent Variable: I				
Method: Least Squares				
Date: 03/25/20 Time: 14:48				
Sample (adjusted): 2 27				
Included observations: 26 after adjustments				
Variable	Coefficient	Std. Error	t-Statistic	Prob.
C	33.40308	10.97821	3.042670	0.0058
I(-1)	0.106163	0.173358	0.612393	0.5463
U	-9.281618	3.690712	-2.514859	0.0193
R-squared	0.288695	Mean dependent var	9.427434	
Adjusted R-squared	0.226843	S.D. dependent var	11.12205	
S.E. of regression	9.779547	Akaike info criterion	7.506630	
Sum squared resid	2199.710	Schwarz criterion	7.651795	
Log likelihood	-94.58619	Hannan-Quinn criter.	7.548433	
F-statistic	4.667471	Durbin-Watson stat	1.984457	
Prob(F-statistic)	0.019890			

2.7.2 The VAR model

In the above discussed models, we have clearly identified a dependent and several independent variables. According to Sims (1980), this is inappropriate, because economic variables are mostly not influencing each other in one direction. For example, inflation affects unemployment, but unemployment also affects inflation. Therefore, Sims developed the Vector Autoregression (VAR) model. The VAR model is a system of equations, which means that several functions, and thus several dependent variables, are estimated simultaneously. We will discuss simultaneous equation modeling more in-depth for cross-sectional data in chapter 3. This section gives a relatively simple example of a bivariate VAR model, because it is a popular model and also frequently used by undergraduate thesis students.

$$y_t = \alpha_1 + \sum_{i=1}^n \beta_i x_{t-i} + \sum_{j=1}^m \gamma_j y_{t-j} + e_{1t} \quad (31)$$

Table 2.7.5

The output of the VAR model

Vector Autoregression Estimates		
Date: 03/26/20 Time: 14:55		
Sample (adjusted): 2000M03 2020M02		
Included observations: 240 after adjustments		
Standard errors in () & t-statistics in []		
highlighted=significantly different from 0 with $\alpha=0.05$		
	DPI	SPI
DPI(-1)	1.475635 (0.05555) [26.5624]	0.152899 (0.11178) [1.36788]
DPI(-2)	-0.523871 (0.05521) [-9.48907]	-0.147106 (0.11108) [-1.32429]
SPI(-1)	-0.050337 (0.03049) [-1.65109]	1.283104 (0.06134) [20.9171]
SPI(-2)	0.063283 (0.03051) [2.07432]	-0.330001 (0.06138) [-5.37598]
C	4.776801 (1.84161) [2.59381]	7.101472 (3.70547) [1.91648]
R-squared	0.965360	0.946417
Adj. R-squared	0.964770	0.945505
Sum sq. resids	9159.918	37083.63
S.E. equation	6.243266	12.56195
F-statistic	1637.249	1037.683
Log likelihood	-777.5797	-945.3803
Akaike AIC	6.521497	7.919836
Schwarz SC	6.594011	7.992349
Mean dependent	139.4091	164.2131
S.D. dependent	33.26257	53.81203
Determinant resid covariance (dof adj.)		6123.484
Determinant resid covariance		5870.996

$$x_t = \alpha_2 + \sum_{i=1}^n \theta_i x_{t-i} + \sum_{j=1}^m \delta_j y_{t-j} + e_{2t} \quad (32)$$

Log likelihood	-1722.424
Akaike information criterion	14.43687
Schwarz criterion	14.58189

We assume that both y and x are stationary, and e_{1t} and e_{2t} are uncorrelated white noise error terms. Because both equations are linear in parameters, OLS is used for the estimation. For example, it is assumed that prices of different food commodities are interdependent. To this end, the FAO deflated food price index data is used to specify the following VAR model:

$$dpi_t = \alpha_1 + \beta_1 spi_{t-1} + \beta_2 spi_{t-2} + \gamma_1 dpi_{t-1} + \gamma_2 dpi_{t-2} + e_{1t}$$

$$spi_t = \alpha_2 + \theta_1 spi_{t-1} + \theta_2 spi_{t-2} + \delta_1 dpi_{t-1} + \delta_2 dpi_{t-2} + e_{2t}$$

Where dpi_t = the deflated dairy price index at year t (base year: 2002-2004)

And spi_t = the deflated sugar price index at year t (base year: 2002-2004)

The number of lags (2 for each variable) is determined based on a minimization of the AIC and BIC. The VAR output of Eviews is displayed in Table 2.7.5. Based on the R^2 and the F-statistic of the model, we can judge that the model performs well. Unfortunately, no p-values are given for the t-tests of the individual parameters, but given the t-statistics (the values in the square brackets []) and given that the sample is large (there are 240 observations), we know that the critical t-value for a 95% confidence level is 1.96. So, all parameter estimates with a t-value > 1.96 (or < -1.96) are considered significant. That is for the DPI formula the first and second lag of DPI and the second lag of SPI. For SPI the DPI lags are insignificant, but the SPI lags are significantly different from 0.

VAR Granger Causality/Block Exogeneity Wald Tests			
Date: 03/26/20 Time: 15:17			
Sample: 2000M01 2020M02			
Included observations: 240			
Dependent variable: DPI			
Excluded	Chi-sq	df	Prob.
SPI	6.021271	2	0.0493
All	6.021271	2	0.0493
Dependent variable: SPI			
Excluded	Chi-sq	df	Prob.
DPI	1.871434	2	0.3923
All	1.871434	2	0.3923

Table 2.7.6
Output of the Granger causality test

Though estimation of the VAR model is relatively simple, it is problematic to interpret VAR parameter estimates, because it seems that ‘everything causes everything’. It is not possible to measure the effect of DPI on SPI, leaving all other factors constant. Luckily, it is possible to test for the direction of the causality. Causality refers to the ability of one variable to predict the other variable. If there are only 2 variables, x and y, there are four possibilities:

- y causes x
- x causes y
- x causes y and y causes x (bi-directional feedback)
- x and y are independent

Granger (1969) developed a test that defines causality as follows: if a variable, x , Granger-causes y , y can be predicted with greater accuracy by using past values of the x variables rather than not using such past values. To test if x causes y , the Granger causality test uses the F-test on the β parameters of equation (31). Similarly, to see if y causes x , an F-test must be conducted on the δ parameters of equation (32).

See Table 2.7.6 for the output of the Granger causality test for the sugar and dairy prices. Recall that DPI was estimated according to the following formula:

$$dpi_t = \alpha_1 + \beta_1 spi_{t-1} + \beta_2 spi_{t-2} + \gamma_1 dpi_{t-1} + \gamma_2 dpi_{t-2} + e_{1t}$$

The first part of Table 2.7.6, *Dependent variable: DPI*, tests the H_0 that $\beta_1 = \beta_2 = 0$. H_0 is rejected (Prob<0.05), so the lagged values of SPI are able to explain the variation in DPI.

The second part of Table 2.7.6, *Dependent variable: SPI*, tests the H_0 that $\delta_1 = \delta_2 = 0$. In other words: the lagged values of DPI cannot explain the variation in SPI. This H_0 is not rejected (Prob.>0.05).

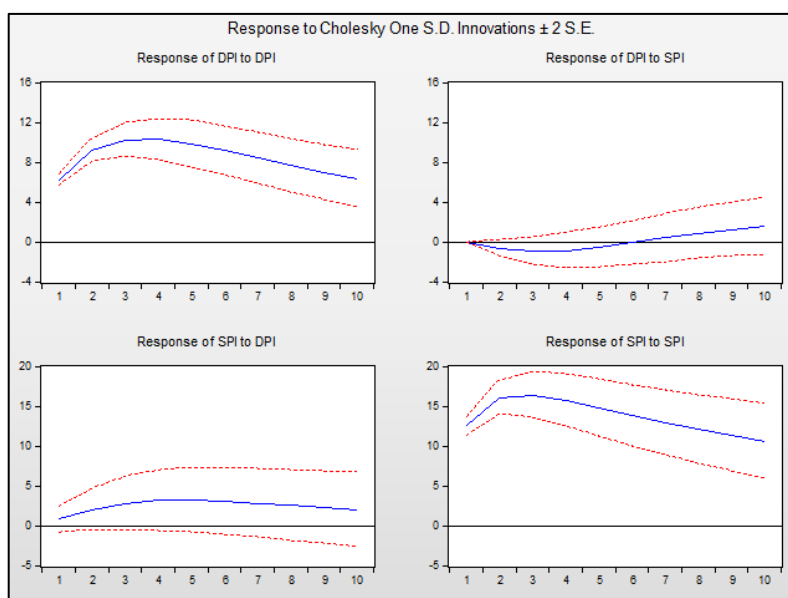
Figure 2.7.2
Response graphs

Concluding, this output reveals that SPI Granger-causes DPI, but DPI does not Granger-cause SPI.

Next to knowing the parameter estimates, and knowing which variable Granger-causes the other variable(s), it may be interesting to create response graphs. In the response graphs you see what a shock to one variable, for example SPI, does through the fitted VAR model to the other variable, in this case DPI. Figure 2.7.2 shows the response graphs from Eviews for the VAR presented above. The right upper graph displays the impact of a change in DPI on SPI. The value is around 0, which means that there is no big impact. This is in line with the Granger causality test which could not reject the $H_0 \delta_1 = \delta_2 = 0$. The lower left graph shows that the response of a change in SPI does positively affect the DPI for some lags. This is in line with the conclusion of the Granger causality test that rejected the H_0 that $\beta_1 = \beta_2 = 0$.

Note that we only explored the equilibrium relationship between DPI and SPI in this assignment. It is also possible to extend the VAR model by exploring the co-integration relationship, as we discussed in the ECM. It goes beyond the scope of this course to

extend the model in this way, but it may be a good option to do this for your thesis to further improve the power of your model.



2.7.3 Closing comments

In this chapter, much attention was paid to characteristics of time series. These characteristics may be troublesome while using time series data in regression. We have also seen some solutions, which allow us to use time series data in regression. The ones we have discussed are the error correction model (ECM), the autoregressive distributed lag model and the variance autoregression model. There are many more models that deal with time series data, which go beyond the scope of this course. It may be that you need to use these models for your thesis next year. If this is the case, we advise you to read the reference materials. The titles of these works can be found in the course outline.

INTRODUCTION TO SIMULTANEOUS EQUATION MODELS

Short notes only (consult your teacher for a softcopy of longer lecture notes with more explanation)

3.1. The Nature of Simultaneous Equation Models

So far, we were concerned exclusively with single equation models, i.e., models in which there was a single dependent variable Y and one or more explanatory variables, the X 's. In such models the emphasis was on estimating and/or predicting the average value of Y conditional upon the fixed values of the X variables. In many situations, such a one-way or unidirectional cause-and-effect relationship is not meaningful. This occurs if Y is determined by the X 's, and some of the X 's are, in turn, determined by Y . In short, there is a twoway, or simultaneous, relationship between Y and (some of) the X 's, which makes the distinction between *dependent* and *explanatory* variables doubtful.

Recall that one of the crucial assumptions of the method of OLS is that the explanatory X variables are either nonstochastic or, if stochastic (random), are distributed independently of the stochastic disturbance term. If this is not the case, as you can see in the below examples, application of the method of OLS is inappropriate.

Some Examples of Simultaneous-Equation Models

Example 1: Demand-and-Supply Models

In a perfectly competitive market setting, the price P of a commodity and the quantity Q sold are determined by the intersection of the demand-and-supply curves for that commodity. Thus, assuming for simplicity that the demand-and-supply curves are linear and adding the stochastic disturbance terms U_1 and U_2 , we may write the empirical demand-and-supply functions as:

$$\begin{aligned} \text{Demand Function: } Q_t^d &= \alpha_0 + \alpha_1 P_t + U_{1t} ; \\ \alpha_1 &< 0 \quad (1) \end{aligned}$$

$$\begin{aligned} \text{Supply Function: } Q_t^s &= \beta_0 + \beta_1 P_t + U_{2t} ; \\ \beta_1 &> 0 \quad (2) \end{aligned}$$

$$\text{Equilibrium condition: } Q_t^d = Q_t^s$$

where Q^d = quantity demanded

Q^s = quantity supplied, t = time and

the α 's and β 's are the parameters.

Now it is easy to see that P and Q are jointly dependent (=interdependent) variables. If, for example, U_{1t} in (1) changes because of changes in other variables affecting Q_t^d (such as income, consumer confidence and tastes), the demand curve will shift upward if U_{1t} is positive and downward if U_{1t} is negative. These shifts are shown in Figure 3.1.

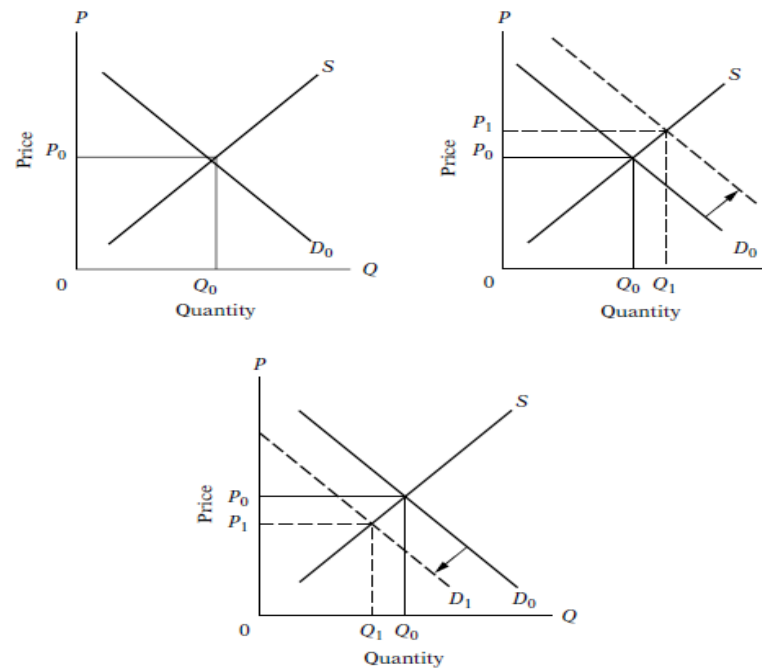


Figure 3.1
Interdependence between price and quantity

As the figure shows, a shift in the demand curve changes both P and Q . Similarly, a change in U_{2t} (because of weather condition, import or export restrictions, etc.) will shift the supply curve, again affecting both P and Q . Because of this simultaneous dependence between Q and P , U_{1t} and P_t in (1) and U_{2t} and P_t in (2) cannot be independent. Therefore, a regression of Q on P as in (1) would violate an important assumption of the classical linear regression model, namely, the assumption of no correlation between the explanatory variable(s) and the disturbance term.

Example 2: Wage–Price Model

Consider the following model of wage and price determination:

$$W_t = \beta_0 + \beta_1 UN_t + \beta_2 P_t + U_{1t} \quad (3)$$

$$P_t = \beta_0 + \beta_1 W_t + \beta_2 M_t + U_{2t} \quad (4)$$

Where, W_t = Nominal Wage Rate (in Birr)

UN_t = Unemployment rate (in %)

P_t = Rate of change of average prices of goods and services

M_t = Rate of change of price of imported raw material

t = time

U_{1t} & U_{2t} = Stochastic disturbances

Since the price variable, P , enters into the wage equation and the wage variable, W , enters into the price equation, the two variables are jointly dependent. Therefore, these stochastic explanatory variables are expected to be correlated with the relevant stochastic disturbances. The classical OLS method is therefore inappropriate to estimate the parameters of the two equations individually.

Example 3: Keynesian Model of Income Determination

Consider the simple Keynesian model of income determination given as:

$$C_t = \beta_0 + \beta_1 Y_t + U_t ; \quad 0 \leq \beta_1 \leq 1 \quad (5)$$

$$Y_t = C_t + I_t \quad (6)$$

Where, C = consumption expenditure

Y = income

I = investment (assumed as exogenous)

U = stochastic disturbance term,

Equation (5) is the consumption function; and (6) is the national income identity, signifying that total income is equal to total consumption expenditure plus total investment expenditure.

From the consumption function, it is clear that C and Y are interdependent and that Y_t in (5) is not expected to be independent of the disturbance term, because when U_t shifts (*for example, because people consume more around Addis Amed*), then the consumption function also shifts, which, in turn, affects Y_t . Therefore, once again the classical least-squares method is inapplicable to (5).

In general, when a relationship is a part of a system, then some explanatory variables are stochastic and are correlated with the disturbances. So, the basic assumption of a linear regression model that the explanatory variable and disturbance are uncorrelated, or explanatory variables are fixed, is violated.

Endogenous and Exogenous Variables

Variables in simultaneous equation models are classified as endogenous variables and exogenous variables.

◆ Endogenous Variables (Jointly determined variables)

Endogenous variables are variables which are influenced by one or more variables in the model. They are explained by the functioning of the system. Their values are determined by the simultaneous interaction of the relations in the model. We call them **endogenous variables**, **interdependent variables** or **jointly determined variables**.

◆ Exogenous Variables (Predetermined variables)

The variables that influence endogenous variables are called **exogenous** or **predetermined variables**. The values of these exogenous variables are determined outside the model. Exogenous variables influence the endogenous variables but are not themselves influenced by them. A variable which is endogenous for one model can be exogenous variable for another model.

Example: Consider our previous model of wage and price determination (equation (3) and (4)):

$$\begin{aligned} W_t &= \beta_0 + \beta_1 UN_t + \beta_2 P_t + U_{1t} \\ P_t &= \beta_0 + \beta_1 W_t + \beta_2 M_t + U_{2t} \end{aligned}$$

In this model, W_t and P_t are endogenous variables while UN_t and M_t are exogenous or predetermined variables.

3.2. Simultaneity Bias

The problem explained above, that there is correlation between the explanatory variable(s) and the error term, is at the core of Simultaneous Equation Modeling. For example, in the Wage-Price Model, does wage affect prices or do prices affect the wage? If OLS is used to estimate the equations individually, the problem of **simultaneity bias** occurs. That means that the least-squares estimators are **biased** (for small/finite samples) and **inconsistent** (for infinite samples). That means: as the sample size increases indefinitely, the estimators do not converge to their true (population) values. If you want to see the proof for this, it is recommended to read the appendix on page 138-139 in Gujarati (Econometrics by example, 2011).

3.3. Order and Rank conditions of Identification

In estimating simultaneous equation models, it is important to see if the model is identified. A simultaneous equation model can either be **overidentified**, **under identified** or **exactly identified**. If a model is **exactly identified**, unique numerical values of the model parameters can be obtained. If a model is **under identified**, no model parameters can be obtained. If a model is **over identified**, more than one numerical value can be obtained for some of the parameters.

Order condition of identification

There are several ways to check if a system of equations is identified. The most common one is the **order condition**.

To understand the order condition, we shall make use of the following notations:

- M = number of endogenous variables in the model
- m = number of endogenous variables in a given equation
- K = number of predetermined variables in the model
- k = number of predetermined variables in a given equation

The order condition:

- In a model of M simultaneous equations in order for an equation to be identified, it must exclude **at least $M - 1$** variables (endogenous as well as predetermined) appearing in the system of equations.
 - If it excludes exactly **$M - 1$** variables, the equation is just identified.
 - If it excludes more than **$M - 1$** variables, it is overidentified.

Or, to define the same thing differently:

- In a model of M simultaneous equations, in order for an equation to be identified, the number of predetermined variables excluded from the equation **must not be less than** the number of endogenous variables included in that equation less 1, that is,

$$K - k \geq m - 1$$

- If $K - k = m - 1$, the equation is just identified
- If $K - k > m - 1$, it is overidentified.

Let's consider some examples for illustration of the order condition of identification.

Example1: Consider our previous example of the demand function and the supply function.

$$\text{Demand Function: } Q_t^d = \alpha_0 + \alpha_1 P_t + U_{1t}; \quad \alpha_1 < 0 \quad (7)$$

$$\text{Supply Function: } Q_t^s = \beta_0 + \beta_1 P_t + U_{2t}; \quad \beta_1 > 0 \quad (8)$$

This model has two endogenous variables P and Q and no predetermined variables. To be identified, each of these equations must exclude at least $M - 1 = 1$ variable. Since this is not the case, **both equations are not identified**.

Example2: Consider the following demand and supply equations

$$\text{Demand Function: } Q_t^d = \alpha_0 + \alpha_1 P_t + \alpha_2 Y_t + U_{1t} \quad (9)$$

$$\text{Supply Function: } Q_t^s = \beta_0 + \beta_1 P_t + U_{2t} \quad (10)$$

In this model, Q and P are endogenous and Y_t (*consumer's income*) is exogenous. Applying the order condition given, we see that the **demand function is unidentified**. On the other hand, the **supply function is just identified** because it excludes exactly $M - 1 = 1$ variable, Y_t .

Example3: Consider the following demand and supply equations

$$\text{Demand Function: } Q_t^d = \alpha_0 + \alpha_1 P_t + \alpha_2 Y_t + \alpha_3 P_{St} + U_{1t} \quad (11)$$

$$\text{Supply Function: } Q_t^s = \beta_0 + \beta_1 P_t + \beta_2 P_{t-1} + U_{2t} \quad (12)$$

In this model, P_t and Q_t are endogenous and Y_t , P_{St} (= Price of a substitute good), and P_{t-1} are predetermined. The **demand function** excludes exactly one variable P_{t-1} , and hence by the order condition it is **exactly identified**. But the **supply function** excludes two variables Y_t and P_{St} , and hence it is **overidentified**. Hence, there is a possibility to have many solutions for β_1 , the coefficient of the price variable of the supply model.

Notice at this juncture that as the previous examples show, identification of an equation in a model of simultaneous equations **is possible if that equation excludes one or more variables that are present elsewhere in the system**. This situation is known as the exclusion (of variables) criterion, or **zero restrictions criterion** (the coefficients of variables not appearing in an equation are assumed to have zero values). This criterion is by far the most commonly used method of securing or determining identification of an equation. In using this method, the researcher should always consider economic theory and judge on whether it is correct that the variable(s) are excluded from or included in the equation.

The order condition is necessary for identification, but unfortunately there are some special cases in which the order condition is insufficient. A more complex, but both necessary and sufficient method is the **rank condition**, which will be shortly discussed below.

The Rank Condition of Identification

The order condition discussed previously is a *necessary but not sufficient* condition for identification; that is, even if it is satisfied, it may happen that an equation is not identified. We need both a necessary and sufficient condition for identification. This is provided by the *rank⁴ condition* of identification.

Rank condition of identification: In a model containing M equations in M endogenous variables, an equation is **identified** if a **rank of size $M-1$** can be found in the matrix from the coefficients of the variables (both endogenous and predetermined) excluded from that particular equation but included in the other equations of the model. Remember that a **rank** is the largest-order square matrix (contained in the given matrix) whose **determinant is nonzero**.

As this is quite theoretical, and your matrix algebra insight might need refreshment, please check the example below.

Example of checking the rank condition of identification

As an illustration of the rank condition of identification, consider the following hypothetical system of simultaneous equations in which the Y variables are endogenous and the X variables are predetermined or exogenous.

$$Y_{1t} - \beta_{10} - \beta_{12}Y_{2t} - \beta_{13}Y_{3t} - \alpha_{11}X_{1t} = U_{1t} \dots \dots \dots (13)$$

$$Y_{2t} - \beta_{20} - \beta_{23}Y_{3t} - \alpha_{21}X_{1t} - \alpha_{22}X_{2t} = U_{2t} \dots \dots \dots (14)$$

$$Y_{3t} - \beta_{30} - \beta_{31}Y_{1t} - \alpha_{31}X_{1t} - \alpha_{32}X_{2t} = U_{3t} \dots \dots \dots (15)$$

$$Y_{4t} - \beta_{40} - \beta_{41}Y_{1t} - \beta_{42}Y_{2t} - \alpha_{43}X_{3t} = U_{4t} \dots \dots \dots (16)$$

◆ Steps in the Rank Condition

To apply the rank condition of identifiability, one may follow the following steps:

1. Write down the system of equations in a tabular form.
2. Cross out the coefficients of the row in which the equation under consideration appears.
3. Also cross out the columns corresponding to the coefficients which are nonzero for the equation under consideration.
4. The entries left in the table will then give only the coefficients of the variables included in the system but not in the equation under consideration. From these entries form all possible matrices of order $M - 1$ and obtain the corresponding determinants. **If at least one nonzero determinant can be found, the equation in question is (just or over) identified.** If all the possible matrices of order $M - 1$ have a determinant of zero, the rank of the matrix is less than $M - 1$ and the equation under investigation is not identified.

Following the above procedure, let's find out whether equation (13) is identified.

Step 1: Table 3.1 displays the system of equations in tabular form.

Step 2: The coefficients for row (13) have been crossed out, because (13) is the equation under consideration.

Step 3: The coefficients for columns 1, Y_1 , Y_2 , Y_3 and X_1 have been crossed out, because they appear in (13).

Table 3.1 The tabular form of the systems of equations (Step 1), equation (13) erased (Step 2) and the coefficients of the variables included in (13) erased (Step 3)

Coefficients of the variables								
Equation No.	1	Y_1	Y_2	Y_3	Y_4	X_1	X_2	X_3

⁴The term **rank** refers to the rank of a matrix and is given by the largest-order square matrix (contained in the given matrix) whose determinant is nonzero. Alternatively, the rank of a matrix is the largest number of linearly independent rows or columns of that matrix.

13	$-\beta_{10}$	1	$-\beta_{12}$	$-\beta_{13}$	0	$-\alpha_{11}$	0	0
14	$-\beta_{20}$	0	1	$-\beta_{23}$	0	$-\alpha_{21}$	$-\alpha_{22}$	0
15	$-\beta_{30}$	$-\beta_{31}$	0	1	0	$-\alpha_{31}$	$-\alpha_{32}$	0
16	$-\beta_{40}$	$-\beta_{41}$	$-\beta_{42}$	0	1	0	0	$-\alpha_{43}$

Step 4: Matrix A (17) is created from the remaining coefficients Table 3.1. For this equation to be identified, we must obtain at least one nonzero determinant of order 3×3 from the coefficients of the variables excluded from this equation but included in other equations.

$$A = \begin{bmatrix} 0 & -\alpha_{22} & 0 \\ 0 & -\alpha_{32} & 0 \\ 1 & 0 & -\alpha_{43} \end{bmatrix} \dots \dots \dots (17)$$

It can be seen that the determinant of this matrix is zero:

$$|A| = \begin{vmatrix} 0 & -\alpha_{22} & 0 \\ 0 & -\alpha_{32} & 0 \\ 1 & 0 & -\alpha_{43} \end{vmatrix} = 0$$

Since the determinant is zero, the rank of matrix A (17), is less than 3, (i.e., $M-1$). **Therefore, Eq. (13) does not satisfy the rank condition and hence is not identified.**

Therefore, although the order condition shows that Eq. (13) is identified, the rank condition shows that it is not. Apparently, the columns or rows of the matrix A given in (17) are not (linearly) independent, meaning that there is some relationship between the variables Y_4, X_2 , and X_3 . As a result, we may not have enough information to estimate the parameters of equation (13).

Our discussion of the order and rank conditions of identification leads to the following general principles of identifiability of a structural equation in a system of M simultaneous equations:

1. If $K - k > m - 1$ and the rank of the A matrix is $M - 1$, the equation is **overidentified**.
2. If $K - k = m - 1$ and the rank of the matrix A is $M - 1$, the equation is **exactly identified**.
3. If $K - k \geq m - 1$ and the rank of the matrix A is less than $M - 1$, the equation is **not identified**.
4. If $K - k < m - 1$, the structural equation is **not identified**. The rank of the A matrix in this case is bound to be less than $M - 1$.

Which condition should one use in practice: Order or rank? For large simultaneous-equation models, applying the rank condition is a formidable task. Therefore, as Harvey notes: “Fortunately, the order condition is usually sufficient to ensure identifiability, and although it is important to be aware of the rank condition, a failure to verify it will rarely result in disaster”.

3.4 Indirect squares and 2SLS estimation of structural equations

If an equation is identified, we can estimate it by using by Indirect Least Squares (ILS) or Two-Stage Least Squares (2SLS)

Indirect Least Squares (ILS)

ILS can be used for an exactly identified structural equation. This section explains the steps involved in ILS.

Step 1: Specify the Structural Model

Consider, for instance, the following demand and supply equations

$$\text{Demand Function: } Q_t = \alpha_0 + \alpha_1 P_t + \alpha_2 Y_t + u_{1t} \quad (18)$$

$$\text{Supply Function: } Q_t = \beta_0 + \beta_1 P_t + \beta_2 P_{t-1} + u_{2t} \quad (19)$$

Where, Q_t, P_t, Y_t and P_{t-1} are quantity, price, consumer's income and lagged price, respectively.

Step 2: Find the reduced-form equations.

A **reduced-form equation** is one that expresses an endogenous variable solely in terms of the predetermined variables and the stochastic disturbances. The reduced form expresses every endogenous variable as a function of (an) exogenous variable(s).

Based on the equilibrium condition,

$$\alpha_0 + \alpha_1 P_t + \alpha_2 Y_t + u_{1t} = \beta_0 + \beta_1 P_t + \beta_2 P_{t-1} + u_{2t}.$$

Solving this equation, we obtain the following equilibrium price:

$$P_t = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} - \frac{\alpha_2}{\alpha_1 - \beta_1} Y_t + \frac{\beta_2}{\alpha_1 - \beta_1} P_{t-1} + \frac{u_{2t} - u_{1t}}{\alpha_1 - \beta_1}$$

We simplify this formula by stating that $\pi_0 = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1}$; $\pi_1 = -\frac{\alpha_2}{\alpha_1 - \beta_1}$; $\pi_2 = \frac{\beta_2}{\alpha_1 - \beta_1}$ and $v_{1t} = \frac{u_{2t} - u_{1t}}{\alpha_1 - \beta_1}$

$$P_t = \pi_0 + \pi_1 Y_t + \pi_2 P_{t-1} + v_{1t} \dots \dots \dots (20)$$

Equation (20) is the first reduced form equation in this model. Because the model has another endogenous variable, Q_t , another reduced form equation must be made. Substituting the equilibrium price into either the demand or supply equation, we obtain the following equilibrium quantity:

$$Q_t = \pi_3 + \pi_4 Y_t + \pi_5 P_{t-1} + v_{2t} \dots \dots \dots (21)$$

$$\text{Where, } \pi_3 = \frac{\alpha_1 \beta_0 - \alpha_0 \beta_1}{\alpha_1 - \beta_1} ; \pi_4 = -\frac{\alpha_2 \beta_1}{\alpha_1 - \beta_1} ; \pi_5 = \frac{\alpha_1 \beta_2}{\alpha_1 - \beta_1} ; v_{2t} = \frac{\alpha_1 u_{2t} - \beta_1 u_{1t}}{\alpha_1 - \beta_1}$$

Step 3: Estimate each of the reduced form equations by OLS individually. This operation is permissible since the explanatory variables in these equations are predetermined and hence uncorrelated with the stochastic disturbances. The estimates obtained are thus consistent.

$$P_t = \pi_0 + \pi_1 Y_t + \pi_2 P_{t-1} + v_{1t}$$

$$Q_t = \pi_3 + \pi_4 Y_t + \pi_5 P_{t-1} + v_{2t}$$

Suppose that the estimation results of the reduced form models are given as follows:

$$P_t = 6 + 10Y_t + 6P_{t-1}$$

$$Q_t = 96 + 50Y_t + 36P_{t-1}$$

Step 4: Determine the coefficients of the Structural model (i.e., $\alpha_0, \alpha_1, \alpha_2, \beta_0, \beta_1$ and β_2) from the estimated reduced-form coefficients. As noted before, if an equation is **exactly identified**, there is a one-to-one correspondence between the structural and reduced-form coefficients; that is, one can derive unique estimates of the former from the latter.

$$\pi_1 = -\frac{\alpha_2}{\alpha_1 - \beta_1} = 10 \text{ and } \pi_4 = -\frac{\alpha_2 \beta_1}{\alpha_1 - \beta_1} = 50$$

$$\alpha_1 - \beta_1 = -\frac{\alpha_2}{10} \text{ and } \alpha_1 - \beta_1 = -\frac{\alpha_2 \beta_1}{10} \\ \therefore \beta_1 = 5$$

$$\pi_2 = \frac{\beta_2}{\alpha_1 - \beta_1} = 6 \text{ and } \pi_5 = \frac{\alpha_1 \beta_2}{\alpha_1 - \beta_1} = 36$$

$$\alpha_1 - \beta_1 = \frac{\beta_2}{6} \text{ and } \alpha_1 - \beta_1 = \frac{\alpha_1 \beta_2}{36} \\ \therefore \alpha_1 = 6$$

$$\pi_0 = \frac{\beta_0 - \alpha_0}{\alpha_1 - \beta_1} = 6 \text{ and } \pi_3 = \frac{\alpha_1 \beta_0 - \alpha_0 \beta_1}{\alpha_1 - \beta_1} = 96$$

Given $\alpha_1 = 6$ & $\beta_1 = 5$, we obtain the following equations

$$\beta_0 - \alpha_0 = 6$$

$$6\beta_0 - 5\alpha_0 = 96$$

Solving this system of 2 equations, we obtain

$$\alpha_0 = 60 \text{ \& } \beta_0 = 66$$

Finally, we can obtain, $\alpha_2 = 10 \text{ \& } \beta_2 = 6$

Therefore, the estimated Structural equations will be:

$$\text{Demand Function: } Q_t = 60 + 6P_t + 10Y_t$$

$$\text{Supply Function: } Q_t = 66 + 5P_t + 6P_{t-1}$$

As this four-step procedure indicates, the name **Indirect Least Squares (ILS)** derives from the fact that structural coefficients (the object of primary enquiry) are obtained **indirectly** from the OLS estimates of the reduced-form coefficients.

The Method of Two-Stage Least Squares (2SLS)

If a structural equation is over identified, ILS is consistent, but does not give a unique estimate. Therefore, it is better to use 2SLS estimation method.

Consider the following income and money supply model:

$$\text{GDP: } Y_{1t} = \alpha_0 + \alpha_1 Y_{2t} + \alpha_3 X_{1t} + \alpha_4 X_{2t} + u_{1t} \quad (22)$$

$$\text{Money Supply: } Y_{2t} = \beta_0 + \beta_1 Y_{1t} + u_{2t} \quad (23)$$

Where, $Y_1 = \text{GDP}$

$Y_2 = \text{Money Supply}$

$X_1 = \text{Investment spending}$

$X_2 = \text{Government expenditure}$

The variables X_1 and X_2 are exogenous.

The GDP equation states that national income is determined by money supply, investment expenditure, and government expenditure. The money supply function states that the stock of money is determined (by the Monetary Authority) on the basis of the level of income.

Applying the order condition of identification, we can see that the income equation is not identified whereas the money supply equation is overidentified. To estimate the overidentified money supply model, one can use the method of 2SLS. As its name indicates, the method of 2SLS involves two successive applications of OLS.

◆ The procedure is as follows:

Stage 1: To get rid of the likely correlation between Y_1 and u_2 , regress first Y_1 on *all* the predetermined variables in the *whole system*, not just that equation. In the present case, this means regressing Y_1 on X_1 and X_2 as follows:

$$Y_{1t} = \hat{\pi}_0 + \hat{\pi}_1 X_{1t} + \hat{\pi}_2 X_{2t} + \hat{u}_t \quad (24)$$

Where, $\hat{u}_t$ are the usual OLS residuals

From Eq. (24) we obtain $\hat{Y}_{1t}$,

$$\hat{Y}_{1t} = \hat{\pi}_0 + \hat{\pi}_1 X_{1t} + \hat{\pi}_2 X_{2t} \quad (25)$$

Note that (25) is nothing but a reduced-form regression because only the exogenous or predetermined variables appear on the right-hand side. Equation (25) can now be expressed as

$$Y_{1t} = \hat{Y}_{1t} + \hat{u}_t \quad (26)$$

which shows that the stochastic Y_1 consists of two parts: $\hat{Y}_{1t}$, which is a linear combination of the nonstochastic X 's, and a random component $\hat{u}_t$. Following the OLS theory, $\hat{Y}_{1t}$ and $\hat{u}_t$ are uncorrelated. (Why?)

Stage 2: The overidentified money supply equation can now be written as:

$$\begin{aligned} Y_{2t} &= \beta_0 + \beta_1(\hat{Y}_{1t} + \hat{u}_t) + u_{2t} \\ &= \beta_0 + \beta_1\hat{Y}_{1t} + (u_{2t} + \beta_1\hat{u}_t) \\ Y_{2t} &= \beta_0 + \beta_1\hat{Y}_{1t} + v_t \quad (27) \\ \text{where } v_t &= u_{2t} + \beta_1\hat{u}_t \end{aligned}$$

Comparing equation (27) with the original money supply model, we see that they are very similar in appearance, the only difference being that Y_1 is replaced by $\hat{Y}_{1t}$. What is the advantage of (27)? It can be shown that although Y_1 in the original money supply equation is correlated or likely to be correlated with the disturbance term u_{2t} (hence making OLS inappropriate), $\hat{Y}_{1t}$ in (27) is uncorrelated with v_t . Therefore, OLS can be applied to (27), which will give consistent estimates of the parameters of the money supply function.

As this two-stage procedure indicates, the basic idea behind 2SLS is to “purify” the stochastic explanatory variable Y_{1t} of the influence of the stochastic disturbance u_2 . This goal is accomplished by performing the reduced-form regression of Y_1 on all the predetermined variables in the system (Stage 1). These predetermined variables are called the **instrumental variables**. Obtaining the estimates $\hat{Y}_{1t}$, based on the instrumental variables, and replacing Y_{1t} in the original equation by the estimated $\hat{Y}_{1t}$, and then applying OLS to the equation thus transformed (Stage 2). The estimators thus obtained are consistent; that is, they converge to their true values as the sample size increases indefinitely.

Note the following **features of 2SLS**.

1. It can be applied to an individual equation in the system without directly taking into account any other equation(s) in the system. Hence, for solving econometric models involving a large number of equations, 2SLS offers an economical method. For this reason, the method has been used extensively in practice.
2. Unlike ILS, which provides multiple estimates of parameters in the **overidentified** equations, 2SLS provides only one estimate per parameter.
3. Although specially designed to handle overidentified equations, the method can also be applied to exactly identified equations. But then ILS and 2SLS will give identical estimates. (Why?)
4. If the R^2 values in the reduced-form regressions (that is, Stage 1 regressions) are very high, say, in excess of 0.8, the classical OLS estimates and 2SLS estimates will be very close to OLS estimates. This is because if the R^2 value in the first stage is very high, it means that the estimated values of the endogenous variables are very close to their actual values, and hence the latter are less likely to be correlated with the stochastic disturbances in the original structural equations.

1.6. Testing Simultaneity

If there is no simultaneous equation, or **simultaneity problem**, the OLS estimators produce consistent and efficient estimators. On the other hand, **if there is simultaneity**, OLS estimators are not even consistent. In the presence of simultaneity, the methods of **Indirect Least Squares (ILS)** and **Two Stage Least Squares (2SLS)** will give estimators that are consistent and efficient. Oddly, if we apply these alternative methods when there is in fact **no simultaneity**, these methods yield estimators that are consistent but not efficient (i.e., with smaller variance). Therefore, we should check for the simultaneity problem before we discard OLS in favor of the alternatives.

As we showed earlier, the simultaneity problem arises because some of the regressors are endogenous and are, therefore, likely to be correlated with the disturbance term. Therefore, a test of simultaneity is essentially a test of whether (an endogenous) regressor is correlated with the error term. If a simultaneity

problem exists, alternatives to OLS must be found; if it is not, we can use OLS. To find out which is the case in a concrete situation, we can use **Hausman's specification test**.

For illustration, consider the following national income and money supply model:

$$\text{National Income Function: } Y_{1t} = \alpha_0 + \alpha_1 Y_{2t} + \alpha_3 X_{1t} + \alpha_4 X_{2t} + u_{1t} \quad (28)$$

$$\text{Money Supply Function: } Y_{2t} = \beta_0 + \beta_1 Y_{1t} + u_{2t} \quad (29)$$

Where, Y_1 = Real Gross Domestic Product

Y_2 = Money Supply

X_1 = Investment spending

X_2 = Government expenditure

Note that applying the order condition, the national income function is under identified while the money supply function is overidentified (which can be estimated by 2SLS).

If there is **no simultaneity problem** (i.e., Y_2 and Y_1 are mutually independent), Y_{1t} and u_{2t} should be uncorrelated. On the other hand, if there is simultaneity, Y_{1t} and u_{2t} will be correlated. To find out which is the case, the Hausman test can be used:

◆ The Hausman test involves the following steps:

Step-1: Regress Y_{1t} on X_{1t} and X_{2t} to obtain $\hat{v}_t$ (i.e., we estimate the reduced-form equation)

$$Y_{1t} = \hat{\pi}_0 + \hat{\pi}_1 X_{1t} + \hat{\pi}_2 X_{2t} + \hat{v}_t \quad (30)$$

Step-2: Regress Y_{2t} on $\hat{Y}_{1t}$ and $\hat{v}_t$ and perform a t test on the coefficient of $\hat{v}_t$. That is,

$$\hat{Y}_{2t} = \hat{\pi}_0 + \hat{\pi}_1 \hat{Y}_{1t} + \hat{\delta}_2 \hat{v}_t \quad (31)$$

If the coefficient of $\hat{v}_t$ is significant (p-value ≤ 0.05), there is simultaneity, and ILS or 2SLS must be used. If the coefficient of $\hat{v}_t$ is insignificant (p-value > 0.05), there is no simultaneity, and it is better to use OLS.

◆ Numerical Example:

Suppose that data is given on GDP, money supply, private investment and government spending. We are interested to estimate the money supply model by 2SLS (since the equation is overidentified). However, we need to make sure that there is indeed a simultaneity problem that makes OLS inappropriate. Otherwise it makes no sense to use the 2SLS method. To this end, we considered the **Hausman's specification error test** and obtained the following results.

- First we estimate the reduced-form regression given in equation (30). From this regression we obtain the estimated GDP and the residuals $\hat{v}_t$.
- Second we regress money supply on estimated GDP and $\hat{v}_t$ to obtain the following results:

$$\begin{aligned} \hat{Y}_{2t} &= -2198.3 + 0.79\hat{Y}_{1t} + 0.6984\hat{v}_t \dots \dots \dots (32) \\ t &= (-17.03) \quad (36.70) \quad (2.35)^{**} \end{aligned}$$

Since the t value of the coefficient of $\hat{v}_t$ is statistically significant at 5% significance level (also visible in the **), we must conclude that there is simultaneity between money supply and GDP. 2SLS is the appropriate procedure to be used.

Worksheet

Exam training true/false

State whether each of the following statements is true or false:

- In a simultaneous-equation model, there must at least be three dependent variables.

- If two variables in a simultaneous-equation model are mutually dependent, they are endogenous variables.
- In SEM, a predetermined variable is the same as an exogenous variable.
- An exogenous variable can be mutually dependent with an endogenous variable.
- The method of OLS is not applicable to estimate a structural equation in a simultaneous-equation model.
- The reduced form equation satisfies all the assumptions needed for the application of OLS.
- If a model is under-identified, the ILS procedure will give more than one parameter estimate for the same parameter.
- Even though a model is not identified, it is possible that one of the model's equations is identified.
- In case an equation is not identified, 2SLS is not applicable.
- If an equation is exactly identified, ILS and 2SLS give identical results.

Examples of Simultaneous Equation Models

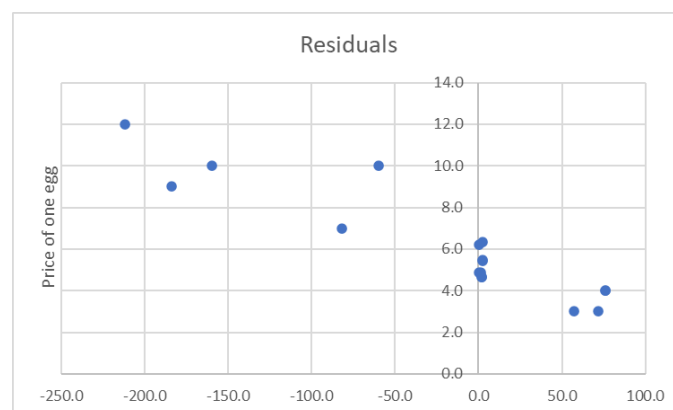
- What are the three examples of SEMs given in the lecture notes?
- Describe which assumption of OLS is violated in all three cases.
- For the first and the second example, indicate which variables are exogenous and which variables are endogenous.

Simultaneity bias

Simultaneity bias means that the independent variable influences the dependent variable, but also vice versa. If an external shock influences the dependent variable, this has effect on the independent variable, and therefore you can observe a correlation between the error term and the explanatory variable. In this simple model, the demand for eggs is estimated based on the price of eggs.

$$D_t = \beta_0 + \beta_1 P_t + U_t$$

- Knowing economic theory, you can expect simultaneity bias in this model. Why?
- Observe the following scatterplot, displaying the relation between P and U. Is P an exogenous or an endogenous variable in this model? Explain your answer.



- Alternative to running an OLS on this simple equation, it may be better to build a system of equations. Build a system of equations for this case.

Order condition of identification

Use the order condition of identification to see if the following systems of equations are over-, under- or exactly identified.

Model 1: Interest rate vs. GDP I (source: Gujarati & Porter)

$$R_t = A_1 + A_2 M_t + A_3 Y_t + u_{1t}$$

$$Y_t = B_1 + B_2 R_t + u_{2t}$$

Where Y= income (GDP), R=interest rate and M=money supply.

Model 2: Interest rate vs. GDP II (source: Gujarati & Porter)

$$R_t = A_1 + A_2 M_t + A_3 Y_t + u_{1t}$$

$$Y_t = B_1 + B_2 R_t + B_3 I_t + u_{2t}$$

Where Y= income (GDP), R=interest rate, M=money supply and I=gross private domestic investment.

Model 3: Demand and supply for loans (source: Gujarati & Porter)

$$\text{Demand: } Q_t = A_1 + A_2 R_t + A_3 RD_t + A_4 IPI_t + u_{1t}$$

$$\text{Supply: } Q_t = B_1 + B_2 R_t + B_3 RS_t + B_4 TBD_t + u_{2t}$$

Where Q = total commercial bank loans (\$ billion), R = average prime rate, RS = 3-month Treasury bill rate, RD = AAA corporate bond rate, IPI = Index of Industrial Production and TBD = total bank deposits.

Model 4: Openness and inflation

$$I_i = \alpha_0 + \alpha_1 IMP_i + \alpha_2 INC_i + U_{1i}$$

$$IMP_i = \beta_0 + \beta_1 I_i + \beta_2 INC_i + \beta_3 LAND_i + U_{2i}$$

Where I=inflation rate, IMP=imports as % of GDP (measure of openness), INC=GDP per capita, LAND=log of land area in square miles.

Rank condition of identification

Verify that by the rank condition, equations (14) and (15) of the lecture notes (page 6) are unidentified but equation (16) is identified.

Indirect least squares

There are two major types of coffee beans: the expensive and high-quality Arabica, and the cheaper and easier-to-produce Robusta. It is stated that the demand and supply for raw coffee beans of type Arabica is determined by the price of Arabica and Robusta beans:

$$\text{Demand Function: } QA_t = \alpha_0 + \alpha_1 PA_t + \alpha_2 PR_t + u_{1t}$$

$$\text{Supply Function: } QA_t = \beta_0 + \beta_1 PA_t + \beta_2 PA_{t-1} + u_{2t}$$

$$\text{Equilibrium condition: } QA_t = QA_t$$

Where QA_t =quantity Arabica beans demanded and supplied (in tonnes), PA_t = price of Arabica coffee beans, PR_t =price of Robusta coffee beans, PA_{t-1} =lagged price of Arabica coffee beans.

- A. Do you expect a positive or a negative value for $\alpha_1, \alpha_2, \beta_1$ and β_2 ? Base your answer on economic theory.
- B. Why does individual estimation of the equations lead to inconsistent parameters?
- C. Is the model identified, according to the order condition? Explain your answer.
- D. Rewrite the system of above structural equations in the reduced form. In order to do so, follow these steps:
 - I. Equalize QA demanded and QA supplied.
 - II. Rewrite, so that the equilibrium price, PA_t , is the only dependent variable, with PA_{t-1} and PR_t as explanatory variables. (solve for PA_t)
 - III. Substitute this PA_t in the demand or the supply function, so that the equilibrium quantity (QA_t) is the only dependent variable, with PA_{t-1} and PR_t as explanatory variables.
- E. Suppose these are the outcomes of the OLS regression on the reduced form equations:

$$PA_t = \pi_0 + \pi_1 PR_t + \pi_2 PA_{t-1} + v_{1t}$$

$$QA_t = \pi_3 + \pi_4 PR_t + \pi_5 PA_{t-1} + v_{2t}$$

Parameter	Parameter estimate from OLS
π_0	8
π_1	0.2
π_2	0.4
π_3	44
π_4	0.6
π_5	-0.8

Use the above parameter estimates of the reduced model, to identify the structural parameters.

- I. Solve for β_1 & α_1 (tip: use the estimates and the formulas of π_1 & π_4 for solving for β_1 , and π_2 & π_5 for solving for α_1)
 - II. Solve for β_2 & α_2
 - III. Solve for β_0 & α_0
 - IV. Write the structural equations including the ILS parameter estimates⁵
- F. For each parameter estimate, check if the sign (- or +) makes sense from an economic point of view (law of demand, law of supply, cross-elasticity of demand).

2SLS

- a. Describe the steps of the 2SLS procedure.
- b. Explain how the 2SLS procedure solves the simultaneity bias problem.
- c. For a system of two equations (one predicting the price (P) and one predicting the wages (W)), the following structural parameter estimates were obtained from running OLS and 2SLS:

$$OLS: \hat{W}_t = 0.276 + 0.258 \hat{P}_t + 0.046 \hat{P}_{t-1} + 4.959 V_t \quad R^2 = 0.924$$

$$OLS: \hat{P}_t = 2.693 + 0.232 \hat{W}_t - 0.544 \hat{X}_t + 0.247 \hat{M}_t + 0.064 \hat{M}_{t-1} \quad R^2 = 0.982$$

$$2SLS: \hat{W}_t = 0.272 + 0.257 \hat{P}_t + 0.046 \hat{P}_{t-1} + 4.966 V_t \quad R^2 = 0.920$$

$$2SLS: \hat{P}_t = 2.686 + 0.233 \hat{W}_t - 0.544 \hat{X}_t + 0.246 \hat{M}_t + 0.046 \hat{M}_{t-1} \quad R^2 = 0.981$$

W_t , P_t , M_t , and X_t are percentage changes in earnings, prices, import prices, and labor productivity (all percentage changes are over the previous year), respectively, and where V_t represents unfilled job vacancies (percentage of total number of employees).

As can be observed, the differences between the OLS and the 2SLS outcomes are very small.

Indicate for the following statements if they are true or false:

- I. Since the OLS and 2SLS results are practically identical, the 2SLS results are meaningless.
- II. We will always see similar outcomes for OLS and 2SLS if the equation is exactly identified.
- III. Since the OLS and 2SLS results are practically identical, the correlation between W and u and between P and u was insignificant.
- IV. There is little/no simultaneity bias in this system of equations.

⁵ If you did well, you've obtained the following structural parameter estimates: $\alpha_0=60$; $\alpha_1=-2$; $\alpha_2=1$; $\beta_0=20$; $\beta_1=3$; $\beta_2=-2$

Lab class

Open the datafile wages.dta in Stata.

Consider the following system of equations:

$$W_t = \beta_0 + \beta_1 UN_t + \beta_2 P_t + U_{1t} \dots \dots \dots (1)$$

$$P_t = \alpha_0 + \alpha_1 W_t + \alpha_2 I_t + U_{2t} \dots \dots \dots (2)$$

W_t =Nominal Wage Rate (in Birr)

UN_t = Unemployment rate (in %)

P_t =Consumer price index

I_t =Import price index

t = time

U_{1t} & U_{2t} = Stochastic disturbances

A person without knowledge of SEM decides to run just two simple OLS models to estimate the parameters. Please do so (use the command **regress**)

- A. Run the OLS for equation (1)
- B. Save the residuals of the regression (use the command **predict res1, res**)
- C. Run the OLS for equation (2)
- D. Save the residuals of the regression (use the command **predict res2, res**)
- E. Proof that “the assumption of no correlation between the explanatory variable(s) and the disturbance term is violated” by making a scatterplot of the residuals vs. the independent variable(s). Use the command **twoway (scatter Pres1)** and **twoway (scatter W res2)**.

- 1) Based on the outcome of f, is there simultaneity bias? Explain your answer

Let's investigate the world cotton market. The following variables are available in the datafile cotton.dta:

T=time

QC= quantity of cotton supplied and demanded in million tons

PC= Cotton, CIF Liverpool, US cents per pound

PW=Wool, coarse, 23-micron, Australian Wool Exchange spot quote, US cents per kilogram

PH= Hides, wholesale dealer's price, US, Chicago, fob Shipping Point, US cents per pound

We will investigate the following system of equations:

Demand Function: $QC_t^d = \alpha_0 + \alpha_1 PC_t + \alpha_2 PW_t + \alpha_3 PH_t + u_{1t}$

Supply Function: $QC_t^s = \beta_0 + \beta_1 PC_t + \beta_2 T + u_{2t}$

Of course, $QC_t^d = QC_t^s$

- 2) What are the endogenous and what are the predetermined variables?
 - 3) Are the equations over-, under-, or exactly identified, according to the order condition for identification?
 - 4) Why may it be useful to include the variable T in the supply function? Refer to what you learned in chapter 2.
 - 5) Why is 2SLS in this case a better estimation method than ILS?
- F. Estimate the **first stage** of the 2 stages of 2SLS (in other words, run an OLS regression with PC as dependent variables, and all the predetermined variables as explanatory variables) by using the command **regress**.
- G. Save the predicted values for PC (use the command: **predict PChat, xb**)

- H. Estimate the **second stage** of the 2 stages of 2SLS. Estimate the supply and the demand function as described above, using $\widehat{PC}_t$ instead of PC_t . Use the command **regress** two times (one per equation).
- 6) List the parameter estimates of $QC_t^d = \alpha_0 + \alpha_1 \widehat{PC}_t + \alpha_2 PW_t + \alpha_3 PH_t + u_{1t}$
 - 7) List the parameter estimates of $QC_t^s = \beta_0 + \beta_1 \widehat{PC}_t + \beta_2 T + u_{2t}$
 - 8) Which of the listed parameter estimates are significantly different from zero at the 5% significance level?
- I. It is also possible to run 2SLS directly in Stata. Use the command **ivregress 2sls QC PW PH (PC = T)** for the demand equation, and **ivregress 2sls QC T (PC = PW PH)** for the supply equation. Give it a try.
- 9) Compare the outcomes of g and f. Is there any difference? Why (not)?

4. INTRODUCTION TO PANEL DATA REGRESSION MODELS

4.1. Introduction

Up until now, we have covered cross-sectional and time series regression analysis using pure cross-sectional or pure time series data. As you know, in cross-sectional data, values of one or more variables are collected for several sample units, or entities, at the same point in time (e.g. Crime rates in Arba Minch for a given year). In time series data we observe the values of one or more variables over a period of time (e.g. GDP for several quarters or years). While these two cases arise often in applications, data sets that have both cross-sectional and time series dimensions, commonly called **panel data**, are being used more and more often in empirical research as such data can often shed light on important policy questions. **In panel data**, the same cross-sectional unit (for example: individuals, families, firms, cities, states, etc) is surveyed over time. That is, panel data have space as well as time dimensions.

For example, a panel data set on individual farmer's (household heads) farm productivity (quintals per hectare), fertilizer applied (DAP in Kg), labour days (in hours), whether the farmer takes farm related training is collected by randomly selecting farm households from a population at a given point in time (2005). Then, these same people are re-interviewed at several subsequent points in time (say 2010 and 2015). This gives us data on productivity, fertilizer, labor, and training, for the same group of farm households in different years (see Table-4.1).

There are other names for panel data, such as pooled data (pooling of time series and cross-sectional observations), combination of time series and cross-section data, micro panel data, longitudinal data (a study over time of a variable or group of subjects), event history analysis (e.g. studying the migration over time of subjects through rural/urban/international areas) and cohort analysis (e.g. following the career path of 1965 graduates of AMU). Although there are subtle variations, all these are about **a movement over time of cross-sectional units**.

Why Panel Data?

What are the **advantages of panel data over cross-section or time series data**? The following advantages are described by Baltagi (1995):

1. Since panel data relate to individuals, firms, states, countries, etc., over time, there is bound to be heterogeneity in these units. The techniques of panel data estimation can take such heterogeneity explicitly into account by allowing for individual-specific variables, as we shall show shortly. We use the term *individual* in a generic sense to include microunits such as individuals, firms, states, and countries.
2. By combining time series of cross-section observations, panel data give more informative data, more variability, less collinearity among variables, more degrees of freedom and more efficiency.

3. By studying the repeated cross section of observations, panel data are better suited to study the *dynamics of change*. Spells of unemployment, job turnover, and labor mobility are better studied with panel data.
4. Panel data can better detect and measure effects that simply cannot be observed in pure cross-section or pure time series data. For example, the effects of minimum wage laws on employment and earnings can be better studied if we include successive waves of minimum wage increases in the federal and/or state minimum wages.
5. Panel data enable us to study more complicated behavioral models. For example, phenomena such as economies of scale and technological change can be better handled by panel data than by pure cross-section or pure time series data.
6. By making data available for several thousand units, panel data can minimize the bias that might result if we aggregate individuals or firms into broad aggregates.

If each cross-sectional unit has the same number of time series observations, then such a panel (data) is called a **balanced panel** or **complete panel** (see for instance **Table-4.1**). If the number of observations differs among panel members, where some cross-sectional elements have fewer observations, we call such a panel an **unbalanced panel**. The same estimation techniques are used in both cases; however, we will be concerned with a **balanced panel** in this chapter. Initially, we assume that the X 's are nonstochastic and that the error term follows the classical assumptions, namely, $E(u_{it}) \sim N(0, \sigma^2)$.

4.2. Estimation of Panel Data Models: The Fixed Effects Approach

A simple solution to estimating panel data would be to pool all the observations. That means that all observation together, no matter the time (t) or the individual (i), will be analyzed as a group, by using OLS. This is called **pooled OLS regression**. Doing this, all individual characteristics that are not included in the explanatory variables are ending up in the error term (i.e. the error term captures differences over time and individuals). In other words, we lose the heterogeneity that may exist between the individuals. Despite its simplicity, the pooled OLS regression may distort the true picture of the relationship between the dependent variable and the regressors across the cross-sectional units; individuals or companies. As this method is unrealistic and undesirable, the **Fixed Effects Model (FEM)** of panel data analysis is preferred over OLS.

This is because the **FEM** accounts for the “individuality” of each cross-sectional unit by letting the intercept, α , to vary for each cross-sectional unit, but assumes that the slope coefficients, β , are constant across cross-sectional units. Assuming that the intercept term varies for cross-sectional units, we can specify a model as:

$$Y_{it} = \alpha_i + X'_{it}\beta + u_{it} \quad (4.1)$$

- Notice that we have put the subscript i on the intercept term to suggest that the intercepts of the cross-sectional units (perhaps firms) may be different; the differences may be due to special features of each firm.

- Note that if we were to write the intercept as α_{it} , it will suggest that the intercept of each cross-sectional unit or firm is **time variant**.
- Note also that the matrix of β of equation (4.1) is the same for all cross-sectional units, and for all time periods. Only α - the intercept, varies per cross-sectional unit (i).

Equation (4.1) is known as the **Fixed Effects Model (FEM)**. The term “**Fixed Effects**” is due to the fact that each cross-sectional-unit’s intercept, although different from the intercepts of other cross-sectional units, does not change over time. The intercept for each cross-sectional unit is **time-invariant**.

How do we actually allow for the (fixed effect) intercept to vary between cross-sectional units? We can easily do that by the dummy variable technique that we learned in Chapter 1, particularly, the **differential intercept dummies**. Remember from chapter 1 that we have to prevent the dummy variable trap. This means that we must include M-1 dummies. If the number of individuals is 50, we need 50-1=49 dummy variables to represent the 50 different individuals in the datafile. This approach to the fixed effects model is called the **least-squares dummy variable (LSDV) model**.

A disadvantage of the dummy variable approach is that the R^2 is usually very high, because all the dummy variables are explanatory variables and the R^2 always increases if more (dummy) explanatory variables are introduced in the model. Also it requires a lot of memory on the computer, which is not always available. An alternative approach to estimating the fixed effects model is the **within estimation method**. This method is used by Stata. In the within estimation method the variables (Y and X’s) are all demeaned, i.e. the time-averaged values are deducted from each value.

$$Y_{it} - \bar{Y}_i = (\alpha_i - \bar{\alpha}_i) + \beta(X_{it} - \bar{X}_i) + (u_{it} - \bar{u}_i)$$

Note that α_i is time invariant, so $(\alpha_i - \bar{\alpha}_i) = 0$. Therefore the within estimation for a model with one explanatory variable (X) can be specified as:

$$Y_{it} - \bar{Y}_i = \beta_1 + \beta_2(X_{it} - \bar{X}_i) + (u_{it} - \bar{u}_i) \quad (4.2)$$

Where:

β_1 is the average of the individual effects (α_i)

$\bar{Y}_i$ is the time-average Y value of individual i

$\bar{X}_i$ is the time-average X value of individual i.

$\bar{u}_i$ is the time-average error term for individual i.

▪ Example FEM

In order to estimate the effect of diet on study results, the university collected data from 50 students that eat university food for 6 semesters. For all 50 students all information is available for the 6 semesters, so we work with a **balanced panel**. The expectation is that eating proteins (present in meat and eggs) has a positive effect on brain capacity, and thus improves the study results. The variables in the model include:

Y_{it} the average study results of student i in semester t

P_{it} how many meals with egg and meat student i ate during semester t

G_i a dummy variable equal to 1 if the student is female, and 0 if the student is male
 A_{it} the age in years of student i during the end of semester t
 S_i the average state exam results of student i (before joining the university)
 H_{it} the average weekly study hours (class attendance) of student i during semester t

$$Y_{it} - \bar{Y}_i = \beta_1 + \beta_2(P_{it} - \bar{P}_i) + \beta_3(G_i - \bar{G}_i) + \beta_4(A_{it} - \bar{A}_i) + \beta_5(S_i - \bar{S}_i) + \beta_6(H_{it} - \bar{H}_i) + \dots + (u_{it} - \bar{u}_i) \quad (4.3)$$

Note that the **time-invariant** variables (in this case G_i and S_i) will be omitted, because their mean values ($\bar{G}_i$ and $\bar{S}_i$) will be the same as the values for each t ; there is no change over time. Therefore, the **FEM** does not provide parameter estimates for time-invariant variables.

Let's look at sample output of this **FEM**:

Fixed-effects (within) regression			Number of obs		=	300
Group variable: i			Number of groups		=	50
R-sq:			Obs per group:			
within	=	0.3818	min		=	6
between	=	0.5217	avg		=	6.0
overall	=	0.4177	max		=	6
corr(u_i, Xb) = 0.1335			F(3,247)		=	50.86
			Prob > F		=	0.0000

Yit	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
Pt	.0584188	.0394064	1.48	0.139	-.0191966	.1360342
Gi	0	(omitted)				
Ait	-1.026449	.9252394	-1.11	0.268	-2.848814	.7959164
Si	0	(omitted)				
Hit	3.782075	.3089015	12.24	0.000	3.173658	4.390492
_cons	51.11189	17.7705	2.88	0.004	16.11085	86.11293
sigma_u	5.1439599					
sigma_e	9.0448019					
rho	.2443948	(fraction of variance due to u_i)				

F test that all u_i=0: F(49, 247) = 1.68			Prob > F = 0.0058		
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Indeed, the individual differential intercept coefficients are not presented in the output, but they are taken into account in estimating the model (as this is a FEM). **This output shows the following:**

- The F-test that tests the mutual significance of all the coefficients in the model is highly significant ($F(3, 247) = 50.86, p = 0.000$). This means that the model is able to explain the variance in Y_{it} .
- The parameter coefficients, and their t-tests reveal that, when controlled for individual differences, the study hours (H_{it}) have a significant effect on study results (Y_{it}), because $p > |t| < 0.05$. The effects of protein intake (P_t) and age (A_{it}) on study results (Y_{it}) are insignificant, because $p > |t| > 0.05$. No estimates for the effect of gender (G_i) and state exam result (S_i) are given, because the fixed effects model can only estimate the effect of time variant variables, and G_i and S_i do not change over time.

- σ_u (5.14) is the standard deviation of the time-invariant error term (u_i), which is available for every individual (i). In this example, it expresses the variety in the time-constant error term of the 50 students. If σ_u is large, it means that students are very different with regards to Y when controlled for the X's. σ_e (9.04) is the standard deviation of the residuals (the over-all error term; e_i). ρ (0.244) is the share of the estimated variance of the overall error accounted for by the individual effect, u_i . A high ρ indicates that the fixed effect model is much better than the pooled regression, because much of the error term can be explained by differences in individuals.
- The F-test that all u_i 's are 0 ($F=1.68$) reveals that there are significant differences between students ($\text{Prob}>F=0.0058$). This supports the use of the Fixed Effects model.

▪ Types of FEM

The above-discussed model is known as a **one-way fixed effects model**, because we have allowed the intercept to differ per individuals (i.e. cross-sectional units). We can also allow the time periods to have different intercepts, by adding $t - 1 = 6 - 1 = 5$ dummies for the different semesters. If we consider fixed effects both for individuals and time periods, the model is called a **two-way fixed effects model**. In that case, the within estimation method can be used for the individual effect and time dummies must be introduced to capture the deterministic trend (chapter 2). However, allowing for individual fixed effects and time fixed effects results in an enormous loss in degrees of freedom. Degrees of freedom are the total number of observations in the sample less the number of independent (linear) constraints or restrictions put on them. In the above model the number of observations is 300. The linear constraints consist of 4 β 's ($\beta_1, \beta_2, \beta_4$ and β_6), and 49 individuals ($I-1$). That means $300-4-49=247$ degrees of freedom are left. If differential intercepts related to time are added, the total degrees of freedom will be $300-4-49-5=242$. Losing too many degrees of freedom usually results in poorer statistical analysis. This is a big disadvantage of the FEM. More disadvantages are discussed below.

▪ Disadvantages of FEM

The disadvantages of the fixed effects model (**FEM**) are as follows:

- Per dummy variable added, a degree of freedom is lost. Therefore, if the sample is not very large, introducing too many dummies will leave few observations to do meaningful statistical analysis.
- It is impossible to obtain parameter estimates for time-invariant variables like gender and ethnicity.
- The model results are valid under the assumption that $\varepsilon_{it} \sim N(0, \sigma^2)$. As u has a cross-sectional component (i) and a time component (t), this assumption may not be realistic.
 - **Regarding the cross-sectional component:** if there is heteroskedasticity, we can obtain heteroscedasticity-corrected standard errors.
 - **Regarding the time component:** if there is autocorrelation we can adjust the model and assume autocorrelation. This is, however, beyond the scope of this course.

An alternative to the **Fixed Effects Model(FEM)** is the **Random Effects Model (REM)**, and this REM is able to deal with some of the above listed problems.

4.3. Estimation of Panel Data Model: The Random Effects Approach

The loss in the degrees of freedom and the fact that time-invariant variables cannot be considered as explanatory variables in the **Fixed Effect Approach**, made econometricians consider and develop another approach in analyzing panel data. In the FEM, dummy variables compensated for all the information of the cross-sectional unit (the individual, the region, the firm or the country) which was not considered in the explanatory variables. In the Random Effects Model (REM) this individual-specific element is not included as dummy variable, but included in the error term. A classical **Random Effects Model** is called the **Error Component Model (ECM)**, which assumes that the error of the model consists of an individual and a common component.

Instead of assuming that all the cross-sectional units are purposively selected, the ECM assumes that the cross-sectional units are randomly selected from a pool of possible cross-sectional units. Let us consider the example of panel data from 30 local restaurants with monthly data from 2016-2017. For each restaurant, the monthly profit (Y) and the average number of customers on weekday evenings (X) were described.

$$Y_{it} = \beta_{1i} + \beta_2 X_{it} + \varepsilon_{it} \quad (4.4)$$

The ECM assumes that the 30 restaurants included are a drawing from a much larger universe of all the local restaurants in the study area. All these restaurants (the whole population of restaurants) have a common mean value for the intercept (β_1) and individual differences in the intercept values of each local restaurant are reflected in the error term u_i . Therefore,

$$\beta_{1i} = \beta_1 + u_i \quad (4.5)$$

Substituting (4.5) in (4.4), we get

$$\begin{aligned} Y_{it} &= \beta_1 + \beta_2 X_{it} + u_i + \varepsilon_{it} \\ Y_{it} &= \beta_1 + \beta_2 X_{it} + w_{it} \end{aligned} \quad (4.6)$$

$$\text{Where, } w_{it} = u_i + \varepsilon_{it}$$

The error term, w_{it} , consists of two components: u_i , which is the individual-specific error (in this case, the restaurant-specific error) and ε_{it} , which is the combined time series and cross-section error component (the error that cannot be explained by the independent variables (X's) and the individual characteristics (u_i)).

For the error terms the ECM assumes (with meaning after $\rightarrow$):

$$u_i \sim N(0, \sigma_u^2)$$

$\rightarrow$ the individual error term is normally distributed with variance σ_u^2

$$\varepsilon_{it} \sim N(0, \sigma_\varepsilon^2)$$

$\rightarrow$ The combined time-series and cross-section error component is normally distributed with variance σ_ε^2

$$E(u_i \varepsilon_{it}) = 0$$

→The error components u_i and ε_{it} are not correlated with each other

$$E(u_i u_j) = 0 \quad (i \neq j)$$

→The individual error components are not correlated with each other

$$E(\varepsilon_{it} \varepsilon_{is}) = E(\varepsilon_{it} \varepsilon_{js}) = E(\varepsilon_{it} \varepsilon_{jt}) = 0 \quad (i \neq j; t \neq s)$$

→The time series and cross section error component is not autocorrelated.

Notice the difference between the FEM and this ECM. In the FEM each individual has its own fixed intercept value. In the ECM, on the other hand, the intercept β_1 represents the mean value of all these individual intercepts and the error component, u_i , represents the random deviation from this mean. We can state that $E(w_{it}) = 0$ and $\text{var}(w_{it}) = \sigma_u^2 + \sigma_\varepsilon^2$. If $\sigma_u^2 = 0$, there are no differences in the individual intercepts, so we can pool all the data and run a pooled OLS regression. But if $\sigma_u^2 > 0$, it is useful to run a panel data model. In the ECM, the disturbance terms, w_{it} for one specific individual (cross-sectional unit) over time are, of course, correlated, because w_{it} contains u_i , which is constant for every individual. Therefore, using OLS to estimate the model will lead to inefficient outcomes. The most appropriate method to estimate an ECM, is the method of Generalized Least Squares (GLS). It goes beyond the scope of this course to explain GLS mathematically, but we can get the outcome from Stata or Eviews also without understanding the mathematics.

▪ Example of an ECM

Small local coffee shops in Arba Minch offer tea, coffee, soft drinks and sometimes snacks like bananas, bombolinos or samosas. Furthermore, some of them offer the opportunity to play checkers (dam). An owner of a small coffee shop wonders if offering checkers is good for the turnover, or bad. On the one hand, customers may choose the shop with a checkers playing board, because they like to play while drinking coffee. On the other hand, customers who play checkers use the seats for a longer time, not leaving space for others who also want to drink coffee. To find an answer to this question, the shop owner collected data from 15 randomly selected small coffee shops ($N=50$) over the period of 10 weeks ($T=10$). For each shop and each week, she collected the following data:

y_{it}	the revenue of the i^{th} shop in week t
d_{it}	if the i^{th} shop offered a dam (checkers) board (1) or not (0) during week t
s_{it}	the number of seats available for guests in the i^{th} shop in week t
c_i	if the i^{th} shop is located in a city center (1) or not (0) (city center: Sikela or Secha)

The ECM model that she ran is the following:

$$y_{it} = \beta_1 + \beta_2 d_{it} + \beta_3 s_{it} + \beta_4 c_i + u_i + \varepsilon_{it}$$

The output GLS of the ECM looks as follows:

```
. xtreg y d s c , re
```

Random-effects GLS regression		Number of obs	=	500
Group variable: shopid		Number of groups	=	50
R-sq:		Obs per group:		
within	= 0.0394	min	=	10
between	= 0.4722	avg	=	10.0
overall	= 0.3449	max	=	10
corr(u_i, X) = 0 (assumed)		Wald chi2(3)	=	60.10
		Prob > chi2	=	0.0000

y	Coef.	Std. Err.	z	P> z	[95% Conf. Interval]	
d	416.9629	92.33132	4.52	0.000	235.9968	597.929
s	335.3458	51.93577	6.46	0.000	233.5535	437.138
c	44.71502	272.5252	0.16	0.870	-489.4246	578.8546
_cons	-940.9676	616.3139	-1.53	0.127	-2148.921	266.9854
sigma_u	915.77763					
sigma_e	846.23356					
rho	.53940719	(fraction of variance due to u_i)				

The output gives parameter estimates both for the time variant (d and s) and the time-invariant (c) variable. Based on this output, the following conclusions can be drawn:

- The Wald Chi² test that tests the mutual significance of all the coefficients in the model is highly significant (Chi²(2)=60.10, p=0.000). This means that the model is able to explain the variance in Y_{it}.
- The parameter coefficients, and their t-tests reveal that the availability of a dam board (d) and the number of seats (s) both have a significant effect on the revenue (y_{it}), because p>|t|<0.05. Presence in the city center has no significant effect on the revenue, because p>|t|>0.05. Also, the constant term, the intercept, is not significantly different from zero.
- Sigma_u (915.78) is the standard deviation of the individual error term (u_i) and sigma_e (846.23) is the standard deviation of the combined time series and cross-section error component (the over-all error term; e_i). Rho (0.539) is the share of the estimated variance of the overall error accounted for by the individual effect, u_i. It would not be good to use pooled OLS regression in this case, because pooled OLS would not consider the individual effect and, therefore, not be able to explain much of the variance in Y.
- To answer the question of the lady who runs the small coffee shop: we can conclude from this model that offering a checkers board has a significant positive effect on the revenue of a local coffee shop. When controlled for the other independent variables, shops that offer dam make, on average, 417 birr extra per week.

▪ ECM vs FEM

Thus far we have seen three possibilities to analyze panel data in a model: pooled OLS regression, the Fixed Effect Model (FEM) and the Random Effect Model, or Error Correction Model (ECM). Which of the three is best? That depends on how the error term behaves.

- Use **pooled OLS regression** if the individuals (cross sectional units) are homogenous and if the time periods are homogenous. We can check this by:
 - Individual homogeneity: If, in the output of the FEM and the ECM σ_u (σ_u) is close to 0, it means that individuals are homogenous (there is no individual heterogeneity). If this is the case, pooled OLS regression can be used.
 - Time homogeneity: Check if the time-variant variables are stationary (chapter 2). If they are stationary, you can use pooled OLS regression. If this is not the case, pooled OLS regression is not optimal and you could use the FEM model with time trend dummies (if the dependent variable has a deterministic trend). Alternatively, you can run time series econometrics models, which are outside the scope of this course.

If there is heterogeneity of the cross-sectional units (or over time), it is better to use the **Fixed Effects Model** or the **Error Correction model**. How to choose between these two? The decision depends on several factors:

- If T (number of time series data) is large and N (number of cross-sectional units/individuals) is small, there is likely to be little difference in the parameter estimates of FEM and ECM. Because FEM is easier, it is preferred to use FEM.
- When N is large and T is small, the estimates obtained by the two methods can differ significantly, because there are many different u_i 's. If the assumptions underlying ECM hold, ECM estimators are more efficient than FEM estimators. Furthermore, the ECM can also give parameter estimates for time-invariant variables.

How to know whether the assumptions underlying ECM hold?

- a. One assumption underlying ECM is that the individuals (cross-sectional components) are randomly selected from the research population. Because the ECM assumes random selection, statistical inference can **only** be done if this is the case. If the selection of i 's is not random, the FEM is appropriate.
- b. Next to the randomness of selected sample units, it is important to see if the individual error component, u_i , and one or more regressors (X 's) are correlated. If the individual error term is not related to any of the regressors, it is likely that the sample is randomly selected, and that β_1 is indeed the common intercept (with a random deviation u for each i). If, however, the X 's and the u_i are correlated, the assumption that the i 's are randomly selected is invalid, so ECM is inappropriate and FEM will give less biased parameter estimates.

A test devised by **Hausman**, which is incorporated also in Stata and Eviews, can be used to check if the ECM is better than the FEM.

Null-hypothesis of the Hausman test

The null hypothesis is that there is no big difference between the parameter estimates of FEM and ECM. If the null hypothesis is correct, the individual errors (u_i) are distributed independently of the explanatory variables (X_{it}), so they do not bias the parameter estimates in the ECM model. If this is correct, both random effects and fixed effects are consistent, but fixed effects will be inefficient because it involves estimating an unnecessary set of dummy variable coefficients.

Alternative hypothesis of the Hausman test

If the null hypothesis is rejected, it means that the parameter estimates of the FEM and the ECM differ significantly. Therefore, the random effects (ECM) estimates will be subject to unobserved heterogeneity bias, because the individual errors (u_i) are related to the explanatory variables (X_{it}). Therefore, if the null hypothesis is rejected, the FEM must be used.

▪ Example Hausman test

Parlasca et al. (2018) collected data to see if a mobile phone can increase household food security in pastoral communities in Northern Kenya.

This information is from their abstract: *Mobile phones could potentially facilitate access to food markets and thus improve food security and nutrition, but research on such types of effects remains scarce. In this study we analyze whether mobile phones improve dietary quality of pastoralists in Northern Kenya. We use six rounds of household panel data covering the period between 2009 and 2015. During this period, mobile phone ownership in the sample increased from less than 30% to more than 70%.*

The authors measured the following variables:

Dependent variable:

HDDS: household dietary diversity score, which is a common tool to assess food security and dietary diversity. The minimum score is 1 (all food is from the same group), the maximum score is 12 (as all food types are classified in 12 categories). This was measured in two ways, representing two different variables:

- Number of food groups where consumption is mainly from self-production (0-12)
- Number of food groups where consumption is mainly from other sources (0-12) (e.g. purchases, food aid or gifts)

Independent variables:

- | | |
|--|--|
| - Mobile phone ownership + use. | - Cooking appliance: 1 if the household uses any form of advanced cooking appliance. |
| - Income | - Household size |
| - Herd size (in TLU= either 1 head of cattle, or 0.7 of a camel, or 10 goats, or 10 sheep) | - Education, gender and age of the household head |
| - Land farmed (hectares) | |
| - Radio ownership | |

- Dummy variables for the years in the sample (from 2009-2013), because of deterministic trend

The results of their Fixed Effects regression are displayed in Table 4.2.

As the results reveal, mobile

ownership especially increased diversity in food consumption from non-self-produced sources. Income positively contributed to the diversity in food consumption, both from self-production and from other sources. The herd size is only (weakly) significant for the diversity of consumption of self-produced foods. The household size is significant for both diversity self-production and food from other sources. The education of the household head contributes positively to the diversity in self-production,

Table 4.2: Results of the regression of Parlasca et al (2018)

Independent variables	Food groups from self-production	Food groups from other sources
Mobile phone ownership & use	-0.011 (0.050)	0.238** (0.097)
Income	0.557* (0.261)	1.812*** (0.378)
Herd size	0.002* (0.001)	0.002 (0.001)
Land farmed	-0.001 (0.010)	0.012 (0.010)
Radio ownership	0.016 (0.078)	-0.221 (0.235)
Cooking source	-0.068 (0.130)	0.186 (0.259)
Household size	0.028*** (0.009)	0.054** (0.024)
Education household head	0.075** (0.026)	-0.121* (0.064)
Gender household head (1=female)	0.045 (0.089)	0.098 (0.179)
Age household head	0.002 (0.002)	-0.003 (0.003)
2009	-0.755*** (0.222)	-0.015 (0.453)
2010	-0.405** (0.179)	0.116 (0.476)
2011	-0.470** (0.214)	0.008 (0.398)
2012	-0.142 (0.129)	-0.068 (0.247)
2013	0.038 (0.065)	-0.024 (0.169)
Constant	0.865** (0.299)	5.701*** (0.664)
Hausman test	96.37***	149.74***
Errors shown in parentheses are robust standard errors. *** p<0.01, ** p<0.05, * p<0.1		

but negatively to the diversity in food obtained from other sources. The seasonal dummies are significant for the diversity in consumption of self-produced foods. The big question is, if this fixed effect model is the most efficient way to estimate the impact of the independent variables on the dependent variable. To this end, the parameter estimates of this FEM must be compared with the parameter estimates of a REM (e.g. the ECM). This was done, and the following hypotheses were formed to choose between FEM and REM:

- H0: the parameter estimates of the FEM do not differ significantly from the parameter estimates of REM (in other words: the individual farmer error is not correlated to any of the independent variables).
- H1: the parameter estimates of the FEM differ significantly from the parameter estimates of REM (in other words: the individual farmer error is correlated to (at least one of) the independent variables, leading to biased parameter estimates).

From Table 4.2 we find that the test result is $\chi^2=96.37$ for the first dependent variable (about food groups from self-production). This is significant at the 0.01 level (visible from ***), so H0 must be rejected. Furthermore, Table 4.2 reveals that the test result is $\chi^2=149.74$ for the second dependent variable (about food groups from other sources). This is significant at the 0.01 level (visible from ***), so H0 must be rejected.

Concluding, for both dependent variables the FEM estimates differ significantly from the REM estimated. Therefore, the parameter estimates in the REM approach (the ECM) are biased, due to correlation between the individual error, u_i , and (at least one of) the independent variable(s). The farmers were, apparently, not very randomly selected from the total population of farmers. Therefore, the FEM is the optimal way to analyze this panel data. The authors Parlasca et al. correctly decided to use the FEM.

Concluding note: we have seen that the ECM can provide parameter estimates for time-invariant variables, such as gender and ethnicity. The FEM does not provide parameter estimates for these variables, but it does control for them by approaching each individual as an individual. To this end, the FEM controls for all possible time-invariant variables, whereas the ECM only considers the time-invariant variables that are explicitly included in the model.

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Worksheet

4.1. Introduction

- Indicate for the following data if it is cross-sectional data, time series data or panel data:
 - Yearly GDP, unemployment and inflation figures of Ethiopia from 2000-2020.
 - Monthly income and savings figures of 3000 households from 2010-2020.
 - Average academic performance of section A, B, C and D in semester 2 of the academic year 2019-2020.
 - Yearly maize production of- and technologies used by 100 farmers from 1995-2015.
 - Monthly unemployment and economic performance statistics for all Ethiopian provinces.
- Describe the advantages of panel data over cross-section or time series data in Amharic (do not copy-paste from Baltagi, but translate to Amharic to get a deep understanding).
- A balanced panel is preferred over an unbalanced panel. How come that still many analyses involve unbalanced panel data?

4.2. Estimation of Panel Data Regression Model: The Fixed Effects Approach

Gujarati (....) collected data on the civilian unemployment rate Y (%) and manufacturing wage index (1992 = 100) for Canada, the United Kingdom, and the United States for the period 1980–1999. Consider the model:

$$Y_{it} = \beta_1 + \beta_2 X_{it} + u_{it}$$

1. From an economic point of view, what is the expected relationship between Y and X ? Why?
2. Explain: how can this data be analyzed by using pooled regression?

The pooled regression gives the following results:

Source	SS	df	MS	Number of obs	=	60
Model	82.4039428	1	82.4039428	F(1, 58)	=	29.42
Residual	162.441891	58	2.80072226	Prob > F	=	0.0000
				R-squared	=	0.3366
				Adj R-squared	=	0.3251
Total	244.845834	59	4.14992939	Root MSE	=	1.6735

unempl	Coef.	Std. Err.	t	P> t	[95% Conf. Interval]	
manuf_wage_index	-.0530767	.0097851	-5.42	0.000	-.0726637	-.0334897
_cons	12.43941	.8182869	15.20	0.000	10.80143	14.07739

3. Interpret the outcome of the model: does the manufacturing wage index have a significant effect on the unemployment rate?
4. What is the disadvantage of using pooled regression?
5. How does the Fixed Effect Model (FEM) solve for this disadvantage?

Because the data is panel data, also a Fixed Effect Model (FEM) was run:

```
. xtreg unempl manuf_wage_index, fe
```

Fixed-effects (within) regression		Number of obs	=	60
Group variable: country		Number of groups	=	3
R-sq:		Obs per group:		
within	= 0.2764	min	=	20
between	= 0.9253	avg	=	20.0
overall	= 0.3366	max	=	20
corr(u_i, Xb) = 0.2906		F(1, 56)	=	21.39
		Prob > F	=	0.0000

unempl	Coef.	Std. Err.	t	P> t	[95% Conf. Interv
manuf_wage_index	-.0410612	.0088786	-4.62	0.000	-.0588472
_cons	11.47026	.739982	15.50	0.000	9.987895
sigma_u	1.118042				
sigma_e	1.4433607				
rho	.37500825	(fraction of variance due to u_i)			

F test that all u_i=0: F(2, 56) = 10.99	Prob > F = 0.0001
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- Report the results of the F-test to indicate if the model, as a whole, is able to explain the variance in unemployment (Y).
- Report the results of the t-test to indicate if the manufacturing wage index is a significant predictor for the unemployment.
- Is the standard deviation of the residual within the countries higher than the standard deviation of the overall error term? Explain your answer by referring to sigma_u and sigma_e.
- Rho is 0.375. What does that mean?

Though the Fixed Effects Model solves the main problem of a pooled OLS regression (loss of heterogeneity between the groups), there are also some disadvantages. The following questions are related to the disadvantages of the Fixed Effects Model:

- An economist wants to predict the effect of corona cases on the import of disinfectant soap. To this end, he collects panel data from December 2019- June 2020 (7 months) for all 54 African countries. For each month and each country, he collects the average active corona cases (X) and the monetary value of imported disinfectant soap (Y). The pooled regression model that he estimates is $Y_{it} = \beta_1 + \beta_2 X_{it} + u_{it}$.
 - How many degrees of freedom are lost in this simple pooled regression model?⁶

⁶ NB: the degrees of freedom equal the number of observations minus the number of parameters you need to calculate during an analysis.

- b. How many degrees of freedom are lost if he uses a one-way FEM (i.e. adding differential intercepts for the countries)?
 - c. How many degrees of freedom are lost if he uses a two-way FEM? Check page 4 of the lecture notes chapter 4, to see what's a two-way FEM.
 - d. Why is losing degrees of freedom a disadvantage of the FEM?
2. Why can time-invariant variables, such as gender and ethnicity, not be estimated in a FEM? Check page 3 of the lecture notes.

4.3. Estimation of Panel Data Regression Model: The Random Effects Approach

There are many banana farmers around Arba Minch. Each harvesting season they face a loss of crop, possibly due to fruit flies, too much or too little water availability, etc. You are asked to design a study in which panel data will be collected and an ECM will be run. To this end, do the following:

1. Identify possible determinants of crop loss (in the case of banana).
2. Indicate how these determinants will be measured (Which question will be asked in the questionnaire? Which variables will be in your econometric model?).
3. Indicate how often the data will be collected over time, and why.
4. Mathematically describe the ECM, including the variables selected above.
5. Explain why the ECM is the appropriate method for analyzing the data.

In 2019, Thanh Phan et al published a research titled: *Does microcredit increase household food consumption? A study of rural Vietnam*⁷. Data was collected in 2008, 2010, 2012, 2014 and 2016. Next to C (food consumption, the dependent variable) and K (microcredit borrowing, the explanatory variable of interest), several other variables were considered:

C	Household food consumption (1000VND)
K	Accumulated microcredit borrowing (1000VND)
S	Social network size (index)
E	Education measured as the sum of schooling gaps relative to expected schooling for household children (years)
Y	Income from sources other than the household business (1000 VND)
EduH	Schooling of household head (years)
HSize	Household members (number)
AgeH	Age of household head (years)
Plot	Land holding (m ²)
Dep	Dependent members (number)
Nfarming	Whether a household is involved in non-farming business activities (1=yes; 0=otherwise)
Shock	Whether a household experienced an income shock (1=yes; 0=otherwise)
Poor	Whether a household is classified as poor by authorities (1=yes; 0=otherwise)
Local	Whether a household is local (1=yes; 0=otherwise)
Irate	Average interest rate of loans obtained within a prefecture (%)
HiE	Highest level of education of a member in the household (number of school years)
DiR	Distance to the nearest all-weather road (km)

In this article, a panel data analysis of an unbalanced panel was presented, by using multiple instrumental variables (remember: chapter 3) and by including several lagged variables (remember: chapter 2). To prepare yourself for thesis or post-graduate work, it is very useful to read the full article. For this exercise, however, the following simplifications are made:

⁷Phan, C. T., Sun, S., Zhou, Z. Y., & Beg, R. (2019). Does microcredit increase household food consumption? A study of rural Vietnam. *Journal of Asian Economics*, 62, 39-51.

- We assume the panel data is balanced (N=1537, T=6)
- We assume that C is the dependent variable, and logK, logY, EduH, HSize, AgeH, Nfarming and Shock are the independent variables of interest. The other variables of the above model, and lagged variables are not considered.

1. What does it mean that we assume that the panel data is balanced?
2. Why would it be wise to include logK and logY rather than K and Y? (remember econometrics I, from last semester)
3. Specify the Error Components Model for the above-described case, including the variables of interest.

The following output, based on the GLS estimation method, is given:

C	Coefficients	Robust SE
logK	0.461***	0.178
logY	-0.0532*	0.029
EduH	-0.0875**	0.037
HSize	-0.302**	0.132
AgeH	-0.0106	0.013
Nfarming	-0.796***	0.231
Shock	-0.367***	0.116
Note: ***, **, and * indicate significance at 1, 5, and 10%, respectively.		

4. Summarize the outcome of this model.
5. Write a conclusion to the above stated research question *Does microcredit increase household food consumption?* based on the above presented model.

Fixed effects or Random effects?

Kumari and Shamar (2017)⁸ published an article in which they presented their panel data model with the following (fragment of) abstract:

Purpose

The purpose of this paper is to identify key determinants of foreign direct investment (FDI) inflows in developing countries by using unbalanced panel data set pertaining to the years 1990-2012. This study considers 20 developing countries from the whole of South, East and South-East Asia.

Design/methodology/approach

Using seven explanatory variables (market size, trade openness, infrastructure, inflation, interest rate, research and development and human capital), the authors have tried to find the best fit model from the two models considered (fixed effect model and random effect model) with the help of Hausman test.

Findings

Fixed effect estimation indicates that market size, trade openness, interest rate and human capital yield significant coefficients in relation to FDI inflow for the panel of developing countries under study. The findings reveal that market size is the most significant determinant of FDI inflow.

The dependent variable is FDI inflow, and the explanatory variables are market size, trade openness, infrastructure, inflation rate, interest rate (IR), research and development (R&D) and human capital (HC). The cross-sectional units are 18 countries in South- East- and South-East Asia with different levels of development. The authors have estimated the Fixed Effects and the Random Effects Model.

⁸Kumari, R., & Sharma, A. K. (2017). Determinants of foreign direct investment in developing countries: a panel data study. *International Journal of Emerging Markets*.

1. Which approach, fixed effects or random effects, do you think is most suitable for this case? Base your answer on the case description and on lecture notes p. 8-11.
2. The results of both approaches is presented in the below table (Table V). Compare the results: what are the biggest differences?

Independent variables	Fixed effects (within) regression	Random effects GLS regression
Constant	-7.84 (-3.58)	5.05 (5.96)
Market size	1.63 (7.39)**	0.33 (3.85)**
Trade openness	0.63 (1.17)	-1.108 (-2.19)
Infrastructure	-0.00012 (-2.22)**	0.000072 (1.48)**
Inflation	-0.00055(-3.67)**	0.0044 (4.66)
Interest rate	-0.0084 (-3.67)**	-0.0096 (-3.91)**
Research and development	-1.09e-06 (-1.27)	5.47e-07 (0.64)
Human capital	0.0037 (2.82)**	0.0046 (3.33)
No of countries	18	18
No of observations (unbalanced sample)	304	304
Overall R squared	0.54	0.48
	F-statistic: 46.88 (p:0.000)	Wald chi-square: 262.88 (p:0.000)
Hausman p-value	0.000	

Notes: FDI is measured as net inflows (BOP current USD). Market size is measured as the GDP (current USD). GDP indicates economic growth and standards of living. Trade openness (asured as an import plus export divided by GDP per capita (current USD))) describes the openness of the economy. Infrastructure measured by proxies of electric power consumption (kWh per capita). Interest rate measured by real interest rate (percent). Research and development is measured as the patent applications residents registered every year. Human capital is measured as the school enrollment secondary (percent gross). 1990-2021. t-statistics are in parentheses. ** significant at the 5% level.

3. Based on the information in the above table, would you still make the same choice between the Fixed Effects Approach and the Random Effects Approach? Refer, at least, to the Hausman test in your answer.

Lab class

A classical key to economic growth is investment. Next to attracting foreign investment, household savings can also be used to fund investment. In order to save, a household needs surplus income. Many panel data studies have been done into the relation between household income and household savings. In this lab class you will analyze (fake) income- and savings data of 221 rich households from Hawassa.

S_{it} = Savings of household i in period t in ETB

I_{it} = Income of household i in period t in ETB

F_{it} = Family size of household i in period t

C_i = If the household owns a car or another expensive asset at the start of the study

First consider the following saving function:

$$S_{it} = \beta_0 + \beta_1 I_{it} + \beta_2 F_{it} + \beta_3 C_i + u_{it}$$

- A. Open Stata, and within Stata the file Householddata.dta.
- B. Use the command **summarize savings**.
 - 1) What is the highest saving? And what is the minimum amount saved?
- C. Run the above presented model using pooled OLS regression. **reg savings income fam_size Car_or_other_asset**
 - 2) Describe the effect of income, family size and asset ownership (like a car) on savings.

- 3) Does this model allow for heterogeneity among the households? Explain your answer.
- 4) Use the command **xtset householdID time** to explain Stata that you are using panel data.
- D. Run the fixed effect model, by using the command **xtreg savings income fam_size Car_or_other_asset,fe**
 - 5) Why is the variable Car_or_other_asset omitted?
 - 6) Describe the effect of income and family size on savings.
 - 7) Is the model able to explain the variation in household saving? Refer to the F-test
 - 8) List the values for σ_u , σ_e and ρ , and explain what they mean.
 - 9) Is it better to use the Fixed Effects Model or Pooled OLS regression for this dataset? Give at least one reason for your answer.
- E. Save the estimates of this fixed effects regression model, so we can use them later in the Hausman test. Use the command **est store fixed**.
- F. Now run the Random Effects Model, by using the command **xtreg savings income fam_size Car_or_other_asset,re**
 - 10) Describe the difference between the fixed effect model and the random effects model
 - 11) Describe the effect of income, family size and car (or other asset) ownership on savings.
 - 12) Is the model able to explain the variation in household saving? Refer to the Wald Chi-squared test.
 - 13) List the values for σ_u , σ_e and ρ , and explain what they mean.
- G. Save the estimates of this random effects model, by using the command **est store random**.
- H. Run the Hausman test which compares the estimated parameters of the FEM and the REM. Use the command **hausman fixed random**.
 - 14) What is the null hypothesis of the Hausman test in this case?
 - 15) Report the chi-square test statistic of the Hausman test, and its p-value.
 - 16) Which model is better to be used? REM or FEM? Base your answer on the outcome of the Hausman test.